MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

COUNTER, BINARY/BCD, DIGITAL, KEY CODE BMG

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic binary/BCD counter module.

* 1.2 Classification. This specification covers two classes of modules.

Class I and class II as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20363, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.398 failures per million operating hours for class I or 4.319 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, failure rate for class I shall be used for reliability prediction (see 6.3).

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I. Class I modules shall conform to the electrical requirements of class I only, and class II modules shall conform to the electrical requirements of class I and class II (see 6.3).

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be BMG.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/275-1 for class I and M28787/275-2 for class II, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

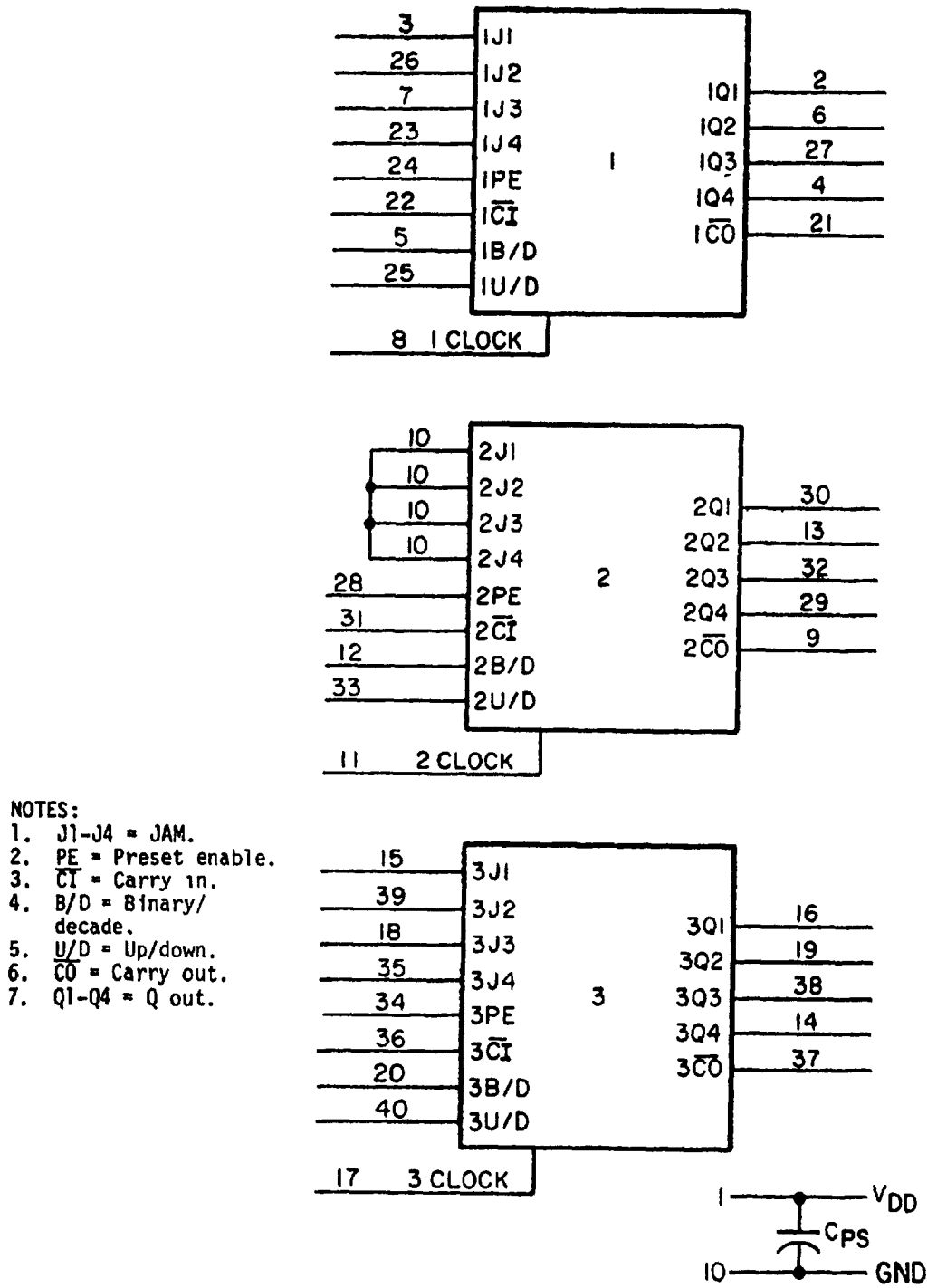
4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

* 4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

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NOTES:
1. J1-J4 = JAM.
2. PE = Preset enable.
3. CI = Carry in.
4. B/D = Binary/ decade.
5. U/D = Up/down.
6. CO = Carry out.
7. Q1-Q4 = Q out.

FIGURE 1. BMG module function diagram, binary/BCD counter.

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TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ± 5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ± 5°C		60°C ± 5°C		-55°C ± 5°C		85°C ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
A	Isolation													
	Initial	III	S	10	∞	10	∞	10	∞	10	∞	10	∞	MΩ
	End-of-life	III	S	1	∞	1	∞	1	∞	1	∞	1	∞	MΩ
B1-B23	Negative input clamp diode voltage	IV	X	-0.4	-6.0	-0.4	-6.0	-0.4	-6.0	-0.4	-6.0	-0.4	-6.0	V
C1-C23	Positive input clamp diode voltage	V	X	0.4	1.5	0.4	1.5	0.4	1.5	0.4	1.5	0.4	1.5	V
D1-D56	Functional	VI	X	← See table VI →										
E1-E15	Output voltage, low level	VII	X	0	50	0	50	0	50	0	50	0	50	mV
F1-F15	Output voltage, high level	VIII	X	4.95	5.05	4.95	5.05	4.95	5.05	4.95	5.05	4.95	5.05	V
G1-G15	Output voltage, low level	IX	X	0	400	0	400	0	400	0	400	0	400	mV
H1-H15	Output voltage, high level	X	X	4.6	5.0	4.6	5.0	4.6	5.0	4.6	5.0	4.6	5.0	V
J1-J15	Output voltage, low level	XI	X	0.0	1.5	0.0	1.5	0.0	1.5	0.0	1.5	0.0	1.5	V
K1-K15	Output voltage, high level	XII	X	13.5	15.0	13.5	15.0	13.5	15.0	13.5	15.0	13.5	15.0	V
L1-L23	Input current, low level	XIII	X	0	-200	---	---	0	-200	---	---	0	-200	nA
M1-M23	Input current, high level	XIV	X	0	200	---	---	0	200	---	---	0	200	nA
N1,N2	Power supply current	XV	X	0	60	0	60	0	1,800	0	60	0	1,800	µA
P1-P15	Transition time (t_{TLH}), C_L = 50 pF	XVI	S	26	260	26	260	29	290	20	200	32	320	ns
P16-P30	Transition time (t_{TLH}), C_L = 150 pF	XVI	Q	74	740	74	740	82	820	56	565	90	905	ns
P31-P45	Transition time (t_{THL}), C_L = 50 pF	XVI	S	26	260	26	260	27	275	23	230	29	290	ns
P46-P60	Transition time (t_{THL}), C_L = 150 pF	XVI	Q	74	740	74	740	78	780	65	650	82	825	ns

See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ± 5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ± 5°C		60°C ± 5°C		-55°C ± 5°C		85°C ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
Q1-Q15	Transition time (t_{TLH}), C_L = 50 pF	XVII	Q	13	130	13	130	14	145	10	100	15	155	ns
Q16-Q30	Transition time (t_{THL}), C_L = 50 pF	XVII	Q	13	130	13	130	14	140	11	115	14	145	ns
R1-R15	Transition time (t_{TLH}), C_L = 50 pF	XVIII	Q	10	100	10	100	11	110	8	80	12	120	ns
R16-R30	Transition time (t_{THL}), C_L = 50 pF	XVIII	Q	10	100	10	100	11	110	8	85	11	115	ns
S1-S4,S6-S9, S11-S14	Propagation delay time (t_{PLH}), C_L = 50 pF	XIX	S	65	650	65	650	66	665	62	620	67	675	ns
S5,S10,S15	Propagation delay time (t_{PLH}), C_L = 50 pF	XIX	S	75	750	75	750	76	765	72	720	77	775	ns
S16-S19,S21-S24,S26-S29	Propagation delay time (t_{PLH}), C_L = 150 pF	XIX	S	85	850	85	850	88	885	77	775	92	920	ns
S20,S25,S30	Propagation delay time (t_{PLH}), C_L = 150 pF	XIX	Q	95	950	95	950	98	985	87	875	102	1,020	ns
S31-S34,S36-S39,S41-S44	Propagation delay time (t_{PHL}), C_L = 50 pF	XIX	S	65	650	65	650	66	665	62	620	67	675	ns
S35,S40,S45	Propagation delay time (t_{PHL}), C_L = 50 pF	XIX	S	75	750	75	750	76	765	72	720	77	775	ns
S46-S49,S51-S54,S56-S59	Propagation delay time (t_{PHL}), C_L = 150 pF	XIX	Q	85	850	85	850	88	885	77	775	92	920	ns
S50,S55,S60	Propagation delay time (t_{PHL}), C_L = 150 pF	XIX	Q	95	950	95	950	98	985	87	875	102	1,020	ns
T1-T4,T6-T9, T11-T14	Propagation delay time (t_{PLH}), C_L = 50 pF	XX	Q	25	250	25	250	26	260	23	235	26	265	ns
T5,T10,T15	Propagation delay time (t_{PLH}), C_L = 50 pF	XX	Q	30	300	30	300	31	310	28	285	31	315	ns

See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ± 5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ± 5°C		60°C ± 5°C		-55°C ± 5°C		85°C ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
T16-T19,T21-T24,T26-T29	Propagation delay time (t_{PHL}), C_L = 50 pF	XX	Q	25	250	25	250	26	260	23	235	26	265	ns
T20,T25,T30	Propagation delay time (t_{PHL}), C_L = 50 pF	XX	Q	30	300	30	300	31	310	28	285	31	315	ns
U1-U4,U6-U9, U11-U14	Propagation delay time (t_{PLH}), C_L = 50 pF	XXI	Q	20	200	20	200	21	210	18	185	21	215	ns
U5,U10,U15	Propagation delay time (t_{PLH}), C_L = 50 pF	XXI	Q	24	240	24	240	25	250	22	225	25	255	ns
U16-U19,U21-U24,U26-U29	Propagation delay time (t_{PHL}), C_L = 50 pF	XXI	Q	20	200	20	200	21	210	18	185	21	215	ns
U20,U25,U30	Propagation delay time (t_{PHL}), C_L = 50 pF	XXI	Q	24	240	24	240	25	250	22	225	25	255	ns
V1-V3	Maximum clock frequency and minimum clock pulse width	XXII	S	4,000	4,400	4,000	4,400	4,000	4,400	4,000	4,400	4,000	4,400	ns
W1-W3	Maximum clock frequency and minimum clock pulse width	XXII	Q	2,280	2,480	2,280	2,480	2,280	2,480	2,280	2,480	2,280	2,480	ns
X1-X3	Maximum clock frequency and minimum clock pulse width	XXII	Q	1,760	1,920	1,760	1,920	1,760	1,920	1,760	1,920	1,760	1,920	ns
Y1-Y3	Propagation delay time (t_{PLH}), $\overline{CI}$ to $\overline{CO}$, C_L = 50 pF	XXIII	S	53	530	53	530	54	545	50	505	55	555	ns
Y4-Y6	Propagation delay time (t_{PHL}), $\overline{CI}$ to $\overline{CO}$, C_L = 50 pF	XXIII	S	53	530	53	530	54	545	50	505	55	555	ns
Z1-Z3	Propagation delay time (t_{PLH}), $\overline{CI}$ to $\overline{CO}$, C_L = 150 pF	XXIII	Q	106	1,060	106	1,060	109	1,095	98	985	113	1,130	ns
Z4-Z6	Propagation delay time (t_{PHL}), $\overline{CI}$ to $\overline{CO}$, C_L = 150 pF	XXIII	Q	106	1,060	106	1,060	109	1,095	98	985	113	1,130	ns

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See footnotes at end of table.

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Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ± 5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ± 5°C		60°C ± 5°C		-55°C ± 5°C		85°C ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
AA1-AA3	Propagation delay time (t_{PLH}), $\overline{CI}$ to $\overline{CO}$, C_L = 50 pF	XXIII	Q	22	220	22	220	23	230	20	205	23	235	ns
AA4-AA6	Propagation delay time (t_{PHL}), $\overline{CI}$ to $\overline{CO}$, C_L = 50 pF	XXIII	Q	22	220	22	220	23	230	20	205	23	235	ns
BB1-BB3	Propagation delay time (t_{PLH}), $\overline{CI}$ to $\overline{CO}$, C_L = 50 pF	XXIII	Q	18	180	18	180	19	190	16	165	19	195	ns
BB4-BB6	Propagation delay time (t_{PHL}), $\overline{CI}$ to $\overline{CO}$, C_L = 50 pF	XXIII	Q	18	180	18	180	19	190	16	165	19	195	ns
CC1-CC12	Propagation delay time (t_{PHL}), PE to Q, C_L = 50 pF	XXIV	S	72	720	72	720	73	735	69	695	74	745	ns
CC13-CC15	Propagation delay time (t_{PLH}), PE to $\overline{CO}$, C_L = 50 pF	XXIV	S	82	820	82	820	83	835	79	795	84	845	ns
DD1-DD12	Propagation delay time (t_{PHL}), PE to Q, C_L = 150 pF	XXIV	Q	92	920	92	920	95	955	84	845	99	990	ns
DD13-DD15	Propagation delay time (t_{PLH}), PE to $\overline{CO}$, C_L = 150 pF	XXIV	Q	102	1,020	102	1,020	105	1,055	94	945	109	1,090	ns
EE1-EE12	Propagation delay time (t_{PHL}), PE to Q, C_L = 50 pF	XXIV	Q	28	280	28	280	29	290	26	265	29	295	ns
EE13-EE15	Propagation delay time (t_{PLH}), PE to $\overline{CO}$, C_L = 50 pF	XXIV	Q	33	330	33	330	34	340	31	315	34	345	ns
FF1-FF12	Propagation delay time (t_{PHL}), PE to Q, C_L = 50 pF	XXIV	Q	22	220	22	220	23	230	20	205	23	235	ns
FF13-FF15	Propagation delay time (t_{PLH}), PE to $\overline{CO}$, C_L = 50 pF	XXIV	Q	26	260	26	260	27	270	24	245	27	275	ns
GG1-GG6	Pulse width setup time, countup and countdown	XXV	S	3,600	4,400	3,600	4,400	3,600	4,400	3,600	4,400	3,600	4,400	ns
HH1-HH6	Pulse width setup time, countup and countdown	XXV	S	1,800	2,200	1,800	2,200	1,800	2,200	1,800	2,200	1,800	2,200	ns

See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ± 5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ± 5°C		60°C ± 5°C		-55°C ± 5°C		85°C ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
JJ1-JJ6	Pulse width setup time, countup and countdown	XXV	Q	1,800	2,200	1,800	2,200	1,800	2,200	1,800	2,200	1,800	2,200	ns
KK1-KK6	Pulse width setup time, binary and decade	XXVI	S	3,600	4,400	3,600	4,400	3,600	4,400	3,600	4,400	3,600	4,400	ns
LL1-LL6	Pulse width setup time, binary and decade	XXVI	Q	1,800	2,200	1,800	2,200	1,800	2,200	1,800	2,200	1,800	2,200	ns
MM1-MM6	Pulse width setup time, binary and decade	XXVI	Q	1,800	2,200	1,800	2,200	1,800	2,200	1,800	2,200	1,800	2,200	ns
NN1-NN3	Preset enable setup and removal time	XXVII	S	900	1,100	900	1,100	900	1,100	900	1,100	900	1,100	ns
PP1-PP3, QQ1-QQ3	Preset enable setup and removal time	XXVII	Q	900	1,100	900	1,100	900	1,100	900	1,100	900	1,100	ns
RR	Power supply filter capacitance	XXVIII	S	0.007	0.013	0.007	0.013	0.007	0.013	0.007	0.013	0.007	0.013	µF
SS1-SS23	Input capacitance	XXIX	Q	2	15	2	15	2	15	2	15	2	15	pF

1/ "X" parameters shall be 100 percent tested during module acceptance tests.
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" parameters shall be tested during qualification testing only.
4/ Unless otherwise specified, limits are for initial and end-of-life.

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4.4.2 Order of testing. Test conditions A through SS shall be performed in any order with the following exceptions. If test condition A is part of a sequence of tests, the order of testing shall be as follows: A where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions B through SS in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions B through RR shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature with the exception of test conditions L and M which shall not be tested at 0°C and -55°C. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column, and test condition RR, shall be performed using the initial limits at 25°C ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests B through RR shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests B through SS shall be performed in the order specified in 4.4.2 using the 25°C ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in a class 1 environment, the failure rate for class I shall be used for reliability prediction (see 3.3 and 3.4).

6.4 Module description. The following electrical parameters reflect the 0°C to 60°C and -55°C to 85°C end-of-life limits from table I.

	Class I	Class II	Unit
Maximum power dissipation, quiescent (V_{DD} = 15 V and outputs not loaded)	27	27	mW
Supply voltage	3 to 18	3 to 18	V

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	Class I	Class II	Unit
Input voltage			
Low level (V_{DD} = 5 V)	0 min., 1.5 max.	0 min., 1.5 max.	V
Low level (V_{DD} = 15 V)	0 min., 4 max.	0 min., 4 max.	V
High level (V_{DD} = 5 V)	3.5 min., V_{DD} max.	3.5 min., V_{DD} max.	V
High level (V_{DD} = 15 V)	11 min., V_{DD} max.	11 min., V_{DD} max.	V
Input current (at 25°C, 60°C, and 85°C only)			
Low level (V_{IL} = 0 V, V_{DD} = 18 V)	0 min., -200 max.	0 min., -200 max.	nA
High level (V_{IH} = 18 V, V_{DD} = 18 V)	0 min., 200 max.	0 min., 200 max.	nA
Output voltage (V_{DD} = 5 V)			
Low level, I_{OL} = 0 (at -55°C, 0°C, 25°C, 60°C, and 85°C)	0.0 min., 0.5 max.	0.0 min., 0.5 max.	V
High level, I_{OH} = 0 (at -55°C, 0°C, 25°C, 60°C, and 85°C)	4.95 min., 5.05 max.	4.95 min., 5.05 max.	V
Low level:			
at -55°C, I_{OL} = 0.64 mA	0.0 min., 0.4 max.	0.0 min., 0.4 max.	V
at 0°C and 25°C, I_{OL} = 0.51 mA	0.0 min., 0.4 max.	0.0 min., 0.4 max.	V
at 60°C, I_{OL} = 0.44 mA	0.0 min., 0.4 max.	0.0 min., 0.4 max.	V
at 85°C, I_{OL} = 0.36 mA	0.0 min., 0.4 max.	0.0 min., 0.4 max.	V
High level			
at -55°C, I_{OH} = 0.64 mA	4.6 min., 5.0 max.	4.6 min., 5.0 max.	V
at 0°C and 25°C, I_{OH} = 0.51 mA	4.6 min., 5.0 max.	4.6 min., 5.0 max.	V
at 60°C, I_{OH} = 0.44 mA	4.6 min., 5.0 max.	4.6 min., 5.0 max.	V
at 85°C, I_{OH} = 0.36 mA	4.6 min., 5.0 max.	4.6 min., 5.0 max.	V
Output voltage (V_{DD} = 15 V)			
Low level, I_{OL} = 0 (at -55°C, 0°C, 25°C, 60°C, and 85°C)	0.0 min., 1.5 max.	0.0 min., 1.5 max.	V
High level, I_{OH} = 0 (at -55°C, 0°C, 25°C, 60°C, and 85°C)	13.5 min., 15.0 max.	13.5 min., 15.0 max.	V
Clock input			
V_{DD} = 5 V			
Maximum frequency	2	2	MHz
Minimum pulse width	340	340	ns
V_{DD} = 10 V			
Maximum frequency	3.5	3.5	MHz
Minimum pulse width	170	170	ns
V_{DD} = 15 V			
Maximum frequency	4.5	4.5	MHz
Minimum pulse width	140	140	ns
U/D input (t_{SETUP} time)			
V_{DD} = 5 V	650 min.	650 min.	ns
V_{DD} = 10 V	230 min.	230 min.	ns
V_{DD} = 15 V	170 min.	170 min.	ns
B/D input (t_{SETUP} time)			
V_{DD} = 5 V	650 min.	650 min.	ns
V_{DD} = 10 V	230 min.	230 min.	ns
V_{DD} = 15 V	170 min.	170 min.	ns
PE input			
V_{DD} = 5 V			
t_{SETUP}	325 min.	325 min.	ns
Pulse width	340 min.	340 min.	ns
$t_{REMOVAL}$	650 min.	650 min.	ns

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	Class I	Class II	Unit
V_{DD} = 10 V			
t_{SETUP}	115 min.	115 min.	ns
Pulse width	170 min.	170 min.	ns
$t_{REMOVAL}$	220 min.	220 min.	ns
V_{DD} = 15 V			
t_{SETUP}	100 min.	100 min.	ns
Pulse width	140 min.	140 min.	ns
$t_{REMOVAL}$	180 min.	180 min.	ns

6.5 Module Function. This module contains 3 independent 4 bit CMOS presettable, up/down, binary/decade counters. A high PRESET ENABLE signal allows information on the JAM INPUTS to preset the counter to any state asynchronously with the clock. A low on each JAM line, when the PRESET ENABLE signal is high, resets the counter to its zero count. The counter is advanced one count at the positive transition of the clock when the CARRY IN and PRESET ENABLE signals are low. Advancement is inhibited when the CARRY IN or PRESET ENABLE signals are high. The CARRY OUT signal is normally high and goes low when the counter reaches its maximum count in the UP mode or the minimum count in the DOWN mode provided the CARRY IN signal is low. The CARRY IN terminal must be connected to V_{SS} when not in use. Binary counting is accomplished when the BINARY/DECADE input is high; the counter counts in the decade mode when the BINARY/DECADE input is low. The counter counts up when the UP/DOWN input is high, and down when the UP/DOWN input is low.

6.5.1 Input to output relationship. Each counter has the input to output relationship of table II for positive logic.

6.5.2 Dynamic power dissipation. The dynamic power dissipation can be approximated by the formula:

P_{TOTAL} = (C V_{DD}2f) N + quiescent power dissipation
where: C = load capacitance
V_{DD} = supply voltage
f = operating frequency
N = number of outputs.

6.5.3 Parallel outputs. Paralleling of outputs is not permissible.

6.6. Module handling considerations. To avoid possible damage to the devices during handling, testing, or actual operation, the following procedures should be followed:

a. Device pins should be in contact with conductive material, except when being tested or in actual operation, to avoid buildup of static charge.

b. Soldering iron tips, metal parts or fixtures and tools, and handling facilities should be grounded.

c. Signals should not be applied to the inputs while the device power supply is off.

d. Input voltages greater than the supply (V_{DD}-V_{SS}) should not be applied as this will forward bias the input protection diodes.

e. All unused input leads must be connected to either V_{SS} (ground) or V_{DD} (device supply), whichever is appropriate, for the logic circuit involved.

6.7 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Agent:
DLA - ES

Preparing activity:
Navy - EC

(Project 5963-N549)

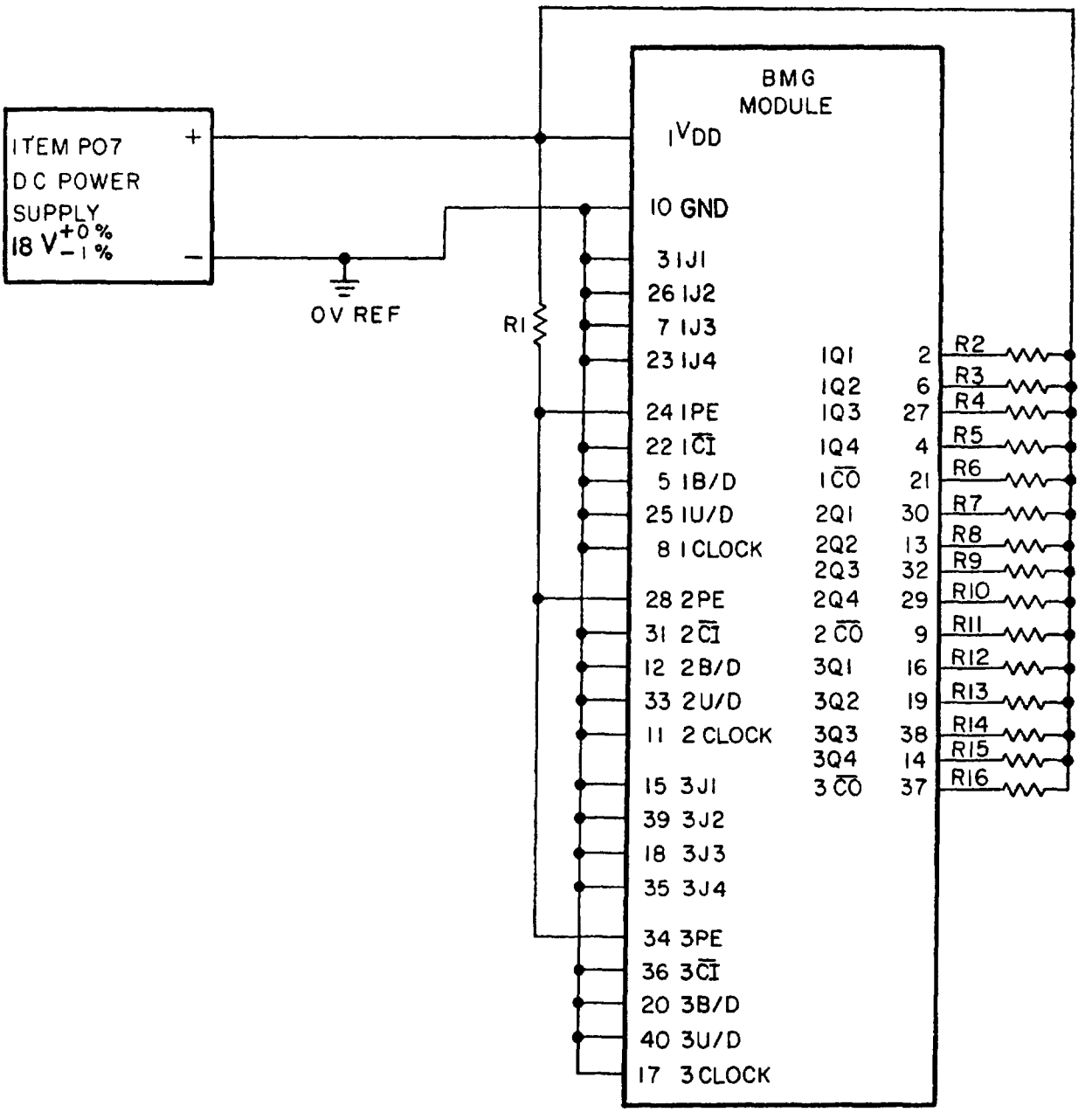
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TABLE II. Input to output relationship. 1/ - 4/

Inputs									Outputs				
PE	B/D	U/D	$\overline{CT}$	CL	J1	J2	J3	J4	Q1	Q2	Q3	Q4	$\overline{CO}$
H	X	X	H	X	H	H	H	H	H	H	H	H	L
H	X	X	H	↑	H	H	H	H	H	H	H	H	L
H	X	X	H	X	L	L	L	L	L	L	L	L	H
H	X	X	H	↑	L	L	L	L	L	L	L	L	H
L	H	H	L	↑	X	X	X	X	H	L	L	L	H
L	H	H	L	↑	X	X	X	X	L	H	L	L	H
L	H	H	L	↑	X	X	X	X	H	H	L	L	H
L	H	H	L	↑	X	X	X	X	L	L	H	L	H
L	H	H	L	↑	X	X	X	X	H	L	H	L	H
L	H	H	L	↑	X	X	X	X	L	H	H	L	H
L	H	H	L	↑	X	X	X	X	H	H	H	L	H
L	H	H	L	↑	X	X	X	X	L	L	L	H	H
L	H	H	L	↑	X	X	X	X	H	L	L	H	H
L	H	H	L	↑	X	X	X	X	L	H	L	H	H
L	H	H	L	↑	X	X	X	X	H	H	L	H	H
L	H	H	L	↑	X	X	X	X	L	L	H	H	H
L	H	H	L	↑	X	X	X	X	H	L	H	H	H
L	H	H	L	↑	X	X	X	X	L	H	H	H	L
L	H	H	L	↑	X	X	X	X	H	H	H	H	L
L	H	L	L	↑	X	X	X	X	L	H	H	H	H
L	H	L	L	↑	X	X	X	X	H	L	H	H	H
L	H	L	L	↑	X	X	X	X	L	L	H	H	H
L	H	L	L	↑	X	X	X	X	H	H	L	H	H
L	H	L	L	↑	X	X	X	X	L	H	L	H	H
L	H	L	L	↑	X	X	X	X	H	L	L	H	H
L	H	L	L	↑	X	X	X	X	L	L	L	H	H
L	H	L	L	↑	X	X	X	X	H	H	H	L	H
L	H	L	L	↑	X	X	X	X	L	H	H	L	H
L	H	L	L	↑	X	X	X	X	H	L	H	L	H
L	H	L	L	↑	X	X	X	X	L	L	H	L	H
L	H	L	L	↑	X	X	X	X	H	H	L	L	H
L	H	L	L	↑	X	X	X	X	L	H	L	L	H
L	H	L	L	↑	X	X	X	X	H	L	L	L	H
L	H	L	L	↑	X	X	X	X	L	L	L	L	L
L	L	H	L	↑	X	X	X	X	H	L	L	L	H
L	L	H	L	↑	X	X	X	X	L	H	L	L	H
L	L	H	L	↑	X	X	X	X	H	H	L	L	H
L	L	H	L	↑	X	X	X	X	L	L	H	L	H
L	L	H	L	↑	X	X	X	X	H	L	H	L	H
L	L	H	L	↑	X	X	X	X	L	H	H	L	H
L	L	H	L	↑	X	X	X	X	H	H	H	L	H
L	L	H	L	↑	X	X	X	X	L	L	L	H	H
L	L	H	L	↑	X	X	X	X	H	L	L	H	L
L	L	L	L	↑	X	X	X	X	L	L	L	H	L
L	L	L	L	↑	X	X	X	X	H	H	H	L	H
L	L	L	L	↑	X	X	X	X	L	H	H	L	H
L	L	L	L	↑	X	X	X	X	H	L	H	L	H
L	L	L	L	↑	X	X	X	X	L	L	H	L	H
L	L	L	L	↑	X	X	X	X	H	H	L	L	H
L	L	L	L	↑	X	X	X	X	L	H	L	L	H
L	L	L	L	↑	X	X	X	X	H	L	L	L	H
L	L	L	L	↑	X	X	X	X	L	L	L	L	L

1/ L = low level.
2/ H = high level.
3/ X = don't care.
4/ ↑ = positive transition.

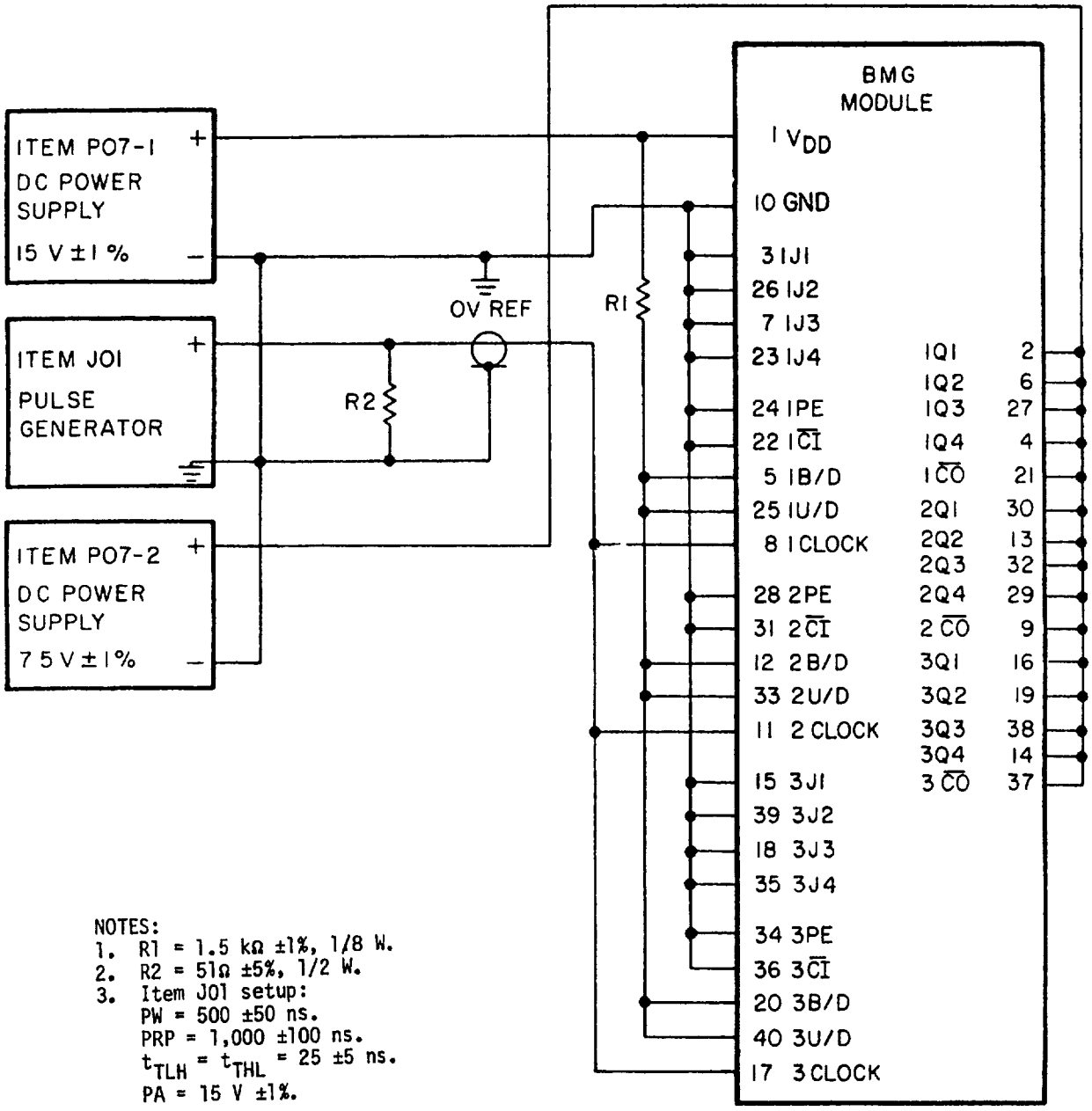
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NOTES:
1. R1 = 1.5 kΩ ±1%, 1/8 W.
2. R2-R16 = 6.81 kΩ ±1%, 1/4 W.

FIGURE 2. Operating temperature and life test circuit.

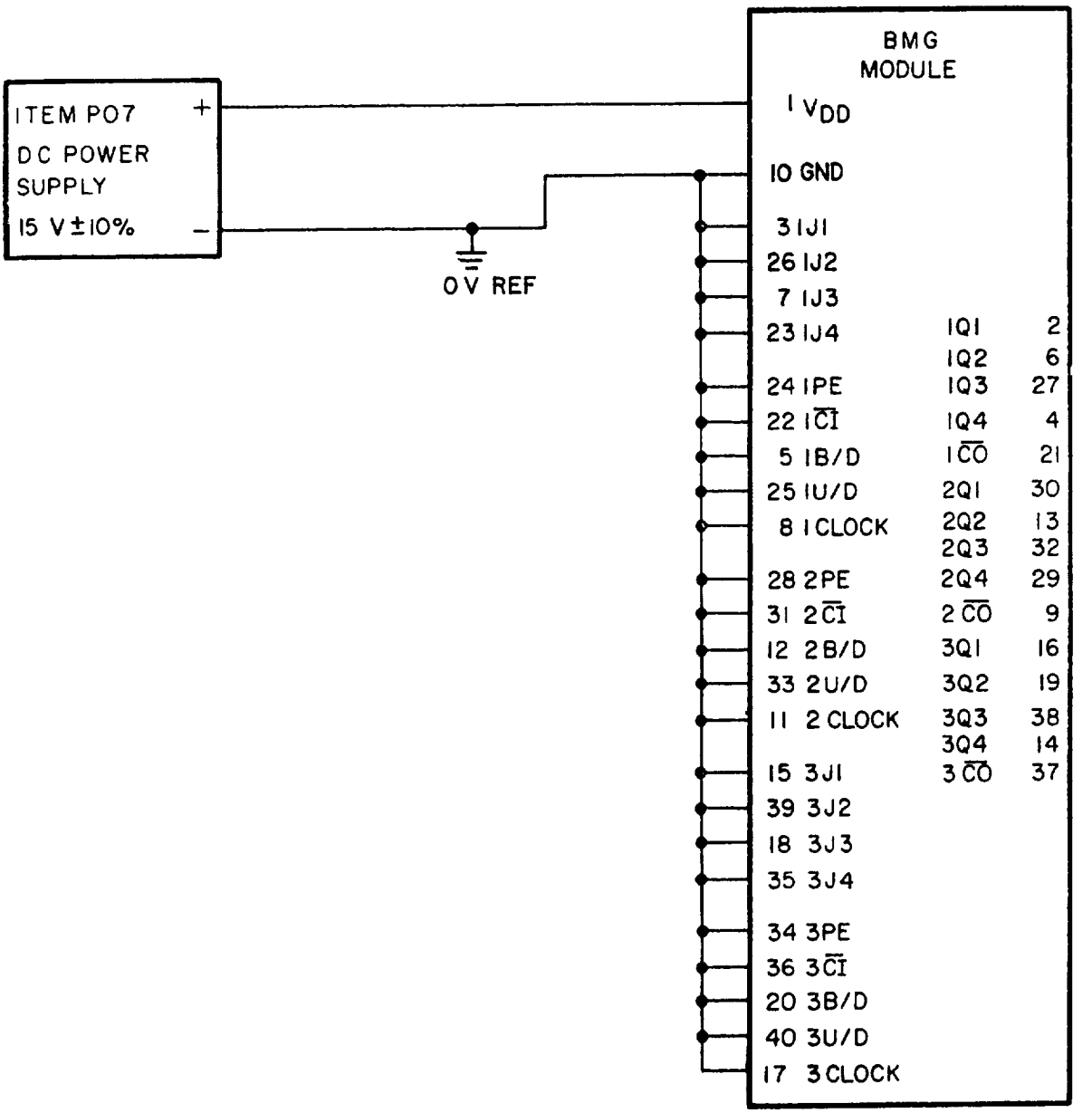
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NOTES:
1. R1 = 1.5 kΩ ±1%, 1/8 W.
2. R2 = 51Ω ±5%, 1/2 W.
3. Item J01 setup:
   PW = 500 ±50 ns.
   PRP = 1,000 ±100 ns.
   $t_{TLH} = t_{THL}$ = 25 ±5 ns.
   PA = 15 V ±1%.

* FIGURE 3. Component temperature test circuit.

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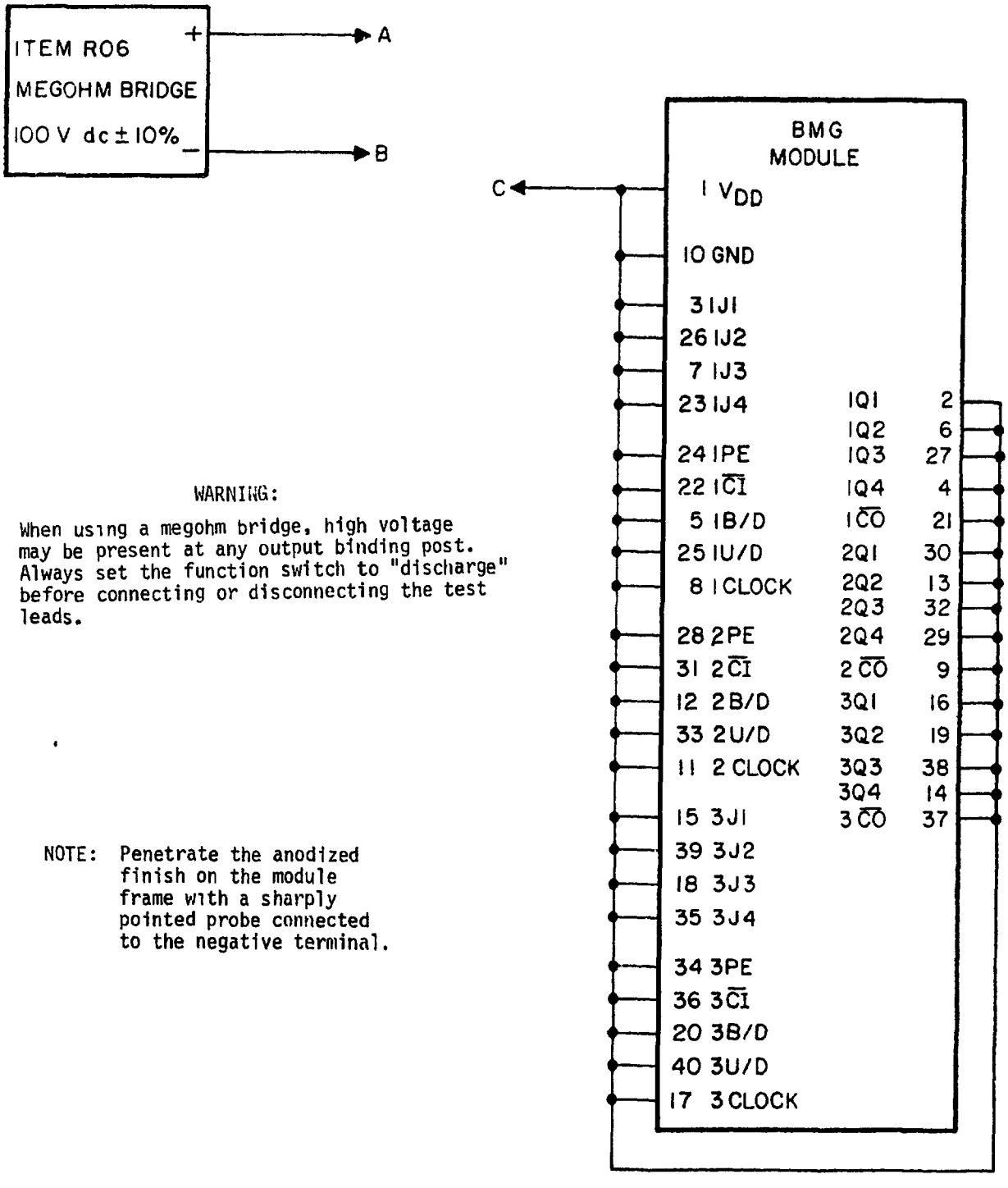
* FIGURE 4. Vibration test circuit.

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TABLE III. Isolation test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A	Isolation	C	Frame	10	∞	MΩ	5

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WARNING:

When using a megohm bridge, high voltage may be present at any output binding post. Always set the function switch to "discharge" before connecting or disconnecting the test leads.

NOTE: Penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

FIGURE 5. Isolation test circuit for table III.

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TABLE IV. Negative input clamp diode voltage test.

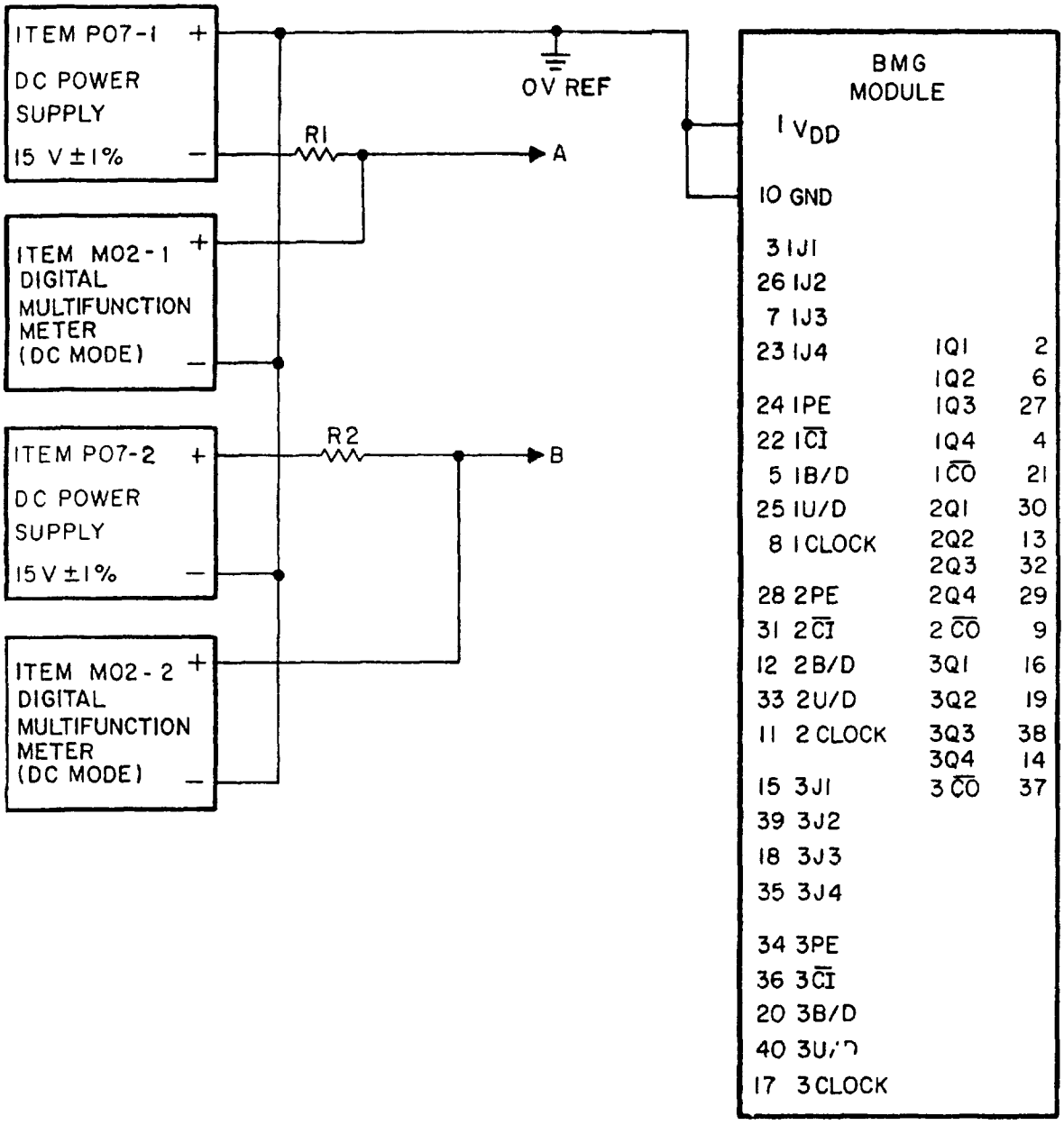
Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
B1	Negative input clamp diode voltage	3	Open	-0.4	-6.0	V	6
B2	"	26	"	"	"	"	"
B3	"	7	"	"	"	"	"
B4	"	23	"	"	"	"	"
B5	"	24	"	"	"	"	"
B6	"	22	"	"	"	"	"
B7	"	5	"	"	"	"	"
B8	"	25	"	"	"	"	"
B9	"	8	"	"	"	"	"
B10	"	28	"	"	"	"	"
B11	"	31	"	"	"	"	"
B12	"	12	"	"	"	"	"
B13	"	33	"	"	"	"	"
B14	"	11	"	"	"	"	"
B15	"	15	"	"	"	"	"
B16	"	39	"	"	"	"	"
B17	"	18	"	"	"	"	"
B18	"	35	"	"	"	"	"
B19	"	34	"	"	"	"	"
B20	"	36	"	"	"	"	"
B21	"	20	"	"	"	"	"
B22	"	40	"	"	"	"	"
B23	"	17	"	"	"	"	"

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TABLE V. Positive input clamp diode voltage test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C1	Positive input clamp diode voltage	Open	3	0.4	1.5	V	6
C2	"	"	26	"	"	"	"
C3	"	"	7	"	"	"	"
C4	"	"	23	"	"	"	"
C5	"	"	24	"	"	"	"
C6	"	"	22	"	"	"	"
C7	"	"	5	"	"	"	"
C8	"	"	25	"	"	"	"
C9	"	"	8	"	"	"	"
C10	"	"	28	"	"	"	"
C11	"	"	31	"	"	"	"
C12	"	"	12	"	"	"	"
C13	"	"	33	"	"	"	"
C14	"	"	11	"	"	"	"
C15	"	"	15	"	"	"	"
C16	"	"	39	"	"	"	"
C17	"	"	18	"	"	"	"
C18	"	"	35	"	"	"	"
C19	"	"	34	"	"	"	"
C20	"	"	36	"	"	"	"
C21	"	"	20	"	"	"	"
C22	"	"	40	"	"	"	"
C23	"	"	17	"	"	"	"

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NOTES:
1. R1 = 9.09 kΩ ±1%, 1/8 W.
2. R2 = 13.7 kΩ ±1%, 1/8 W.
3. Since items P07-1, P07-2, M02-1, and M02-2 are not used simultaneously, only one item P07 and M02 is required.

* FIGURE 6. Negative and positive input clamp diode voltage test circuit for tables IV and V.

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*TABLE VI. Functional test.

Test condition	Parameter	Connect			D to output pins														Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	2	6	27	4	21	30	13	32	29	9	16	19	38	14	37	Min	Max	
D1	Functional, parallel loading	5,25,12,33,20,40,8,11,17	24,28,34,3,26,7,23,15,39,18,35,22,31,36	Open	H	H	H	H	L	L	L	L	L	H	H	H	H	H	L	1/		7,8
D2	"	5,25,12,33,20,40	24,28,34,3,26,7,23,15,39,18,35,22,31,36	8,11,17	H	H	H	H	L	L	L	L	L	H	H	H	H	H	L	"		"
D3	"	5,25,12,33,20,40,3,26,7,23,15,39,18,35	8,11,17,24,28,34,22,31,36	Open	L	L	L	L	H	L	L	L	L	H	L	L	L	L	H	"		"
D4	"	5,25,12,33,20,40,3,26,7,23,15,39,18,35	24,28,34,22,31,36	8,11,17	L	L	L	L	H	L	L	L	L	H	L	L	L	L	H	"		"
D5	Functional, countup, binary	24,28,34,22,31,36,3,26,7,23	5,25,12,33,20,40,15,39,18,35	"	H	L	L	L	H	H	L	L	L	H	H	L	L	L	H	"		"
D6	"	"	"	"	L	H	L	L	H	L	H	L	L	H	L	H	L	L	H	"		"
D7	"	"	"	"	H	H	L	L	H	H	H	L	L	H	H	H	L	L	H	"		"
D8	"	"	"	"	L	L	H	L	H	L	L	H	L	H	L	L	H	L	H	"		"
D9	"	"	"	"	H	L	H	L	H	H	L	H	L	H	H	L	H	L	H	"		"
D10	"	"	"	"	L	H	H	L	H	L	H	H	L	H	L	H	H	L	H	"		"
D11	"	"	"	"	H	H	H	L	H	H	H	H	L	H	H	H	H	L	H	"		"
D12	"	"	"	"	L	L	L	H	H	L	L	L	H	H	L	L	L	H	H	"		"
D13	"	"	"	"	H	L	L	H	H	H	L	L	H	H	H	L	L	H	H	"		"
D14	"	"	"	"	L	H	L	H	H	L	H	L	H	H	L	H	L	H	H	"		"
D15	"	"	"	"	H	H	L	H	H	H	H	L	H	H	H	H	L	H	H	"		"
D16	"	"	"	"	L	L	H	H	H	L	L	H	H	H	L	L	H	H	H	"		"
D17	"	"	"	"	H	L	H	H	H	H	L	H	H	H	H	L	H	H	H	"		"
D18	"	"	"	"	L	H	H	H	H	L	H	H	H	H	L	H	H	H	H	"		"
D19	"	"	"	"	H	H	H	H	L	H	H	H	H	L	H	H	H	H	L	"		"

See footnote at end of table.

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*TABLE VI. Functional test - Continued.

Test condition	Parameter	Connect			D to output pins														Initial limits 25°C ±5°C ambient			Unit	Test figure
		A to pin	B to pin	C to pin	2	6	27	4	21	30	13	32	29	9	16	19	38	14	Min	37	Max		
D20	Functional, countup, binary	24,28,34,22,31,36, 3,26,7,23	5,25,12,33,20,40, 15,39,18,35	8,11 17	L	L	L	L	H	L	L	L	L	H	L	L	L	L	H	1/			7,8
D21	Functional, countdown, binary	24,28,34,22,31,36, 3,26,7,23,25,33,40	5,12,20,15,39,18, 35	"	H	H	H	H	L	H	H	H	H	L	H	H	H	H	L	"			"
D22	"	"	"	"	L	H	H	H	H	L	H	H	H	H	L	H	H	H	H	"			"
D23	"	"	"	"	H	L	H	H	H	H	L	H	H	H	H	L	H	H	H	"			"
D24	"	"	"	"	L	L	H	H	H	L	L	H	H	H	L	L	H	H	H	"			"
D25	"	"	"	"	H	H	L	H	H	H	H	L	H	H	H	H	L	H	H	"			"
D26	"	"	"	"	L	H	L	H	H	L	H	L	H	H	L	H	L	H	H	"			"
D27	"	"	"	"	H	L	L	H	H	H	L	L	H	H	H	L	L	H	H	"			"
D28	"	"	"	"	L	L	L	H	H	L	L	L	H	H	L	L	L	H	H	"			"
D29	"	"	"	"	H	H	H	L	H	H	H	H	L	H	H	H	H	L	H	"			"
D30	"	"	"	"	L	H	H	L	H	L	H	H	L	H	L	H	H	L	H	"			"
D31	"	"	"	"	H	L	H	L	H	H	L	H	L	H	H	L	H	L	H	"			"
D32	"	"	"	"	L	L	H	L	H	L	L	H	L	H	L	L	H	L	H	"			"
D33	"	"	"	"	H	H	L	L	H	H	H	L	L	H	H	H	L	L	H	"			"
D34	"	"	"	"	L	H	L	L	H	L	H	L	L	H	L	H	L	L	H	"			"
D35	"	"	"	"	H	L	L	L	H	H	L	L	L	H	H	L	L	L	H	"			"
D36	"	"	"	"	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	"			"
D37	Functional, countup, decade	24,28,34,22,31,36, 3,26,7,23,5,12,20	25,33,40,15,39,18, 35	"	H	L	L	L	H	H	L	L	L	H	H	L	L	L	H	"			"
D38	"	"	"	"	L	H	L	L	H	L	H	L	L	H	L	H	L	L	H	"			"
D39	"	"	"	"	H	H	L	L	H	H	H	L	L	H	H	H	L	L	H	"			"

See footnote at end of table.

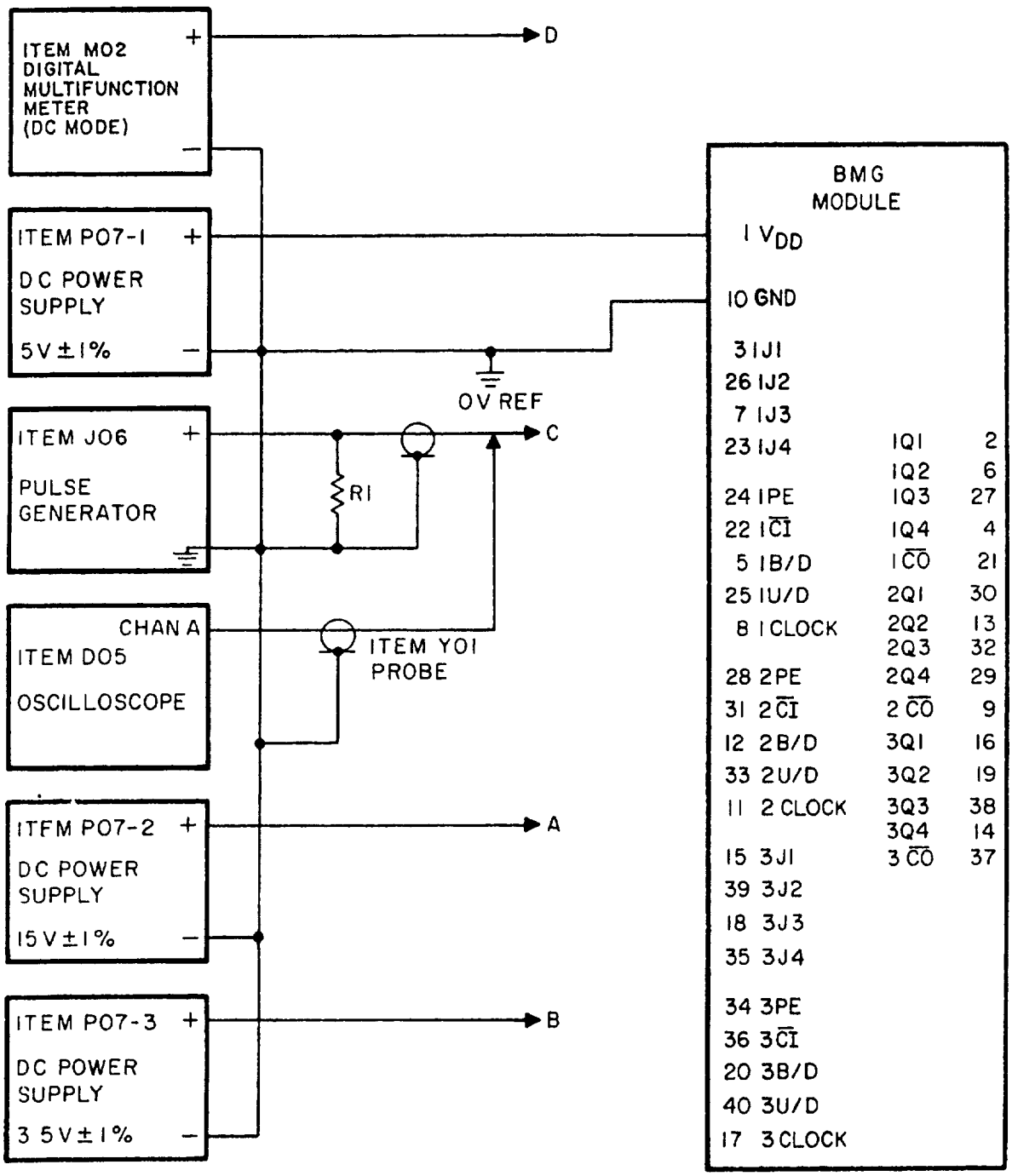
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*TABLE VI. Functional test - Continued

Test condition	Parameter	Connect			D to output pins														Initial limits 25°C ±5°C ambient			Unit	Test figure
		A to pin	B to pin	C to pin	2	6	27	4	21	30	13	32	29	9	16	19	38	14	37				
																		Min		Max			
D40	Functional, countup, decade	24,28,34,22,31,36, 3,26,7,23,5,12,20	25,33,40,15,39,18, 35	8,11, 17	L	L	H	L	H	L	L	H	L	H	L	L	H	L	H	1/		7,8
D41	"	"	"	"	H	L	H	L	H	H	L	H	L	H	H	L	H	L	H	"		"
D42	"	"	"	"	L	H	H	L	H	L	H	H	L	H	L	H	H	L	H	"		"
D43	"	"	"	"	H	H	H	L	H	H	H	H	L	H	H	H	H	L	H	"		"
D44	"	"	"	"	L	L	L	H	H	L	L	L	H	H	L	L	L	H	H	"		"
D45	"	"	"	"	H	L	L	H	L	H	L	L	H	L	H	L	L	H	L	"		"
D46	"	"	"	"	L	L	L	L	H	L	L	L	L	H	L	L	L	L	H	"		"
D47	Functional, countdown, decade	24,28,34,32,31,36, 3,26,7,23,5,12,20, 25,33,40	15,39,18,35	"	H	L	L	H	L	H	L	L	H	L	H	L	L	H	L	"		"
D48	"	"	"	"	L	L	L	H	H	L	L	L	H	H	L	L	L	H	H	"		"
D49	"	"	"	"	H	H	H	L	H	H	H	H	L	H	H	H	H	L	H	"		"
D50	"	"	"	"	L	H	H	L	H	L	H	H	L	H	L	H	H	L	H	"		"
D51	"	"	"	"	H	L	H	L	H	H	L	H	L	H	H	L	H	L	H	"		"
D52	"	"	"	"	L	L	H	L	H	L	L	H	L	H	L	L	H	L	H	"		"
D53	"	"	"	"	H	H	L	L	H	H	H	L	L	H	H	H	L	L	H	"		"
D54	"	"	"	"	L	H	L	L	H	L	H	L	L	H	L	H	L	L	H	"		"
D55	"	"	"	"	H	L	L	L	H	H	L	L	L	H	H	L	L	L	H	"		"
D56	"	"	"	"	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	"		"

1/ Output logic states:
L = Low level = 0.0 to 0.4 V.
H = High level = 4.6 to 5.0 V.

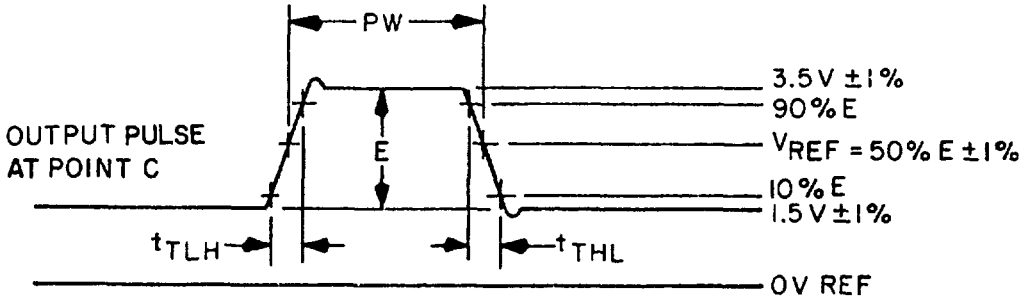
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NOTE: R1 = 51Ω ±5%, 1/2 W.

* FIGURE 7. Functional test circuit for table VI.

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NOTES:
1. Input pulse parameters:
   PW = 200 ±20 ns.
   $t_{TLH} = t_{THL}$ = 10 ±2 ns.
2. Place item J01 in single pulse mode.
3. Pulse the module one time for each test condition D1-D56 when item J01 is connected.
4. Test conditions D1-D56 must be performed in order.

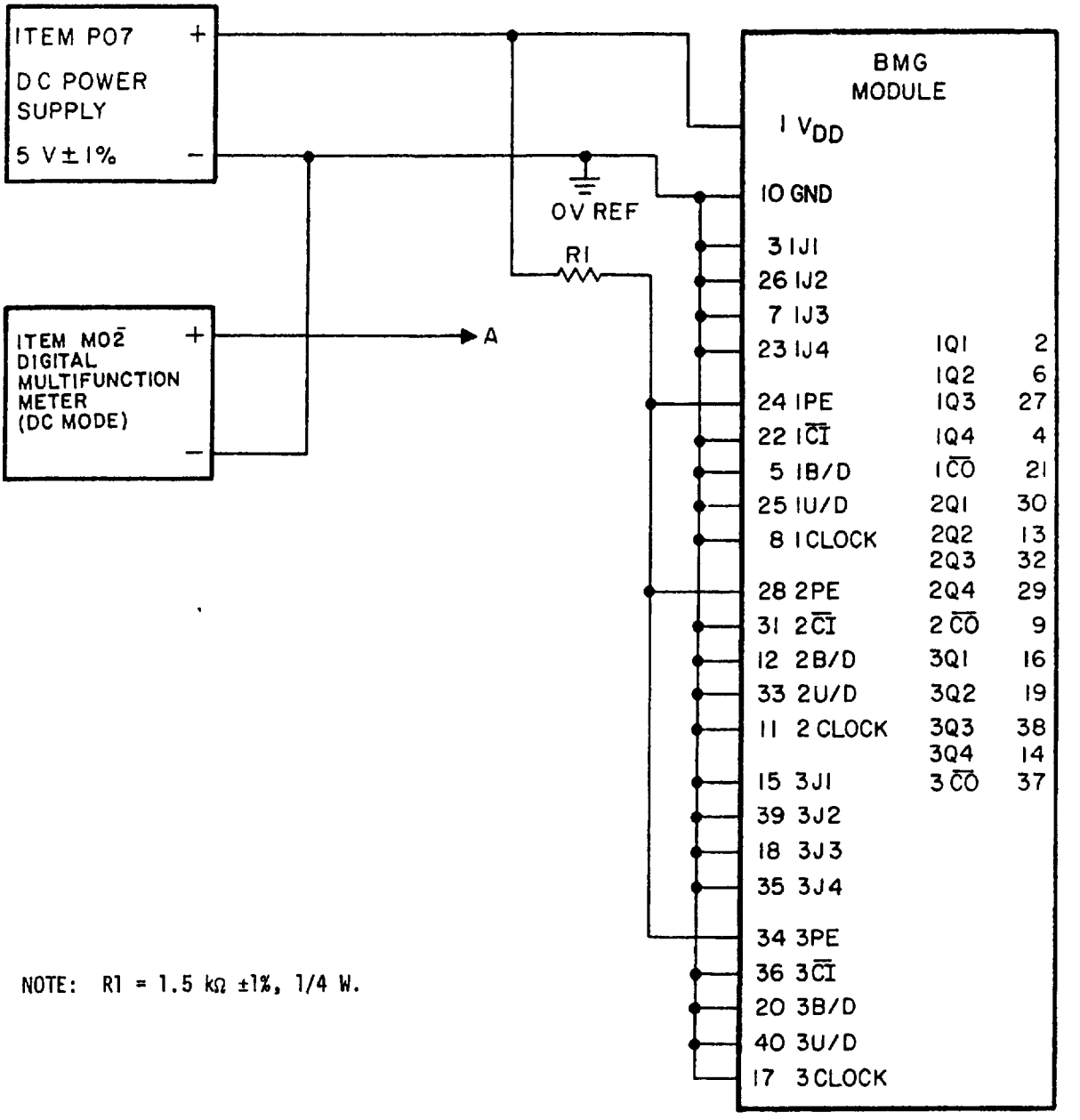
* FIGURE 8. Functional test circuit waveform for table VI.

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TABLE VII. Output voltage (low-level) test.

Test condition	Parameter		Connect A to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
E1	1Q1	Output voltage, low level	2	0	50	mV	9
E2	1Q2	"	6	"	"	"	"
E3	1Q3	"	27	"	"	"	"
E4	1Q4	"	4	"	"	"	"
E5	$1\overline{CO}$	"	21	"	"	"	"
E6	2Q1	"	30	"	"	"	"
E7	2Q2	"	13	"	"	"	"
E8	2Q3	"	32	"	"	"	"
E9	2Q4	"	29	"	"	"	"
E10	$2\overline{CO}$	"	9	"	"	"	"
E11	3Q1	"	16	"	"	"	"
E12	3Q2	"	19	"	"	"	"
E13	3Q3	"	38	"	"	"	"
E14	3Q4	"	14	"	"	"	"
E15	$3\overline{CO}$	"	37	"	"	"	"

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NOTE: R1 = 1.5 kΩ ±1%, 1/4 W.

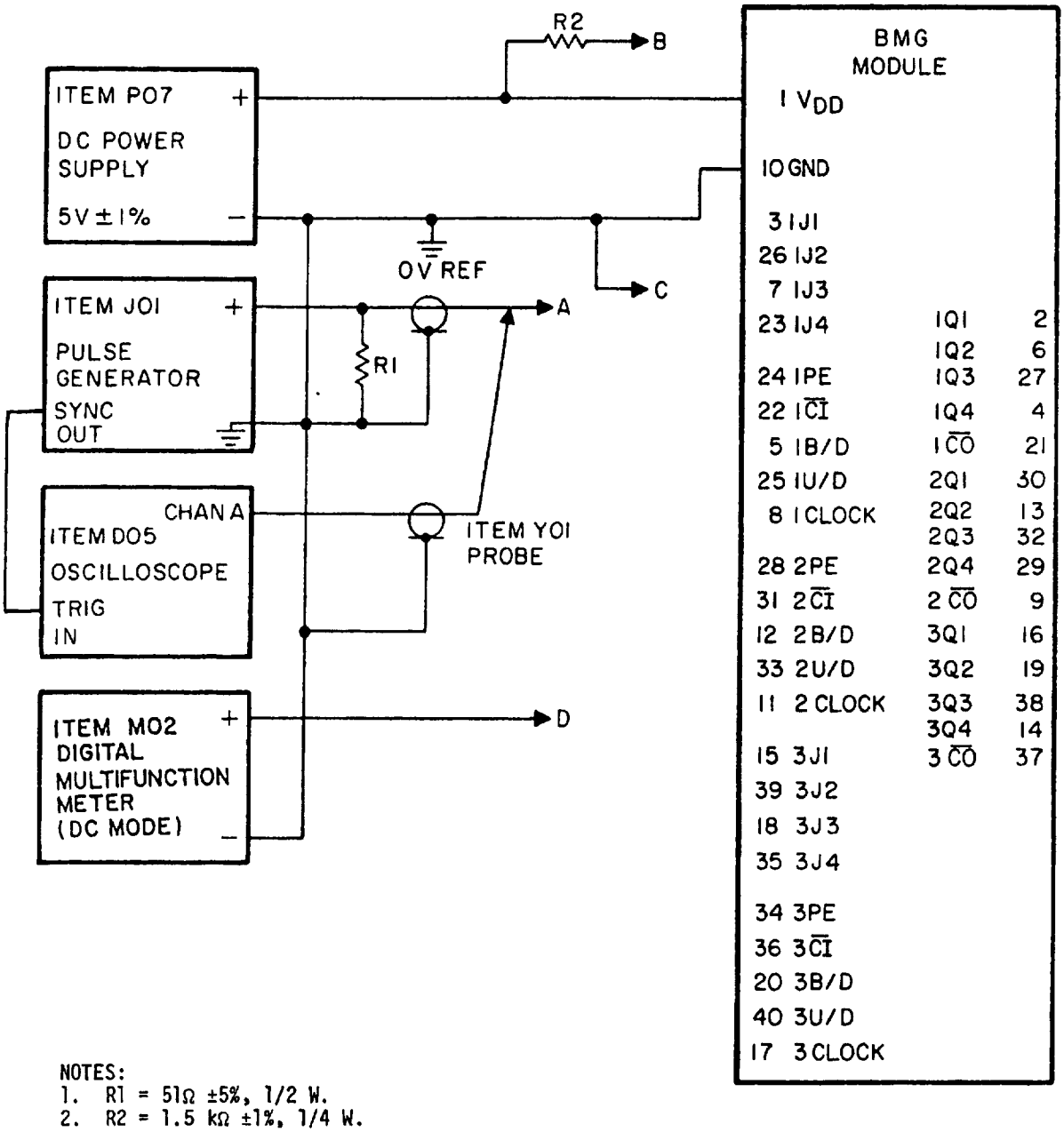
* FIGURE 9. Output voltage (low-level) test circuit for table VII.

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TABLE VIII. Output voltage (high-level) test.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
F1	$1\overline{CO}$ Output voltage, high level	Open	22,31,36	3,26,7,23,24,5,25,8,28,12,33,11,15,39,18,35,34,20,40,17	21	4.95	5.05	V	10,11
F2	$2\overline{CO}$ "	"	"	"	9	"	"	"	"
F3	$3\overline{CO}$ "	"	"	"	37	"	"	"	"
F4	1Q1 "	"	3,26,7,23,15,39,18,35,24,28,34	22,5,25,8,31,12,33,11,36,20,40,17	2	"	"	"	"
F5	1Q2 "	"	"	"	6	"	"	"	"
F6	1Q3 "	"	"	"	27	"	"	"	"
F7	1Q4 "	"	"	"	4	"	"	"	"
F8	3Q1 "	"	"	"	16	"	"	"	"
F9	3Q2 "	"	"	"	19	"	"	"	"
F10	3Q3 "	"	"	"	38	"	"	"	"
F11	3Q4 "	"	"	"	14	"	"	"	"
F12	2Q1 "	11	5,25,12,33,20,40	3,26,7,23,24,22,8,28,31,15,39,18,35,34,36,17	30	"	"	"	"
F13	2Q2 "	"	"	"	13	"	"	"	"
F14	2Q3 "	"	"	"	32	"	"	"	"
F15	2Q4 "	"	"	"	29	"	"	"	"

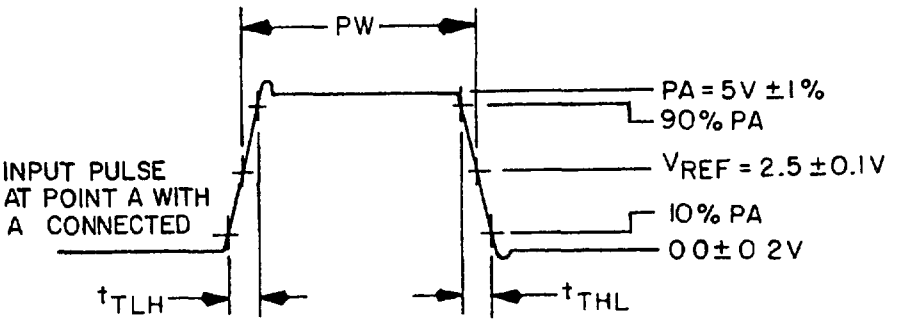
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NOTES:
1. R1 = 51Ω ±5%, 1/2 W.
2. R2 = 1.5 kΩ ±1%, 1/4 W.

* FIGURE 10. Output voltage (high-level) test circuit for table VIII.

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NOTES:
1. Input pulse parameters:
   PW = 200 ±20 ns.
   $t_{TLH} = t_{THL}$ = 10 ±2 ns.
2. Item J01 is in single pulse mode.
3. Before performing test conditions F12 through F15, momentarily connect pin 28 to point B, pulse the module 15 times, then take the measurements for test conditions F12 through F15.

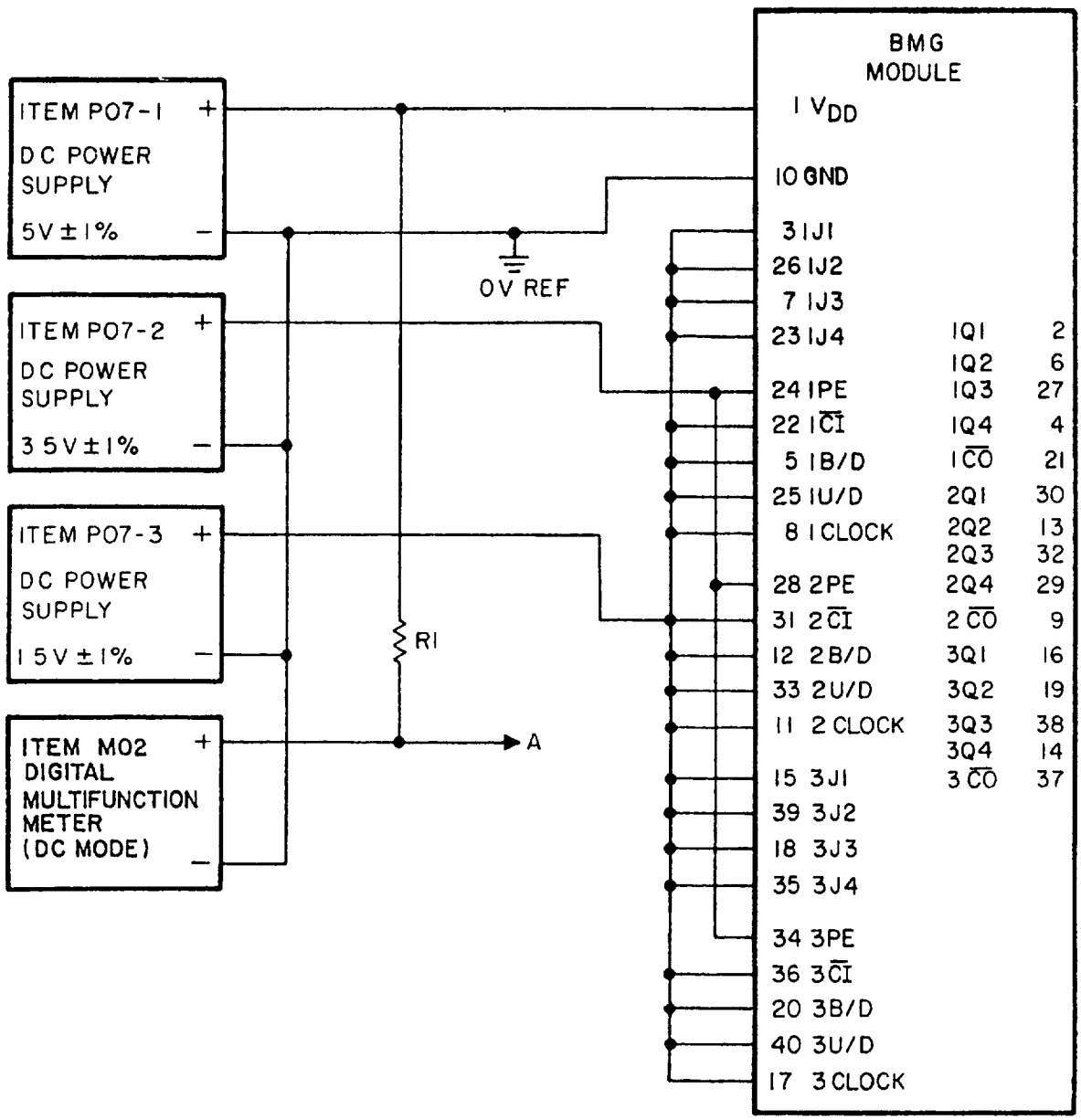
* FIGURE 11. Output voltage (high-level) test circuit waveform for table VIII.

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TABLE IX. Output voltage (low-level) test.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
G1	1Q1 Output voltage, low level	2	0	400	mV	12
G2	1Q2 "	6	"	"	"	"
G3	1Q3 "	27	"	"	"	"
G4	1Q4 "	4	"	"	"	"
G5	1$\overline{CO}$ "	21	"	"	"	"
G6	2Q1 "	30	"	"	"	"
G7	2Q2 "	13	"	"	"	"
G8	2Q3 "	32	"	"	"	"
G9	2Q4 "	29	"	"	"	"
G10	2$\overline{CO}$ "	9	"	"	"	"
G11	3Q1 "	16	"	"	"	"
G12	3Q2 "	19	"	"	"	"
G13	3Q3 "	38	"	"	"	"
G14	3Q4 "	14	"	"	"	"
G15	3$\overline{CO}$ "	37	"	"	"	"

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NOTE: R1 = @ -55°C, 7.19 kΩ ±1%, 1/8 W.
@ 0°C and 25°C, 9.02 kΩ ±1%, 1/8 W.
@ 60°C, 10.5 kΩ ±1%, 1/8 W.
@ 85°C, 11.8 kΩ ±1%, 1/8 W.

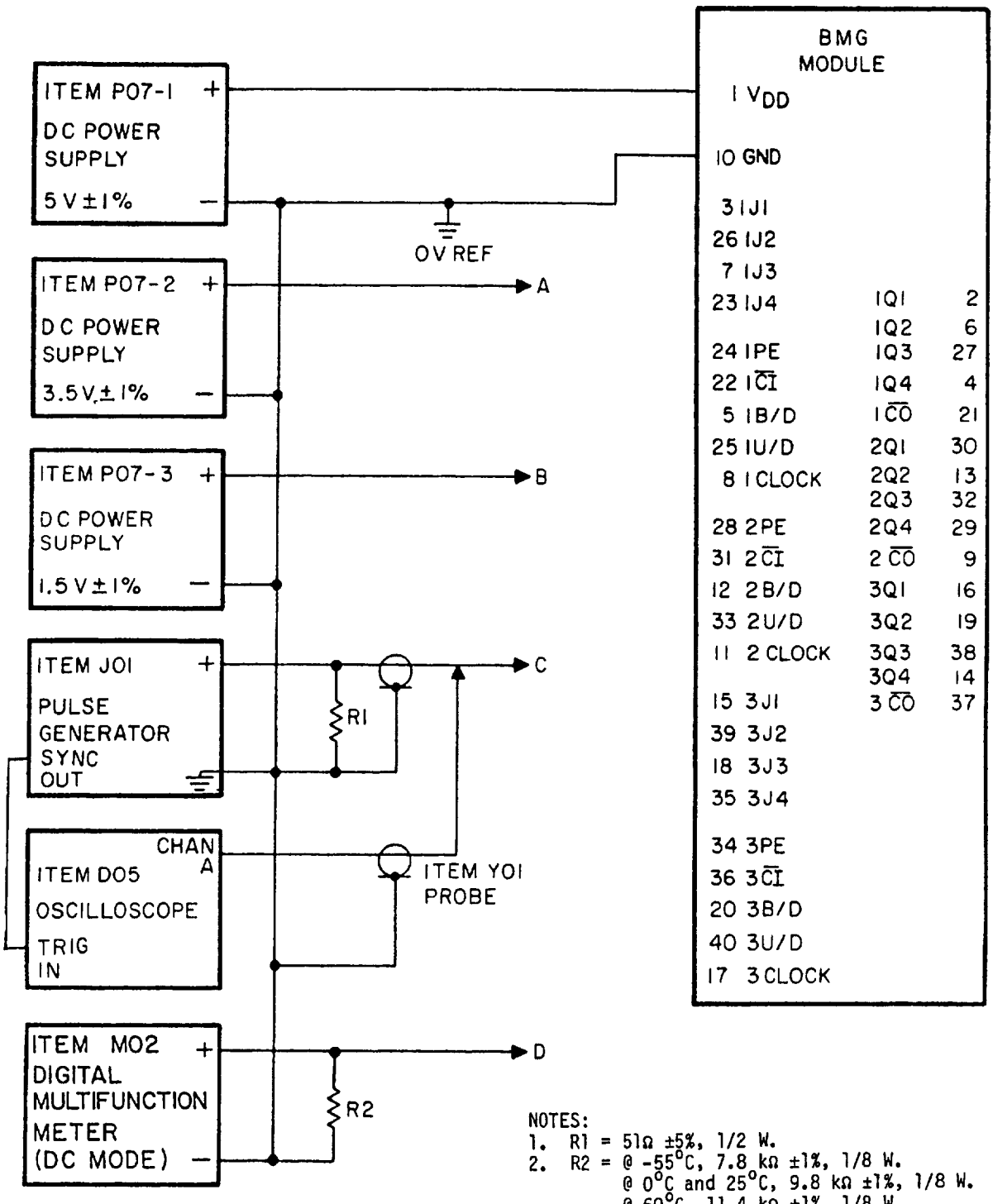
* FIGURE 12. Output voltage (low-level) test circuit for table IX.

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TABLE X. Output voltage (high-level) test.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
H1	$1\overline{CO}$ Output voltage, high level	22,31,36	3,26,7,23,24,5,25,8,28,12,33,11,15,39,18,35,34,20,40,17	Open	21	4.6	5.0	V	13,14
H2	$2\overline{CO}$ "	"	"	"	9	"	"	"	"
H3	$3\overline{CO}$ "	"	"	"	37	"	"	"	"
H4	1Q1 "	3,26,7,23,24,28,15,39,18,35,34	22,5,25,8,31,12,33,11,36,20,40,17	"	2	"	"	"	"
H5	1Q2 "	"	"	"	6	"	"	"	"
H6	1Q3 "	"	"	"	27	"	"	"	"
H7	1Q4 "	"	"	"	4	"	"	"	"
H8	3Q1 "	"	"	"	16	"	"	"	"
H9	3Q2 "	"	"	"	19	"	"	"	"
H10	3Q3 "	"	"	"	38	"	"	"	"
H11	3Q4 "	"	"	"	14	"	"	"	"
H12	2Q1 "	12,33	3,26,7,23,24,22,5,25,8,28,31,15,39,18,35,34,36,20,40,17	11	30	"	"	"	"
H13	2Q2 "	"	"	"	13	"	"	"	"
H14	2Q3 "	"	"	"	32	"	"	"	"
H15	2Q4 "	"	"	"	29	"	"	"	"

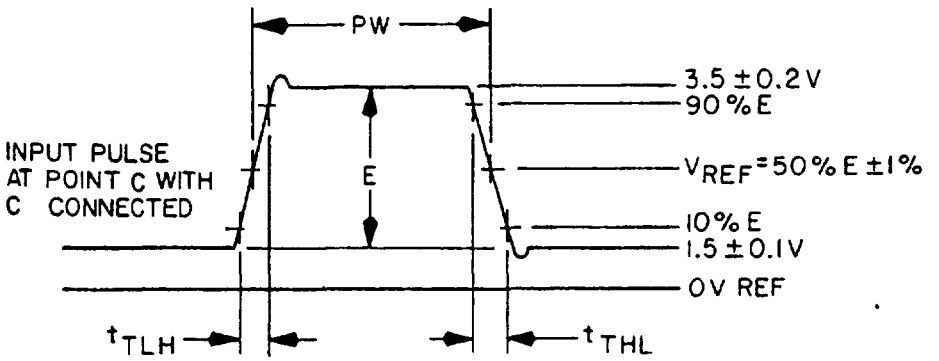
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NOTES:
1. R1 = 51Ω ±5%, 1/2 W.
2. R2 = @ -55°C, 7.8 kΩ ±1%, 1/8 W.
@ 0°C and 25°C, 9.8 kΩ ±1%, 1/8 W.
@ 60°C, 11.4 kΩ ±1%, 1/8 W.
@ 85°C, 13 kΩ ±1%, 1/8 W.

* FIGURE 13. Output voltage (high-level) test circuit for table X.

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NOTES:
1. Input pulse parameters:
   PW = 200 ±20 ns.
   $t_{TLH} = t_{THL}$ = 10 ±2 ns.
2. Place item J01 in single pulse mode.
3. After making the required connections for test conditions H12 through H15, pulse the module 15 times, then take the measurements for test conditions H12 through H15.

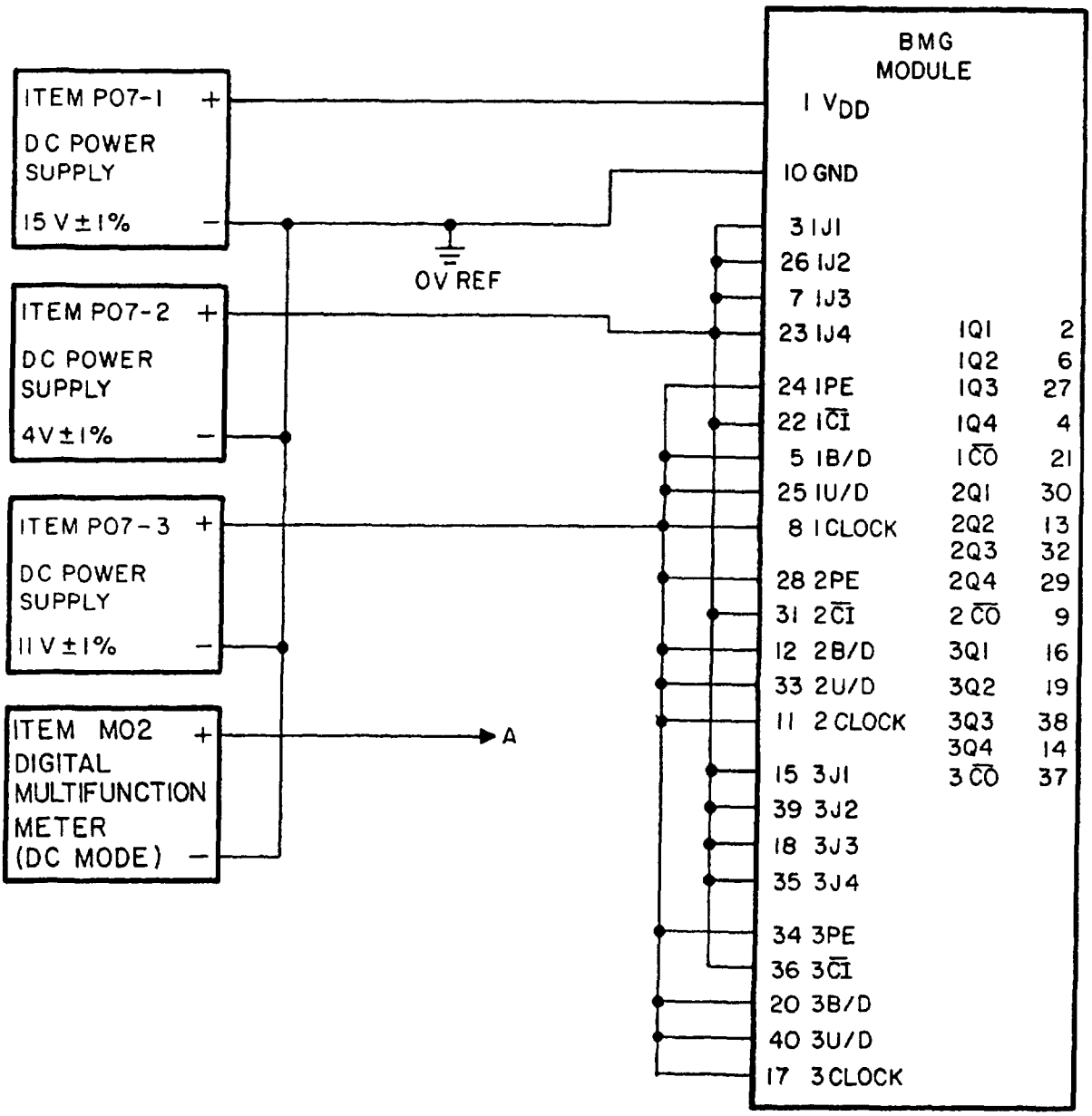
FIGURE 14. Output voltage (high-level) test circuit waveform for table X.

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TABLE XI. Output voltage (low-level) test.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
J1	1Q1 Output voltage, low level	2	0.0	1.5	V	15
J2	1Q2 "	6	"	"	"	"
J3	1Q3 "	27	"	"	"	"
J4	1Q4 "	4	"	"	"	"
J5	$1\overline{CO}$ "	21	"	"	"	"
J6	2Q1 "	30	"	"	"	"
J7	2Q2 "	13	"	"	"	"
J8	2Q3 "	32	"	"	"	"
J9	2Q4 "	29	"	"	"	"
J10	$2\overline{CO}$ "	9	"	"	"	"
J11	3Q1 "	16	"	"	"	"
J12	3Q2 "	19	"	"	"	"
J13	3Q3 "	38	"	"	"	"
J14	3Q4 "	14	"	"	"	"
J15	$3\overline{CO}$ "	37	"	"	"	"

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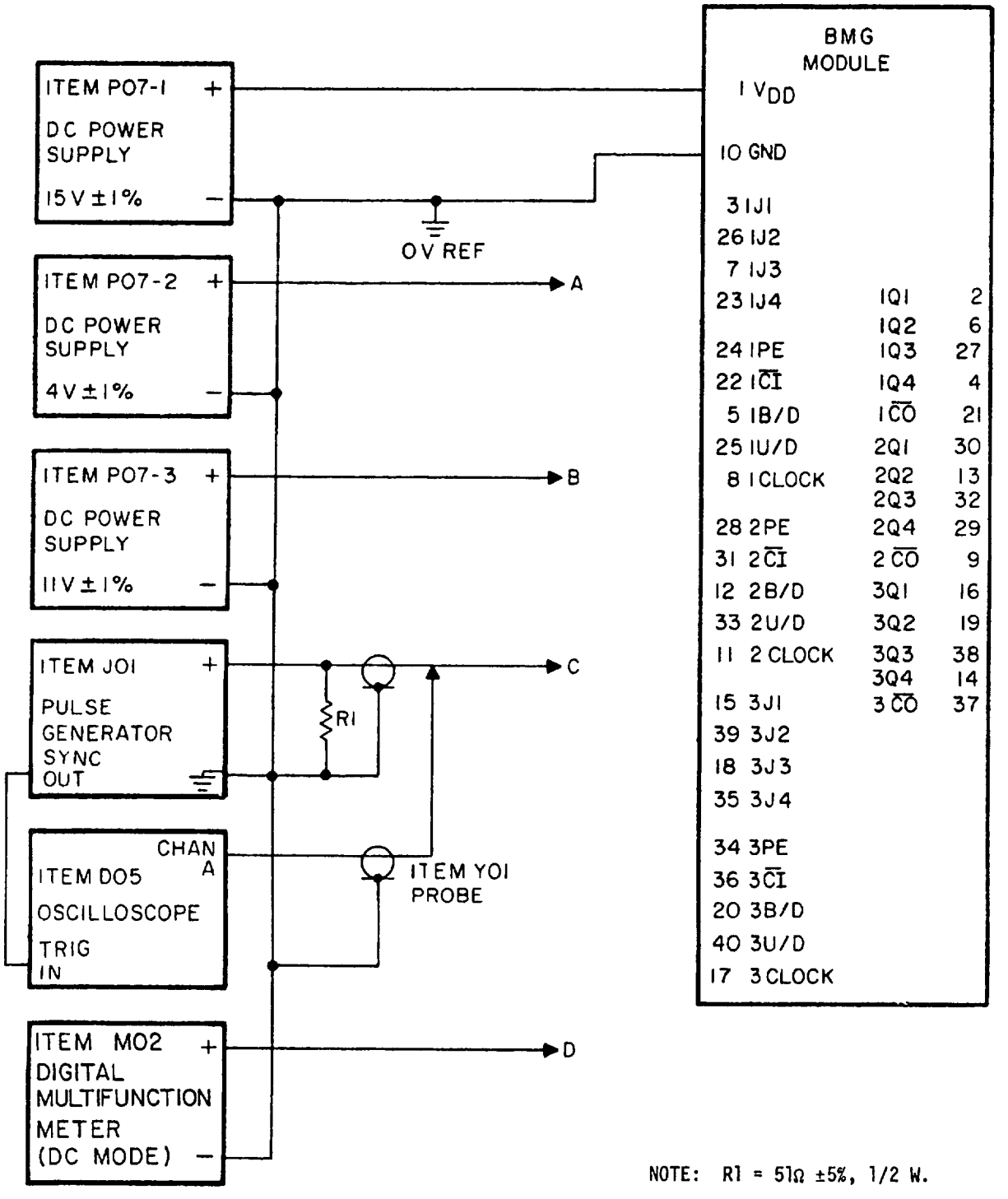
* FIGURE 15. Output voltage (low-level) test circuit for table XI.

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TABLE XII. Output voltage (high-level) test.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
K1	$1\overline{CO}$ Output voltage, high level	3,26,7,23,24,5,25,8,28,12,33,11,15,39,18,35,34,20,40,17	22,31,36	Open	21	13.5	15.0	V	16,17
K2	$2\overline{CO}$ "	"	"	"	9	"	"	"	"
K3	$3\overline{CO}$ "	"	"	"	37	"	"	"	"
K4	1Q1 "	22,5,25,8,31,12,33,11,36,20,40,17	3,26,7,23,24,28,15,39,18,35,34	"	2	"	"	"	"
K5	1Q2 "	"	"	"	6	"	"	"	"
K6	1Q3 "	"	"	"	27	"	"	"	"
K7	1Q4 "	"	"	"	4	"	"	"	"
K8	3Q1 "	"	"	"	16	"	"	"	"
K9	3Q2 "	"	"	"	19	"	"	"	"
K10	3Q3 "	"	"	"	38	"	"	"	"
K11	3Q4 "	"	"	"	14	"	"	"	"
K12	2Q1 "	3,26,7,23,24,22,5,25,8,28,31,15,39,18,35,34,36,20,40,17	12,33	11	30	"	"	"	"
K13	2Q2 "	"	"	"	13	"	"	"	"
K14	2Q3 "	"	"	"	32	"	"	"	"
K15	2Q4 "	"	"	"	29	"	"	"	"

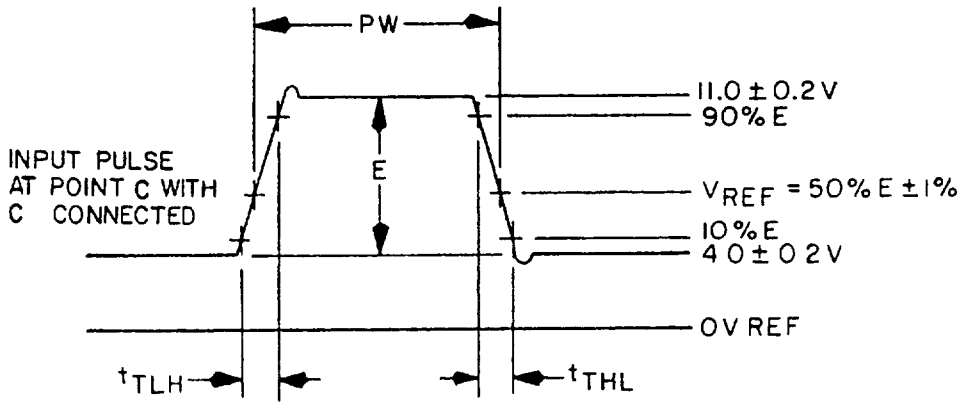
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NOTE: R1 = 51Ω ±5%, 1/2 W.

* FIGURE 16. Output voltage (high-level) test circuit for table XII.

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NOTES:
1. Input pulse parameters:
   PW = 200 ±20 ns.
   $t_{TLH} = t_{THL}$ = 10 ±2 ns.
2. Place item 101 in single pulse mode.
3. After making the connections required for test conditions K12 through K15, pulse the module 15 times, then take the measurements for test conditions K12 through K15.

FIGURE 17. Output voltage (high-level) test circuit waveform for table XII.

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TABLE XIII. Input current (low-level) test.

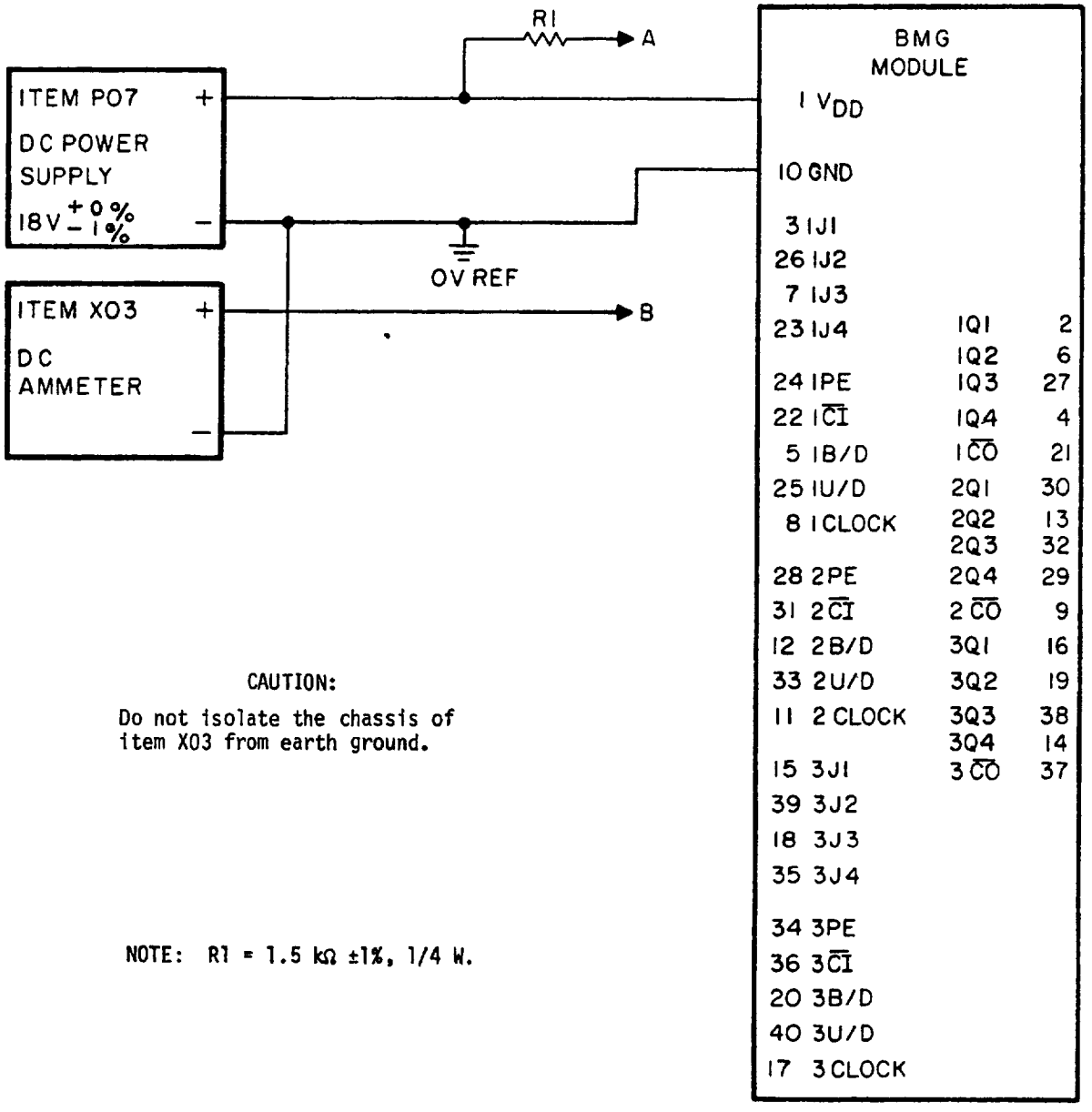
Test condition	Parameter		Connect: A to pin	Connect: B to pin	Initial limits 25°C ±5°C ambient: Min	Initial limits 25°C ±5°C ambient: Max	Unit	Test figure
L1	1J1	Input current, low level	26,7,23,24,22,5,25,8,28,31,12,33, 11,15,39,18,35,34,36,20,40,17	3	0	-200	nA	18
L2	1J2	"	3,7,23,24,22,5,25,8,28,31,12,33, 11,15,39,18,35,34,36,20,40,17	26	"	"	"	"
L3	1J3	"	3,26,23,24,22,5,25,8,28,31,12,33, 11,15,39,18,35,34,36,20,40,17	7	"	"	"	"
L4	1J4	"	3,26,7,24,22,5,25,8,28,31,12,33, 11,15,39,18,35,34,36,20,40,17	23	"	"	"	"
L5	1PE	"	3,26,7,23,22,5,25,8,28,31,12,33, 11,15,39,18,35,34,36,20,40,17	24	"	"	"	"
L6	$1\overline{CT}$	"	3,26,7,23,24,5,25,8,28,31,12,33, 11,15,39,18,35,34,36,20,40,17	22	"	"	"	"
L7	1B/D	"	3,26,7,23,24,22,25,8,28,31,12,33, 11,15,39,18,35,34,36,20,40,17	5	"	"	"	"
L8	1U/D	"	3,26,7,23,24,22,5,8,28,31,12,33, 11,15,39,18,35,34,36,20,40,17	25	"	"	"	"
L9	1 Clock	"	3,26,7,23,24,22,5,25,28,31,12,33, 11,15,39,18,35,34,36,20,40,17	8	"	"	"	"
L10	2PE	"	3,26,7,23,24,22,5,25,8,31,12,33, 11,15,39,18,35,34,36,20,40,17	28	"	"	"	"
L11	$2\overline{CT}$	"	3,26,7,23,24,22,5,25,8,28,12,33, 11,15,39,18,35,34,36,20,40,17	31	"	"	"	"
L12	2B/D	"	3,26,7,23,24,22,5,25,8,28,31,33, 11,15,39,18,35,34,36,20,40,17	12	"	"	"	"
L13	2U/D	"	3,26,7,23,24,22,5,25,8,28,31,12, 11,15,39,18,35,34,36,20,40,17	33	"	"	"	"
L14	2 Clock	"	3,26,7,23,24,22,5,25,8,28,31,12, 33,15,39,18,35,34,36,20,40,17	11	"	"	"	"
L15	3J1	"	3,26,7,23,24,22,5,25,8,28,31,12, 33,11,39,18,35,34,36,20,40,17	15	"	"	"	"

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TABLE XIII. Input current (low-level) test - Continued.

Test condition	Parameter		Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
L16	3J2	Input current, low level	3,26,7,23,24,22,5,25,8,28,31,12,33, 11,15,18,35,34,36,20,40,17	39	0	-200	nA	18
L17	3J3	"	3,26,7,23,24,22,5,25,8,28,31,12,33, 11,15,39,35,34,36,20,40,17	18	"	"	"	"
L18	3J4	"	3,26,7,23,24,22,5,25,8,28,31,12,33, 11,15,39,18,34,36,20,40,17	35	"	"	"	"
L19	3PE	"	3,26,7,23,24,22,5,25,8,28,31,12,33, 11,15,39,18,35,36,20,40,17	34	"	"	"	"
L20	3$\overline{CT}$	"	3,26,7,23,24,22,5,25,8,28,31,12,33, 11,15,39,18,35,34,20,40,17	36	"	"	"	"
L21	3B/D	"	3,26,7,23,24,22,5,25,8,28,31,12,33, 11,15,39,18,35,34,36,40,17	20	"	"	"	"
L22	3U/D	"	3,26,7,23,24,22,5,25,8,28,31,12,33, 11,15,39,18,35,34,36,20,17	40	"	"	"	"
L23	3 Clock	"	3,26,7,23,24,22,5,25,8,28,31,12,33, 11,15,39,18,35,34,36,20,40	17	"	"	"	"

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CAUTION:

Do not isolate the chassis of item X03 from earth ground.

NOTE: R1 = 1.5 kΩ ±1%, 1/4 W.

FIGURE 18. Input current (low-level) test circuit for table XIII.

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TABLE XIV. Input current (high-level) test.

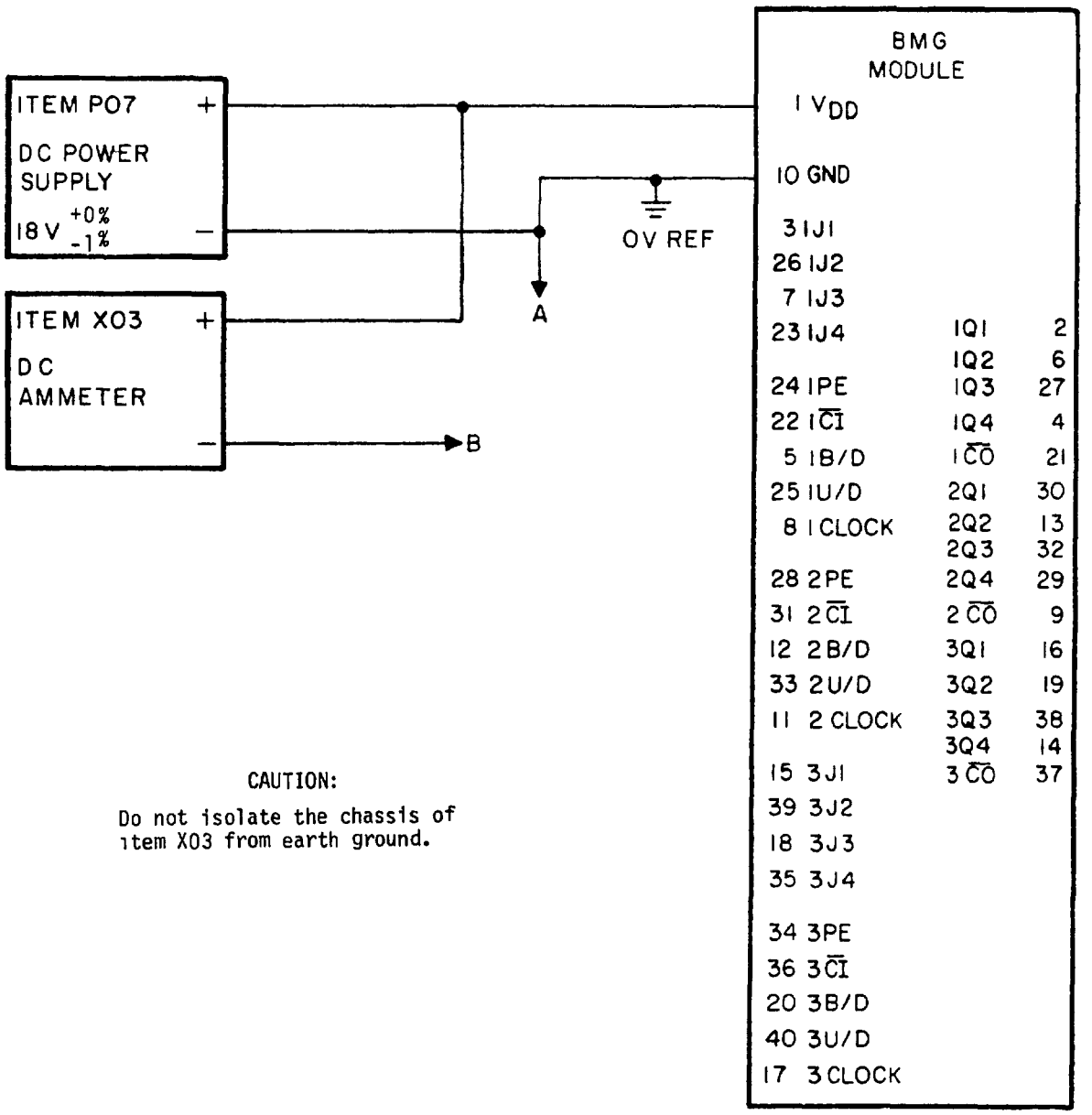
Test condition	Parameter	Connect: A to pin	Connect: B to pin	Initial limits 25°C ±5°C ambient: Min	Initial limits 25°C ±5°C ambient: Max	Unit	Test figure
M1	1J1 Input current, high level	26,7,23,24,22,5,25,8,28,31,12,33, 11,15,39,18,35,34,36,20,40,17	3	0	200	nA	19
M2	1J2 "	3,7,23,24,22,5,25,8,28,31,12,33, 11,15,39,18,35,34,36,20,40,17	26	"	"	"	"
M3	1J3 "	3,26,23,24,22,5,25,8,28,31,12,33, 11,15,39,18,35,34,36,20,40,17	7	"	"	"	"
M4	1J4 "	3,26,7,24,22,5,25,8,28,31,12,33, 11,15,39,18,35,34,36,20,40,17	23	"	"	"	"
M5	1PE "	3,26,7,23,22,5,25,8,28,31,12,33, 11,15,39,18,35,34,36,20,40,17	24	"	"	"	"
M6	1$\overline{CI}$ "	3,26,7,23,24,5,25,8,28,31,12,33, 11,15,39,18,35,34,36,20,40,17	22	"	"	"	"
M7	1B/D "	3,26,7,23,24,22,25,8,28,31,12,33, 11,15,39,18,35,34,36,20,40,17	5	"	"	"	"
M8	1U/D "	3,26,7,23,24,22,5,8,28,31,12,33, 11,15,39,18,35,34,36,20,40,17	25	"	"	"	"
M9	1 Clock "	3,26,7,23,24,22,5,25,28,31,12,33, 11,15,39,18,35,34,36,20,40,17	8	"	"	"	"
M10	2PE "	3,26,7,23,24,22,5,25,8,31,12,33, 11,15,39,18,35,34,36,20,40,17	28	"	"	"	"
M11	2$\overline{CI}$ "	3,26,7,23,24,22,5,25,8,28,12,33, 11,15,39,18,35,34,36,20,40,17	31	"	"	"	"
M12	2B/D "	3,26,7,23,24,22,5,25,8,28,31,33, 11,15,39,18,35,34,36,20,40,17	12	"	"	"	"
M13	2U/D "	3,26,7,23,24,22,5,25,8,28,31,12, 11,15,39,18,35,34,36,20,40,17	33	"	"	"	"
M14	2 Clock "	3,26,7,23,24,22,5,25,8,28,31,12, 33,15,39,18,35,34,36,20,40,17	11	"	"	"	"
M15	3J1 "	3,26,7,23,24,22,5,25,8,28,31,12, 33,11,39,18,35,34,36,20,40,17	15	"	"	"	"

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TABLE XIV. Input current (high-level) test - Continued.

Test condition	Parameter		Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
M16	3J2	Input current, high level	3,26,7,23,24,22,5,25,8,28,31,12,33, 11,15,18,35,34,36,20,40,17	39	0	200	nA	19
M17	3J3	"	3,26,7,23,24,22,5,25,8,28,31,12,33, 11,15,39,35,34,36,20,40,17	18	"	"	"	"
M18	3J4	"	3,26,7,23,24,22,5,25,8,28,31,12,33, 11,15,39,18,34,36,20,40,17	35	"	"	"	"
M19	3PE	"	3,26,7,23,24,22,5,25,8,28,31,12,33, 11,15,39,18,35,36,20,40,17	34	"	"	"	"
M20	3$\overline{CT}$	"	3,26,7,23,24,22,5,25,8,28,31,12,33, 11,15,39,18,35,34,20,40,17	36	"	"	"	"
M21	3B/D	"	3,26,7,23,24,22,5,25,8,28,31,12,33, 11,15,39,18,35,34,36,40,17	20	"	"	"	"
M22	3U/D	"	3,26,7,23,24,22,5,25,8,28,31,12,33, 11,15,39,18,35,34,36,20,17	40	"	"	"	"
M23	3 Clock	"	3,26,7,23,24,22,5,25,8,28,31,12,33, 11,15,39,18,35,34,36,20,40	17	"	"	"	"

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CAUTION:

Do not isolate the chassis of item XO3 from earth ground.

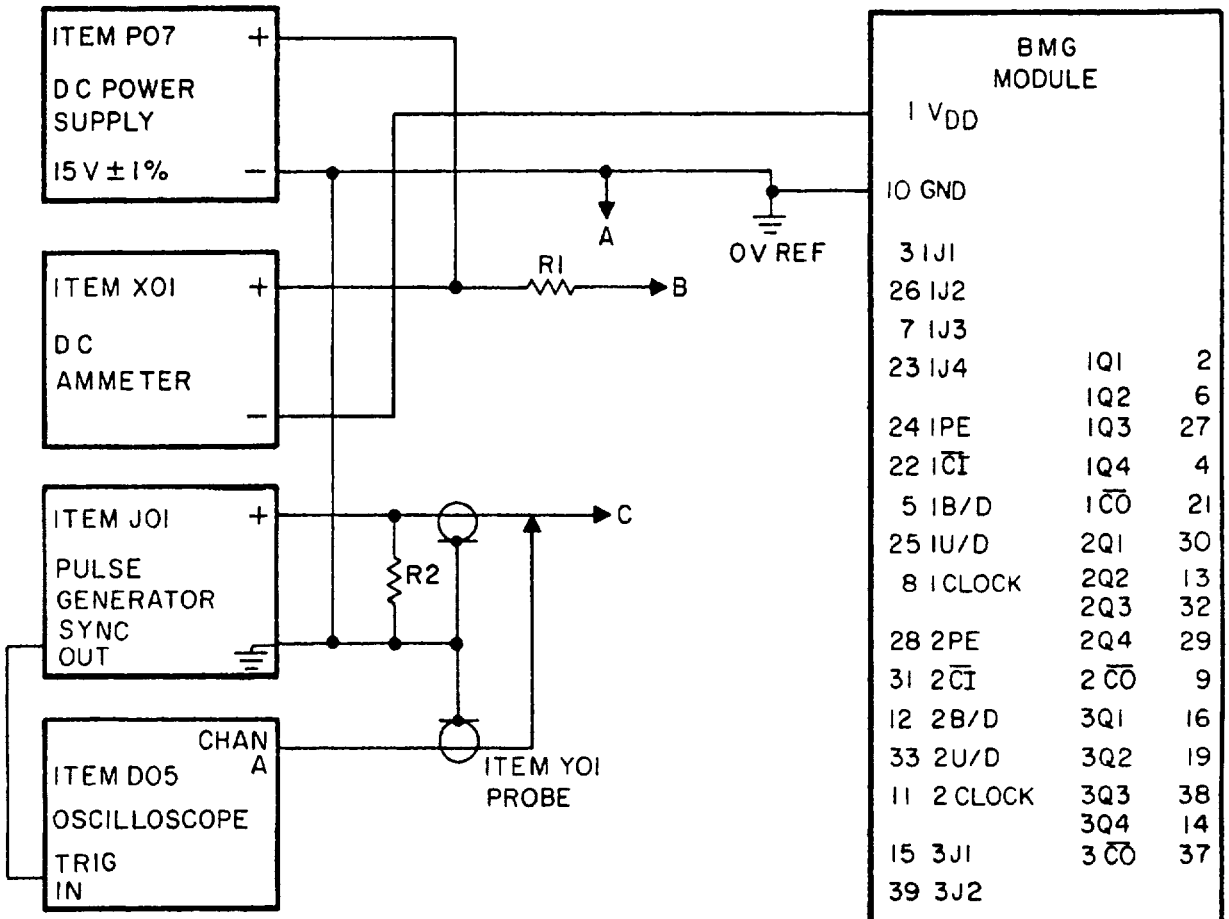
FIGURE 19. Input current (high-level) test circuit for table XIV.

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TABLE XV. Power supply current (low-level and high-level) test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
N1	Power supply current, low level	3,26,7,23,22,5, 25,8,31,12,33,11, 15,39,18,35,36, 20,40,17	24,28,34	Open	0	60	µA	20,21
N2	Power supply current, high level	22,5,25,8,28,31, 36,20,40,17	24,34,3,26,7, 23,15,39,18, 35,12,33	11	0	60	µA	20,21

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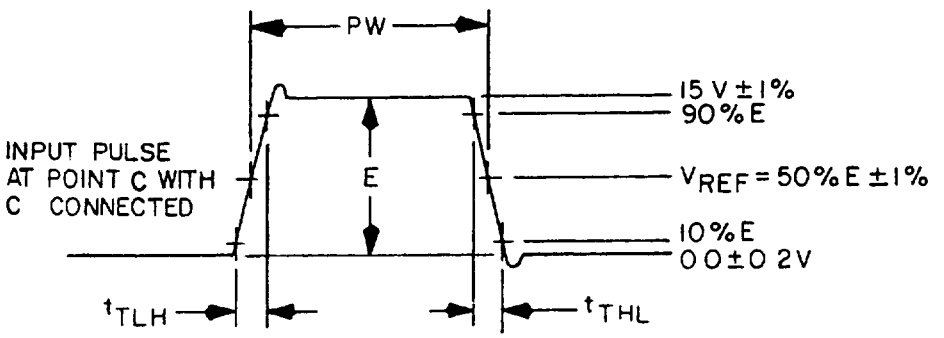
CAUTION:

Do not isolate the chassis of item X01 from earth ground.

NOTES:
1. R1 = 1.5 kΩ ±1%, 1/8 W.
2. R2 = 51Ω ±5%, 1/2 W.
3. Place item J01 in single pulse mode.
4. For test condition N2, pulse the module 15 times then take the measurement.

FIGURE 20. Power supply current (low-level and high-level) test circuit for table XV.

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NOTE: Input pulse parameters:
PW = 200 ±20 ns.
$t_{TLH} = t_{THL} = 10 \pm 2$ ns.

FIGURE 21. Power supply current (low-level and high-level) test circuit waveform for table XV.

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TABLE XVI. Transition time (t_{TLH} and t_{THL}) test.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
P1	1Q1	Transition time (t_{TLH}), C_L = 50 pF	5,25,12,33,20,40	3,26,7,23,24,22, 28,31,15,39,18,35, 34,36	8,11,17	2	26	260	ns	22,23
P2	1Q2	"	"	"	"	6	"	"	"	"
P3	1Q3	"	"	"	"	27	"	"	"	"
P4	1Q4	"	"	"	"	4	"	"	"	"
P5	$1\overline{CO}$	"	"	"	"	21	"	"	"	"
P6	2Q1	"	"	"	"	30	"	"	"	"
P7	2Q2	"	"	"	"	13	"	"	"	"
P8	2Q3	"	"	"	"	32	"	"	"	"
P9	2Q4	"	"	"	"	29	"	"	"	"
P10	$2\overline{CO}$	"	"	"	"	9	"	"	"	"
P11	3Q1	"	"	"	"	16	"	"	"	"
P12	3Q2	"	"	"	"	19	"	"	"	"
P13	3Q3	"	"	"	"	38	"	"	"	"
P14	3Q4	"	"	"	"	14	"	"	"	"
P15	$3\overline{CO}$	"	"	"	"	37	"	"	"	"
P16	1Q1	Transition time (t_{TLH}), C_L = 150 pF	"	"	"	2	74	740	"	"
P17	1Q2	"	"	"	"	6	"	"	"	"
P18	1Q3	"	"	"	"	27	"	"	"	"
P19	1Q4	"	"	"	"	4	"	"	"	"
P20	$1\overline{CO}$	"	"	"	"	21	"	"	"	"
P21	2Q1	"	"	"	"	30	"	"	"	"

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TABLE XVI. Transition time (t_{TLH} and t_{THL}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
P22	2Q2	Transition time (t_{TLH}), C_L = 150 pF	5,25,12,33,20,40	3,26,7,23,24,22, 28,31,15,39,18,35, 34,36	8,11,17	13	74	740	ns	22,23
P23	2Q3	"	"	"	"	32	"	"	"	"
P24	2Q4	"	"	"	"	29	"	"	"	"
P25	$2\overline{CO}$	"	"	"	"	9	"	"	"	"
P26	3Q1	"	"	"	"	16	"	"	"	"
P27	3Q2	"	"	"	"	19	"	"	"	"
P28	3Q3	"	"	"	"	38	"	"	"	"
P29	3Q4	"	"	"	"	14	"	"	"	"
P30	$3\overline{CO}$	"	"	"	"	37	"	"	"	"
P31	1Q1	Transition time (t_{THL}), C_L = 50 pF	"	"	"	2	26	260	"	"
P32	1Q2	"	"	"	"	6	"	"	"	"
P33	1Q3	"	"	"	"	27	"	"	"	"
P34	1Q4	"	"	"	"	4	"	"	"	"
P35	$1\overline{CO}$	"	"	"	"	21	"	"	"	"
P36	2Q1	"	"	"	"	30	"	"	"	"
P37	2Q2	"	"	"	"	13	"	"	"	"
P38	2Q3	"	"	"	"	32	"	"	"	"
P39	2Q4	"	"	"	"	29	"	"	"	"
P40	$2\overline{CO}$	"	"	"	"	9	"	"	"	"
P41	3Q1	"	"	"	"	16	"	"	"	"
P42	3Q2	"	"	"	"	19	"	"	"	"

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TABLE XVI. Transition time (t_{TLH} and t_{THL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
P43	3Q3 Transition time (t_{THL}), C_L = 50 pF	5,25,12,33,20,40	3,26,7,23,24,22, 28,31,15,39,18,35, 34,36	8,11,17	38	26	260	ns	22,23
P44	3Q4 "	"	"	"	14	"	"	"	"
P45	$3\overline{CO}$ "	"	"	"	37	"	"	"	"
P46	1Q1 Transition time (t_{THL}), C_L = 150 pF	"	"	"	2	74	740	"	"
P47	1Q2 "	"	"	"	6	"	"	"	"
P48	1Q3 "	"	"	"	27	"	"	"	"
P49	1Q4 "	"	"	"	4	"	"	"	"
P50	$1\overline{CO}$ "	"	"	"	21	"	"	"	"
P51	2Q1 "	"	"	"	30	"	"	"	"
P52	2Q2 "	"	"	"	13	"	"	"	"
P53	2Q3 "	"	"	"	32	"	"	"	"
P54	2Q4 "	"	"	"	29	"	"	"	"
P55	$2\overline{CO}$ "	"	"	"	9	"	"	"	"
P56	3Q1 "	"	"	"	16	"	"	"	"
P57	3Q2 "	"	"	"	19	"	"	"	"
P58	3Q3 "	"	"	"	38	"	"	"	"
P59	3Q4 "	"	"	"	14	"	"	"	"
P60	$3\overline{CO}$ "	"	"	"	37	"	"	"	"

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TABLE XVII. Transition time (t_{TLH} and t_{THL}) test.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
Q1	1Q1	Transition time (t_{TLH}), C_L = 50 pF	5,25,12,33,20,40	3,26,7,23,24,22, 28,31,15,39,18,35, 34,36	8,11,17	2	13	130	ns	22,23
Q2	1Q2	"	"	"	"	6	"	"	"	"
Q3	1Q3	"	"	"	"	27	"	"	"	"
Q4	1Q4	"	"	"	"	4	"	"	"	"
Q5	$1\overline{CO}$	"	"	"	"	21	"	"	"	"
Q6	2Q1	"	"	"	"	30	"	"	"	"
Q7	2Q2	"	"	"	"	13	"	"	"	"
Q8	2Q3	"	"	"	"	32	"	"	"	"
Q9	2Q4	"	"	"	"	29	"	"	"	"
Q10	$2\overline{CO}$	"	"	"	"	9	"	"	"	"
Q11	3Q1	"	"	"	"	16	"	"	"	"
Q12	3Q2	"	"	"	"	19	"	"	"	"
Q13	3Q3	"	"	"	"	38	"	"	"	"
Q14	3Q4	"	"	"	"	14	"	"	"	"
Q15	$3\overline{CO}$	"	"	"	"	37	"	"	"	"
Q16	1Q1	Transition time (t_{THL}), C_L = 50 pF	"	"	"	2	"	"	"	"
Q17	1Q2	"	"	"	"	6	"	"	"	"
Q18	1Q3	"	"	"	"	27	"	"	"	"
Q19	1Q4	"	"	"	"	4	"	"	"	"
Q20	$1\overline{CO}$	"	"	"	"	21	"	"	"	"
Q21	2Q1	"	"	"	"	30	"	"	"	"

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TABLE XVII. Transition time (t_{TLH} and t_{THL}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
Q22	2Q2	Transition time (t_{THL}), C_L = 50 pF	5,25,12,33,20,40	3,26,7,23,24,22, 28,31,15,39,18,35, 34,36	8,11,17	13	13	130	ns	22,23
Q23	2Q3	"	"	"	"	32	"	"	"	"
Q24	2Q4	"	"	"	"	29	"	"	"	"
Q25	$2\overline{CO}$	"	"	"	"	9	"	"	"	"
Q26	3Q1	"	"	"	"	16	"	"	"	"
Q27	3Q2	"	"	"	"	19	"	"	"	"
Q28	3Q3	"	"	"	"	38	"	"	"	"
Q29	3Q4	"	"	"	"	14	"	"	"	"
Q30	$3\overline{CO}$	"	"	"	"	37	"	"	"	"

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TABLE XVIII. Transition time (t_{TLH} and t_{THL}) test.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
R1	1Q1 Transition time (t_{TLH}), C_L = 50 pF	5,25,12,33,20,40	3,26,7,23,24,22, 28,31,15,39,18,35, 34,36	8,11,17	2	10	100	ns	22,23
R2	1Q2 "	"	"	"	6	"	"	"	"
R3	1Q3 "	"	"	"	27	"	"	"	"
R4	1Q4 "	"	"	"	4	"	"	"	"
R5	$1\overline{CO}$ "	"	"	"	21	"	"	"	"
R6	2Q1 "	"	"	"	30	"	"	"	"
R7	2Q2 "	"	"	"	13	"	"	"	"
R8	2Q3 "	"	"	"	32	"	"	"	"
R9	2Q4 "	"	"	"	29	"	"	"	"
R10	$2\overline{CO}$ "	"	"	"	9	"	"	"	"
R11	3Q1 "	"	"	"	16	"	"	"	"
R12	3Q2 "	"	"	"	19	"	"	"	"
R13	3Q3 "	"	"	"	38	"	"	"	"
R14	3Q4 "	"	"	"	14	"	"	"	"
R15	$3\overline{CO}$ "	"	"	"	37	"	"	"	"
R16	1Q1 Transition time (t_{THL}), C_L = 50 pF	"	"	"	2	"	"	"	"
R17	1Q2 "	"	"	"	6	"	"	"	"
R18	1Q3 "	"	"	"	27	"	"	"	"
R19	1Q4 "	"	"	"	4	"	"	"	"
R20	$1\overline{CO}$ "	"	"	"	21	"	"	"	"
R21	2Q1 "	"	"	"	30	"	"	"	"

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TABLE XVIII. Transition time (t_{TLH} and t_{THL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
R22	2Q2 Transition time (t_{THL}), C_L = 50 pF	5,25,12,33,20,40	3,26,7,23,24,22, 28,31,15,39,18,35, 34,36	8,11,17	13	10	100	ns	22,23
R23	2Q3 "	"	"	"	32	"	"	"	"
R24	2Q4 "	"	"	"	29	"	"	"	"
R25	$2\overline{CO}$ "	"	"	"	9	"	"	"	"
R26	3Q1 "	"	"	"	16	"	"	"	"
R27	3Q2 "	"	"	"	19	"	"	"	"
R28	3Q3 "	"	"	"	38	"	"	"	"
R29	3Q4 "	"	"	"	14	"	"	"	"
R30	$3\overline{CO}$ "	"	"	"	37	"	"	"	"

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TABLE XIX. Propagation delay time (t_{PLH} and t_{PHL}) test.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
S1	1Q1	Propagation delay time (t_{PLH}), C_L = 50 pF	5,25,12,33, 20,40	3,26,7,23,24,22,28,31,15, 39,18,35,34,36	8,11,17	2	65	650	ns	22,23
S2	1Q2	"	"	"	"	6	"	"	"	"
S3	1Q3	"	"	"	"	27	"	"	"	"
S4	1Q4	"	"	"	"	4	"	"	"	"
S5	$1\overline{CO}$	"	"	"	"	21	75	750	"	"
S6	2Q1	"	"	"	"	30	65	650	"	"
S7	2Q2	"	"	"	"	13	"	"	"	"
S8	2Q3	"	"	"	"	32	"	"	"	"
S9	2Q4	"	"	"	"	29	"	"	"	"
S10	$2\overline{CO}$	"	"	"	"	9	75	750	"	"
S11	3Q1	"	"	"	"	16	65	650	"	"
S12	3Q2	"	"	"	"	19	"	"	"	"
S13	3Q3	"	"	"	"	38	"	"	"	"
S14	3Q4	"	"	"	"	14	"	"	"	"
S15	$3\overline{CO}$	"	"	"	"	37	75	750	"	"
S16	1Q1	Propagation delay time (t_{PLH}), C_L = 150 pF	"	"	"	2	85	850	"	"
S17	1Q2	"	"	"	"	6	"	"	"	"
S18	1Q3	"	"	"	"	27	"	"	"	"
S19	1Q4	"	"	"	"	4	"	"	"	"
S20	$1\overline{CO}$	"	"	"	"	21	95	950	"	"
S21	2Q1	"	"	"	"	30	85	850	"	"

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TABLE XIX. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
S22	2Q2 Propagation delay time (t_{PLH}), C_L = 150 pF	5,25,12,33, 20,40	3,26,7,23,24,22,28,31,15, 39,18,35,34,36	8,11,17	13	85	850	ns	22,23
S23	2Q3 "	"	"	"	32	"	"	"	"
S24	2Q4 "	"	"	"	29	"	"	"	"
S25	2$\overline{CO}$ "	"	"	"	9	95	950	"	"
S26	3Q1 "	"	"	"	16	85	850	"	"
S27	3Q2 "	"	"	"	19	85	"	"	"
S28	3Q3 "	"	"	"	38	"	"	"	"
S29	3Q4 "	"	"	"	14	"	"	"	"
S30	3$\overline{CO}$ "	"	"	"	37	95	950	"	"
S31	1Q1 Propagation delay time (t_{PHL}), C_L = 50 pF	"	"	"	2	65	650	"	"
S32	1Q2 "	"	"	"	6	"	"	"	"
S33	1Q3 "	"	"	"	27	"	"	"	"
S34	1Q4 "	"	"	"	4	"	"	"	"
S35	1$\overline{CO}$ "	"	"	'	21	75	750	"	"
S36	2Q1 "	"	"	"	30	65	650	"	"
S37	2Q2 "	"	"	"	13	"	"	"	"
S38	2Q3 "	"	"	"	32	"	"	"	"
S39	2Q4 "	"	"	"	29	"	"	"	"
S40	2$\overline{CO}$ "	"	"	"	9	75	750	"	"
S41	3Q1 "	"	"	"	16	65	650	"	"
S42	3Q2 "	"	"	"	19	65	650	"	"

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TABLE XIX. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
S43	3Q3	Propagation delay time (t_{PHL}), C_L = 50 pF	5,25,12,33, 20,40	3,26,7,23,24,22,28,31,15, 39,18,35,34,36	8,11,17	38	65	650	ns	22,23
S44	3Q4	"	"	"	"	14	65	650	"	"
S45	$3\overline{CO}$	"	"	"	"	37	75	750	"	"
S46	1Q1	Propagation delay time (t_{PHL}), C_L = 150 pF	"	"	"	2	85	850	"	"
S47	1Q2	"	"	"	"	6	"	"	"	"
S48	1Q3	"	"	"	"	27	"	"	"	"
S49	1Q4	"	"	"	"	4	"	"	"	"
S50	$1\overline{CO}$	"	"	"	"	21	95	950	"	"
S51	2Q1	"	"	"	"	30	85	850	"	"
S52	2Q2	"	"	"	"	13	"	"	"	"
S53	2Q3	"	"	"	"	32	"	"	"	"
S54	2Q4	"	"	"	"	29	"	"	"	"
S55	$2\overline{CO}$	"	"	"	"	9	95	950	"	"
S56	3Q1	"	"	"	"	16	85	850	"	"
S57	3Q2	"	"	"	"	19	"	"	"	"
S58	3Q3	"	"	"	"	38	"	"	"	"
S59	3Q4	"	"	"	"	14	"	"	"	"
S60	$3\overline{CO}$	"	"	"	"	37	95	950	"	"

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TABLE XX. Propagation delay time (t_{PLH} and t_{PHL}) test.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
T1	1Q1	Propagation delay time (t_{PLH}), C_L = 50 pF	5,25,12,33, 20,40	3,26,7,23,24,22,28,31,15, 39,18,35,34,36	8,11,17	2	25	250	ns	22,23
T2	1Q2	"	"	"	"	6	"	"	"	"
T3	1Q3	"	"	"	"	27	"	"	"	"
T4	1Q4	"	"	"	"	4	"	"	"	"
T5	$1\overline{CO}$	"	"	"	"	21	30	300	"	"
T6	2Q1	"	"	"	"	30	25	250	"	"
T7	2Q2	"	"	"	"	13	"	"	"	"
T8	2Q3	"	"	"	"	32	"	"	"	"
T9	2Q4	"	"	"	"	29	"	"	"	"
T10	$2\overline{CO}$	"	"	"	"	9	30	300	"	"
T11	3Q1	"	"	"	"	16	25	250	"	"
T12	3Q2	"	"	"	"	19	"	"	"	"
T13	3Q3	"	"	"	"	38	"	"	"	"
T14	3Q4	"	"	"	"	14	"	"	"	"
T15	$3\overline{CO}$	"	"	"	"	37	30	300	"	"
T16	1Q1	Propagation delay time (t_{PHL}), C_L = 50 pF	"	"	"	2	25	250	"	"
T17	1Q2	"	"	"	"	6	"	"	"	"
T18	1Q3	"	"	"	"	27	"	"	"	"
T19	1Q4	"	"	"	"	4	"	"	"	"
T20	$1\overline{CO}$	"	"	"	"	21	30	300	"	"
T21	2Q1	"	"	"	"	30	25	250	"	"

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TABLE XX. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
T22	2Q2	Propagation delay time (t_{PHL}), C_L = 50 pF	5,25,12,33, 20,40	3,26,7,23,24,22,28,31,15, 39,18,35,34,36	8,11,17	13	25	250	ns	22,23
T23	2Q3	"	"	"	"	32	"	"	"	"
T24	2Q4	"	"	"	"	29	"	"	"	"
T25	$2\overline{CO}$	"	"	"	"	9	30	300	"	"
T26	3Q1	"	"	"	"	16	25	250	"	"
T27	3Q2	"	"	"	"	19	"	"	"	"
T28	3Q3	"	"	"	"	38	"	"	"	"
T29	3Q4	"	"	"	"	14	"	"	"	"
T30	$3\overline{CO}$	"	"	"	"	37	30	300	"	"

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TABLE XXI. Propagation delay time (t_{PLH} and t_{PHL}) test.

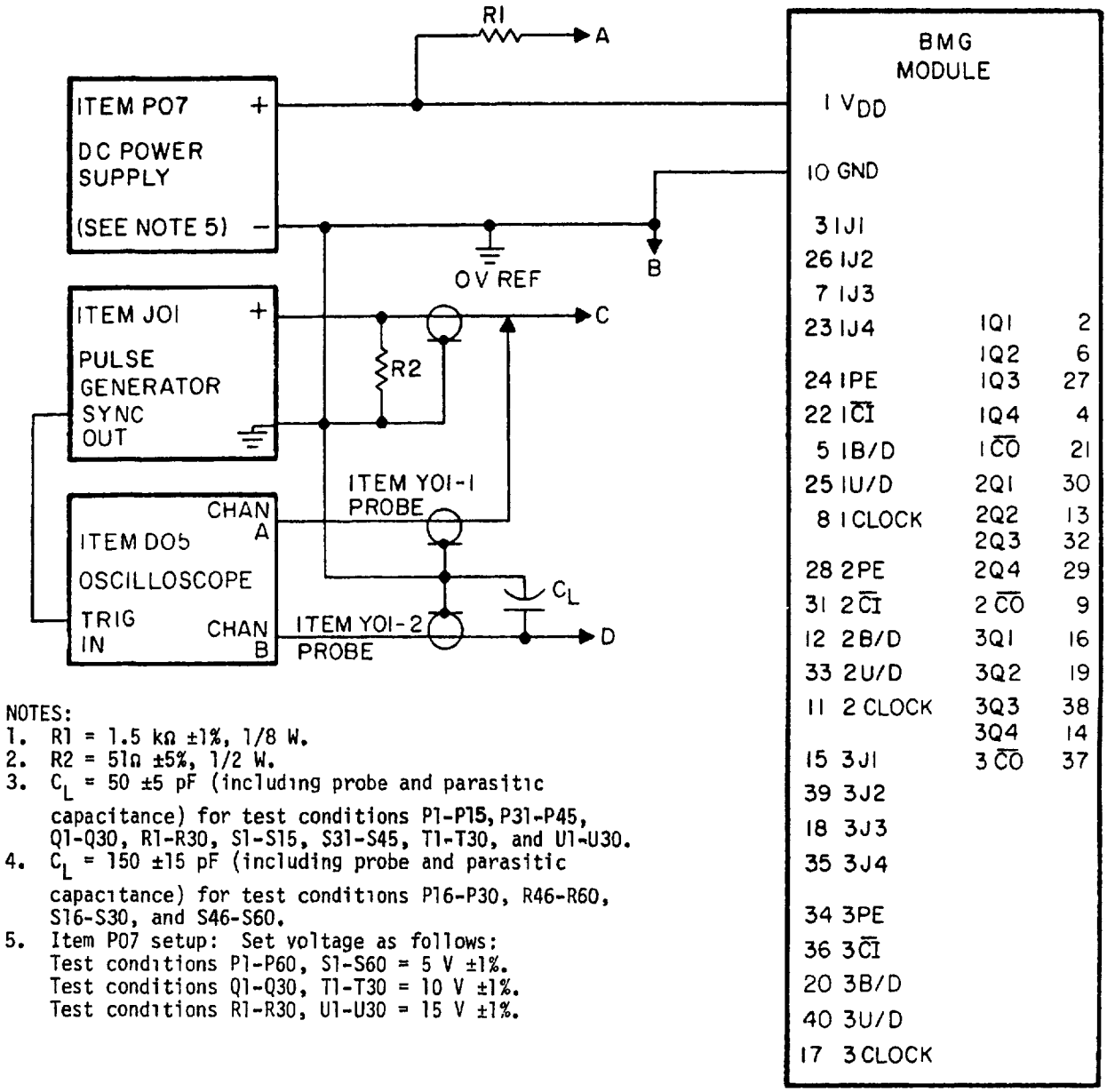
Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
U1	1Q1 Propagation delay time (t_{PLH}), C_L = 50 pF	5,25,12,33, 20,40	3,26,7,23,24,22,28,31,15, 39,18,35,34,36	8,11,17	2	20	200	ns	22,23
U2	1Q2 "	"	"	"	6	"	"	"	"
U3	1Q3 "	"	"	"	27	"	"	"	"
U4	1Q4 "	"	"	"	4	"	"	"	"
U5	1$\overline{CO}$ "	"	"	"	21	24	240	"	"
U6	2Q1 "	"	"	"	30	20	200	"	"
U7	2Q2 "	"	"	"	13	"	"	"	"
U8	2Q3 "	"	"	"	32	"	"	"	"
U9	2Q4 "	"	"	"	29	"	"	"	"
U10	2$\overline{CO}$ "	"	"	"	9	24	240	"	"
U11	3Q1 "	"	"	"	16	20	200	"	"
U12	3Q2 "	"	"	"	19	"	"	"	"
U13	3Q3 "	"	"	"	38	"	"	"	"
U14	3Q4 "	"	"	"	14	"	"	"	"
U15	3$\overline{CO}$ "	"	"	"	37	24	240	"	"
U16	1Q1 Propagation delay time (t_{PHL}), C_L = 50 pF	"	"	"	2	20	200	"	"
U17	1Q2 "	"	"	"	6	"	"	"	"
U18	1Q3 "	"	"	"	27	"	"	"	"
U19	1Q4 "	"	"	"	4	"	"	"	"
U20	1$\overline{CO}$ "	"	"	"	21	24	240	"	"
U21	2Q1 "	"	"	"	30	20	200	"	"

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TABLE XXI. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
U22	2Q2	Propagation delay time (t_{PHL}), C_L = 50 pF	5,25,12,33, 20,40	3,26,7,23,24,22,28,31,15, 39,18,35,34,36	8,11,17	13	20	200	ns	22,23
U23	2Q3	"	"	"	"	32	"	"	"	"
U24	2Q4	"	"	"	"	29	"	"	"	"
U25	$2\overline{CO}$	"	"	"	"	9	24	240	"	"
U26	3Q1	"	"	"	"	16	20	200	"	"
U27	3Q2	"	"	"	"	19	"	"	"	"
U28	3Q3	"	"	"	"	38	"	"	"	"
U29	3Q4	"	"	"	"	14	"	"	"	"
U30	$3\overline{CO}$	"	"	"	"	37	24	240	"	"

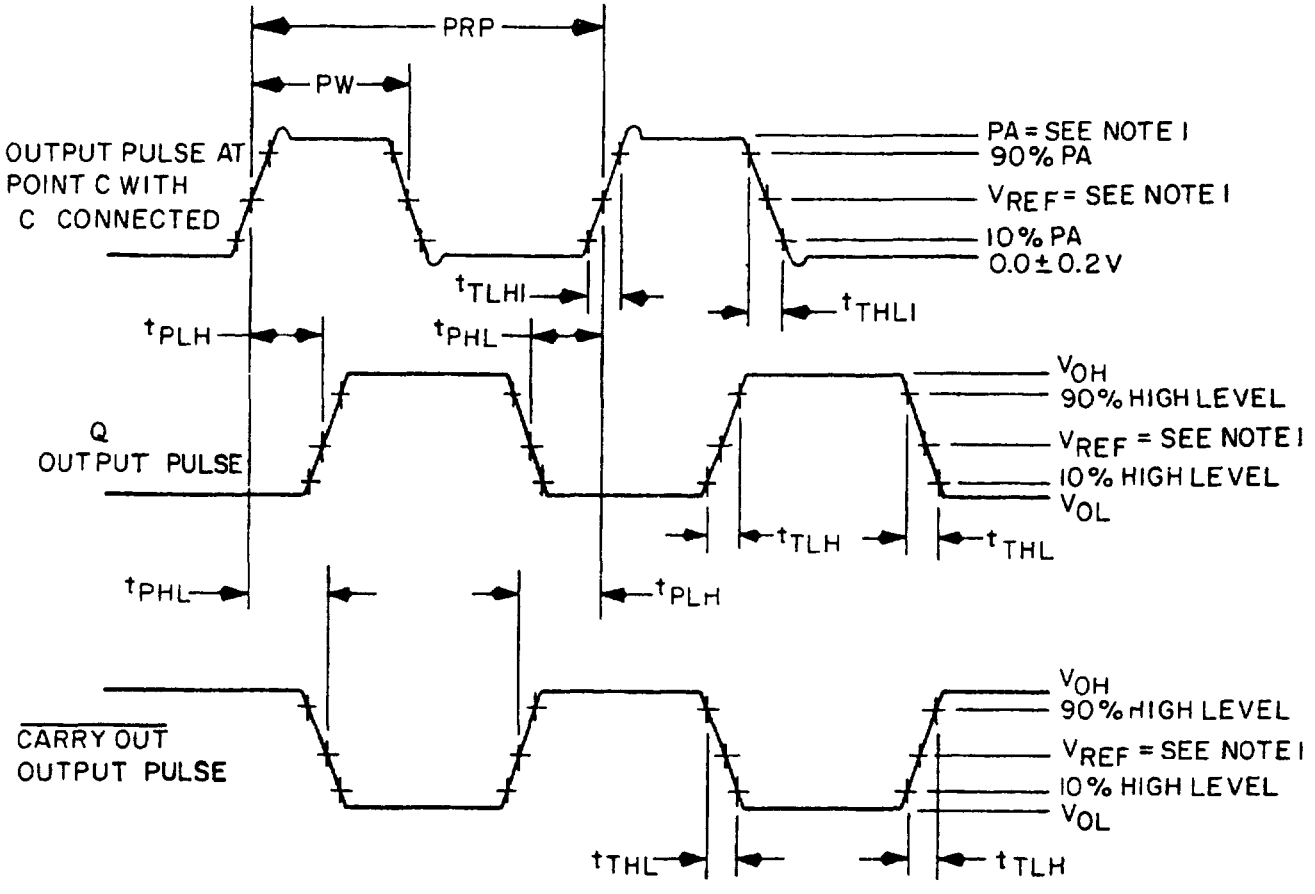
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NOTES:
1. R1 = 1.5 kΩ ±1%, 1/8 W.
2. R2 = 51Ω ±5%, 1/2 W.
3. C_L = 50 ±5 pF (including probe and parasitic capacitance) for test conditions P1-P15, P31-P45, Q1-Q30, R1-R30, S1-S15, S31-S45, T1-T30, and U1-U30.
4. C_L = 150 ±15 pF (including probe and parasitic capacitance) for test conditions P16-P30, R46-R60, S16-S30, and S46-S60.
5. Item P07 setup: Set voltage as follows:
   Test conditions P1-P60, S1-S60 = 5 V ±1%.
   Test conditions Q1-Q30, T1-T30 = 10 V ±1%.
   Test conditions R1-R30, U1-U30 = 15 V ±1%.

FIGURE 22. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit for tables XVI through XXI.

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NOTES:

1. Input pulse parameters:
   PW = 200 ±20 ns.
   PRP = 400 ±40 ns.
   $t_{TLH1} = t_{THL1}$ = 10 ±2 ns.
   PA = For test conditions:
   P1-P60, S1-S60 = 5.0 ±0.1 V; V_{REF} = 2.5 ±0.1 V.
   Q1-Q30, T1-T30 = 10.0 ±0.1 V; V_{REF} = 5.0 ±0.1 V.
   R1-R30, U1-U30 = 15.0 ±0.1 V; V_{REF} = 7.5 ±0.1 V.
2. Before making the measurements specified, verify that the output pulse high-level and the low-level voltage is as follows:

For test conditions:	High-level	Low-level
P1-P60, S1-S60	≥ 4.6 V	≤ 0.4 V
Q1-Q30, T1-T30	≥ 9.5 V	≤ 0.5 V
R1-R30, U1-U30	≥ 13.5 V	≤ 1.5 V

3. The lead length of electrical conductors between resistor R2 and input pins, items Y01-1 and Y01-2 to input and output pins shall be less than 1.00 inch (25.4 mm).
4. All voltages are referenced to ground.

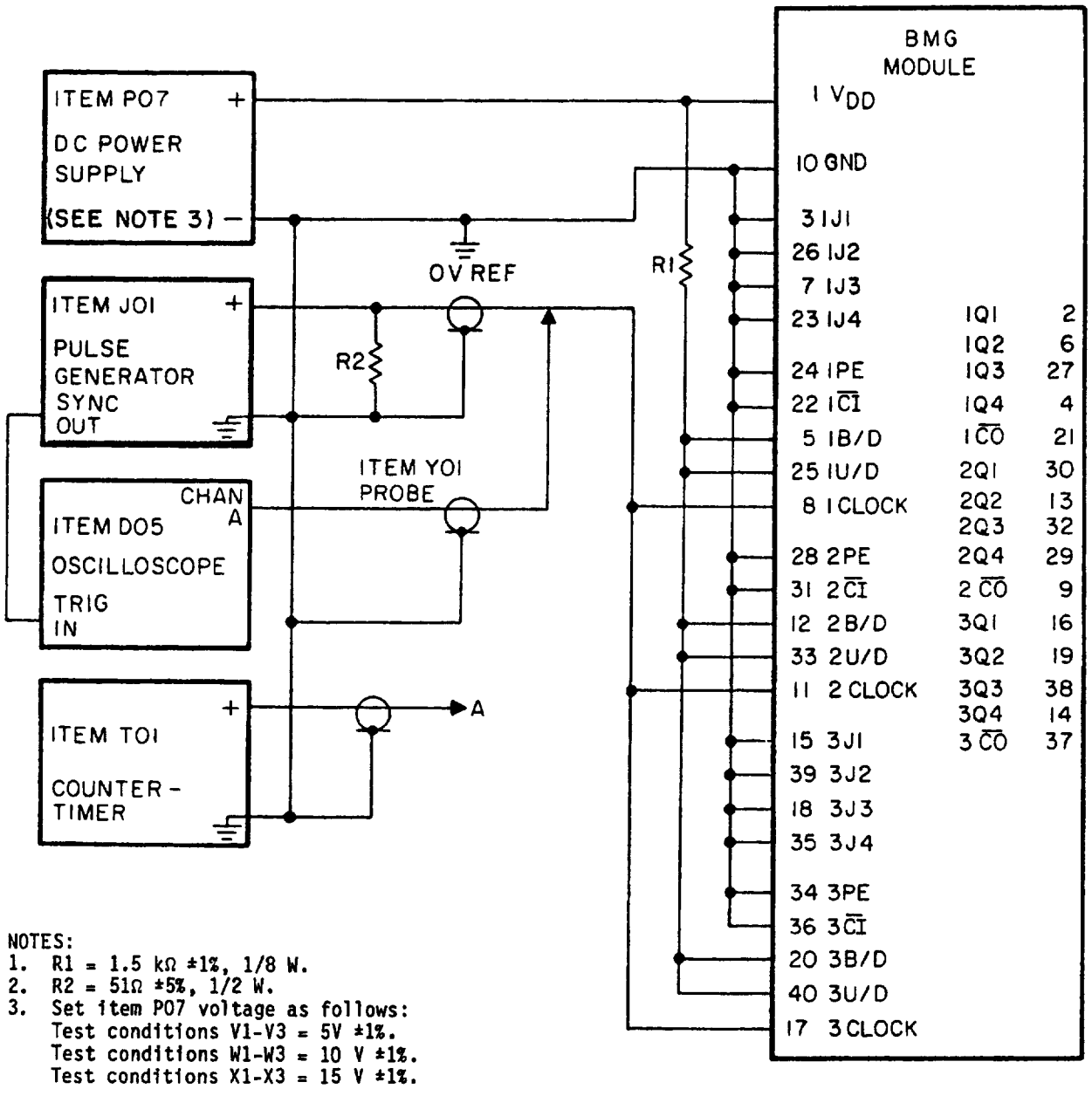
FIGURE 23. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for tables XVI through XXI.

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TABLE XXII. Maximum clock frequency and minimum clock pulse width test.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
V1	Maximum clock frequency and minimum clock pulse width	4	4,000	4,400	ns	24,25
V2	"	29	"	"	"	"
V3	"	14	"	"	"	"
W1	"	4	2,280	2,480	"	"
W2	"	29	"	"	"	"
W3	"	14	"	"	"	"
X1	"	4	1,760	1,920	"	"
X2	"	29	"	"	"	"
X3	"	14	"	"	"	"

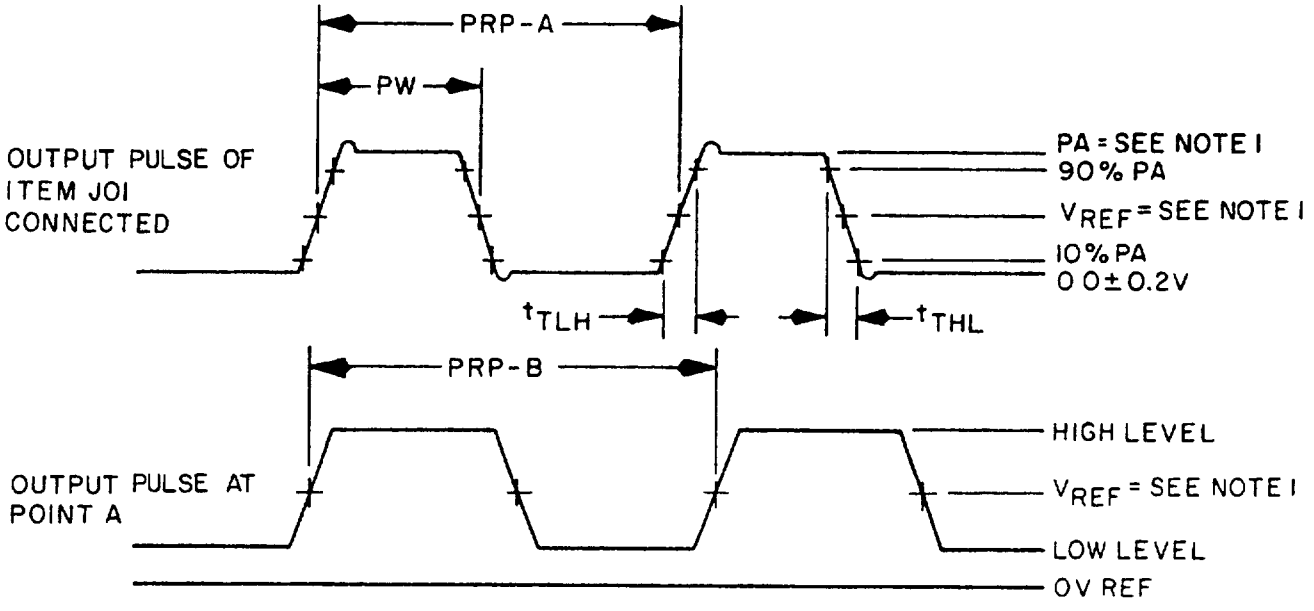
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NOTES:
1. R1 = 1.5 kΩ ±1%, 1/8 W.
2. R2 = 51Ω ±5%, 1/2 W.
3. Set item PO7 voltage as follows:
   Test conditions V1-V3 = 5V ±1%.
   Test conditions W1-W3 = 10 V ±1%.
   Test conditions X1-X3 = 15 V ±1%.

FIGURE 24. Maximum clock frequency and minimum clock pulse width test circuit for table XXII.

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NOTES:
1. Input pulse parameters for item J01:

Test condition V1-V3	Test condition W1-W3	Test condition X1-X3
PA = 5.0 ±0.1 V	PA = 10 ±0.1 V	PA = 15 ±0.1 V
V_{REF} = 2.5 ±0.1 V	V_{REF} = 5.0 ±0.1 V	V_{REF} = 7.5 ±0.1 V
PW = 340 +30, -0 ns	PW = 170 +15, -0 ns	PW = 140 +15, -0 ns
PRP-A = 500 +50, -0 ns	PRP-A = 285 +25, -0 ns	PRP-A = 220 +20, -0 ns
$t_{TLH} = t_{THL}$ = 10 ±2 ns	$t_{TLH} = t_{THL}$ = 10 ±2 ns	$t_{TLH} = t_{THL}$ = 10 ±2 ns

2. Before making the measurements specified, verify that the output pulse high-level and the low-level voltage is as follows:

For test conditions:	High-level	Low-level
V1-V3	≥ 4.6 V	≤ 0.4 V
W1-W3	≥ 9.5 V	≤ 0.5 V
X1-X3	≥13.5 V	≤ 1.5 V

3. The lead length of electrical conductors between resistor R2 and input pin and item Y01 to input pin shall be less than 1 inch (25.4 mm).
4. All voltages are referenced to ground.

FIGURE 25. Maximum clock frequency and minimum clock pulse width test circuit waveforms for table XXII.

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TABLE XXIII. Propagation delay time (t_{PLH} and t_{PHL}) test.

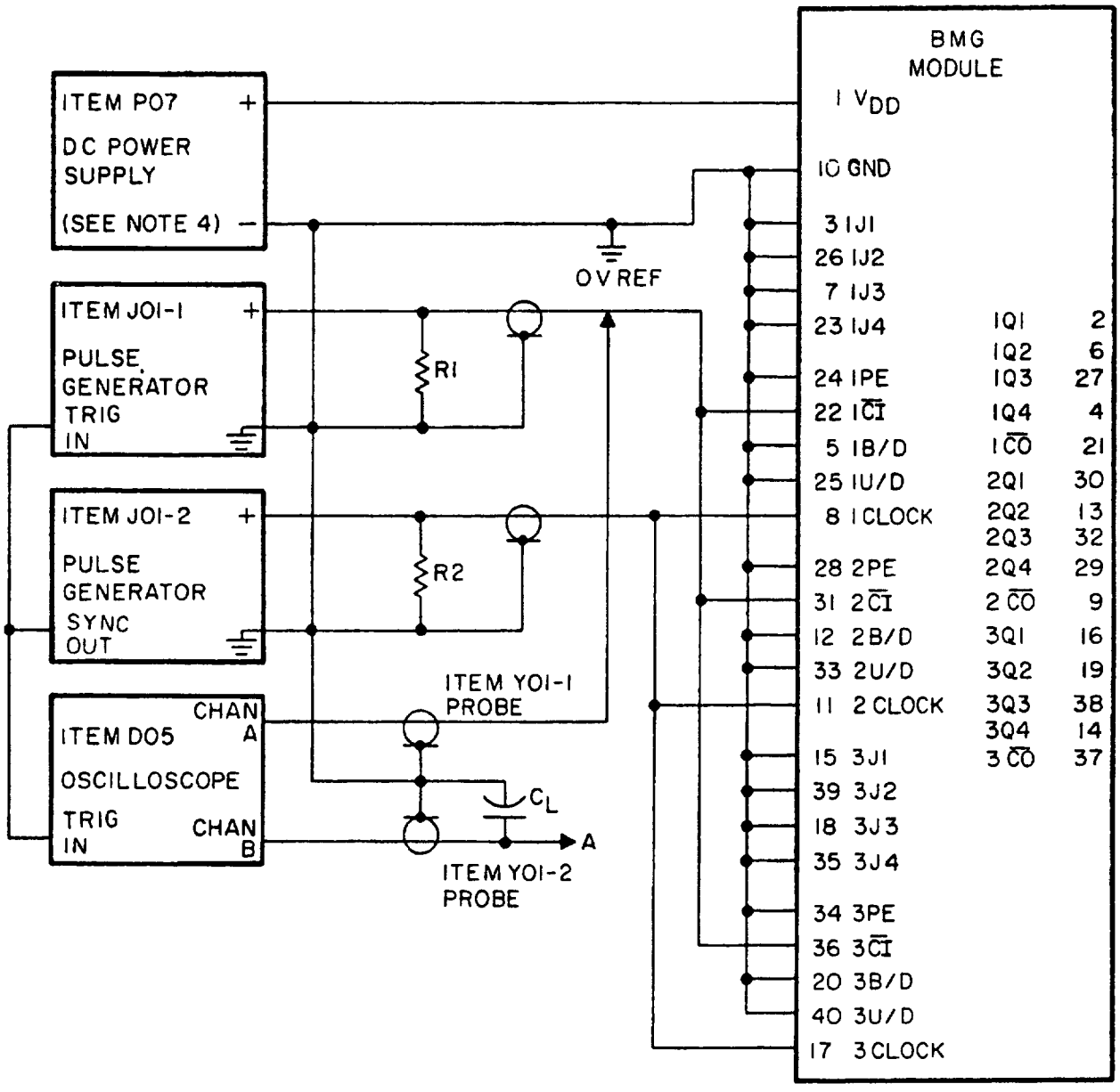
Test condition	Parameter		Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	Min	Max		
Y1	$1\overline{CO}$	Propagation delay time (t_{PLH}), $\overline{CI}$ to $\overline{CO}$, C_L = 50 pF	21	53	530	ns	26,27
Y2	$2\overline{CO}$	"	9	"	"	"	"
Y3	$3\overline{CO}$	"	37	"	"	"	"
Y4	$1\overline{CO}$	Propagation delay time (t_{PHL}), $\overline{CI}$ to $\overline{CO}$, C_L = 50 pF	21	"	"	"	"
Y5	$2\overline{CO}$	"	9	"	"	"	"
Y6	$3\overline{CO}$	"	37	"	"	"	"
Z1	$1\overline{CO}$	Propagation delay time (t_{PLH}), $\overline{CI}$ to $\overline{CO}$, C_L = 150 pF	21	106	1,060	"	"
Z2	$2\overline{CO}$	"	9	"	"	"	"
Z3	$3\overline{CO}$	"	37	"	"	"	"
Z4	$1\overline{CO}$	Propagation delay time (t_{PHL}), $\overline{CI}$ to $\overline{CO}$, C_L = 150 pF	21	"	"	"	"
Z5	$2\overline{CO}$	"	9	"	"	"	"
Z6	$3\overline{CO}$	"	37	"	"	"	"
AA1	$1\overline{CO}$	Propagation delay time (t_{PLH}), $\overline{CI}$ to $\overline{CO}$, C_L = 50 pF	21	22	220	"	"
AA2	$2\overline{CO}$	"	9	"	"	"	"
AA3	$3\overline{CO}$	"	37	"	"	"	"
AA4	$1\overline{CO}$	Propagation delay time (t_{PHL}), $\overline{CI}$ to $\overline{CO}$, C_L = 50 pF	21	"	"	"	"
AA5	$2\overline{CO}$	"	9	"	"	"	"
AA6	$3\overline{CO}$	"	37	"	"	"	"

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TABLE XXIII. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	Min	Max		
BB1	$1\overline{CO}$	Propagation delay time (t_{PLH}), $\overline{CI}$ to $\overline{CO}$, C_L = 50 pF	21	18	180	ns	26,27
BB2	$2\overline{CO}$	"	9	"	"	"	"
BB3	$3\overline{CO}$	"	37	"	"	"	"
BB4	$1\overline{CO}$	Propagation delay time (t_{PHL}), $\overline{CI}$ to $\overline{CO}$, C_L = 50 pF	21	"	"	"	"
BB5	$2\overline{CO}$	"	9	"	"	"	"
BB6	$3\overline{CO}$	"	37	"	"	"	"

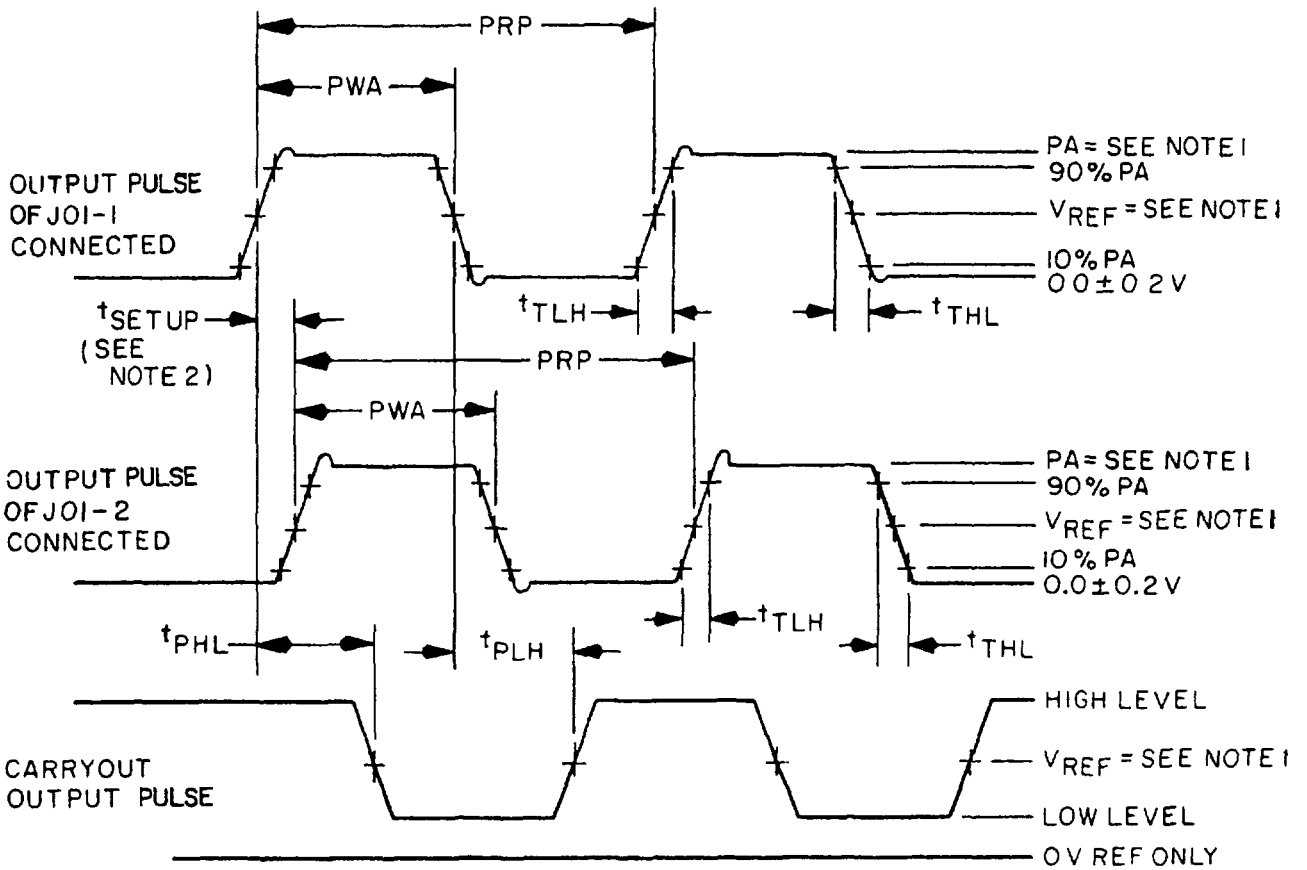
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NOTES:
1. R1 = R2 = 51Ω ±5%, 1/2 W.
2. C_L = 50 ±5 pF (including probe and parasitic capacitance) for test condition Y1-Y6, AA1-AA6, and BB1-BB6.
3. C_L = 150 ±15 pF (including probe and parasitic capacitance) for test conditions Z1-Z6.
4. Set item P07 voltage as follows:
   Test conditions Y1-Y6, Z1-Z6 = 5 V ±1%.
   Test conditions AA1-AA6 = 10 V ±1%.
   Test conditions BB1-BB6 = 15 V ±1%.

FIGURE 26. Propagation delay time (t_{PLH} and t_{PHL}) test circuit for table XXIII.

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NOTES:

1. Input pulse parameters for item J01-1 and J01-2:

For test conditions:	Y1-Y6, Z1-Z6	AA1-AA6	BB1-BB6
	PA = 5.0 ±0.1 V	PA = 10 ±0.1 V	PA = 15 ±0.1 V
	V_{REF} = 2.5 ±0.1 V	V_{REF} = 5.0 ±0.1 V	V_{REF} = 7.5 ±0.1 V.
	PWA = 500 ±50 ns	PWA = 500 ±50 ns	PWA = 500 ±50 ns.
	PRP = 1,000 ±100 ns	PRP = 1,000 ±100 ns	PRP = 1,000 ±100 ns
	$t_{TLH} = t_{THL}$ = 10 ±2 ns	$t_{TLH} = t_{THL}$ = 10 ±2 ns	$t_{TLH} = t_{THL}$ = 10 ±2 ns

2. t_{SETUP} for test conditions:

   Y1-Y6, Z1-Z6 = 360 +3, -0 ns
   AA1-AA6 = 140 +15, -0 ns
   BB1-BB6 = 110 +10, -0 ns

3. Before making the measurements specified, verify tht the output pulse high-level and the low-level voltage is as follows:

For test conditions:	High-level	Low-level
Y1-Y6, Z1-Z6	≥ 4.6 V	≤ 0.4 V
AA1-AA6	≥ 9.5 V	≤ 0.5 V
BB1-BB6	≥13.5 V	≤ 1.5 V

4. The lead length of electrical conductors between resistors R1 and R2 and input pin and items Y01-1 and Y01-2 to input pin shall be less than 1 inch (25.4 mm).
5. All voltages are referenced to ground.

FIGURE 27. Propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for table XXIII.

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TABLE XXIV. Propagation delay time (t_{PHL} and t_{PLH}) test.

Test condition	Parameter		Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	Min	Max		
CC1	1Q1	Propagation delay time (t_{PHL}), PE to Q, C_L = 50 pF	2	72	720	ns	28,29
CC2	1Q2	"	6	"	"	"	"
CC3	1Q3	"	27	"	"	"	"
CC4	1Q4	"	4	"	"	"	"
CC5	2Q1	"	30	"	"	"	"
CC6	2Q2	"	13	"	"	"	"
CC7	2Q3	"	32	"	"	"	"
CC8	2Q4	"	29	"	"	"	"
CC9	3Q1	"	16	"	"	"	"
CC10	3Q2	"	19	"	"	"	"
CC11	3Q3	"	38	"	"	"	"
CC12	3Q4	"	14	"	"	"	"
CC13	1$\overline{CO}$	Propagation delay time (t_{PLH}), PE to $\overline{CO}$, C_L = 50 pF	21	82	820	"	"
CC14	2$\overline{CO}$	"	9	"	"	"	"
CC15	3$\overline{CO}$	"	37	"	"	"	"
DD1	1Q1	Propagation delay time (t_{PHL}), PE to Q, C_L = 150 pF	2	92	920	"	"
DD2	1Q2	"	6	"	"	"	"
DD3	1Q3	"	27	"	"	"	"
DD4	1Q4	"	4	"	"	"	"
DD5	2Q1	"	30	"	"	"	"

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TABLE XXIV. Propagation delay time (t_{PHL} and t_{PLH}) test - Continued.

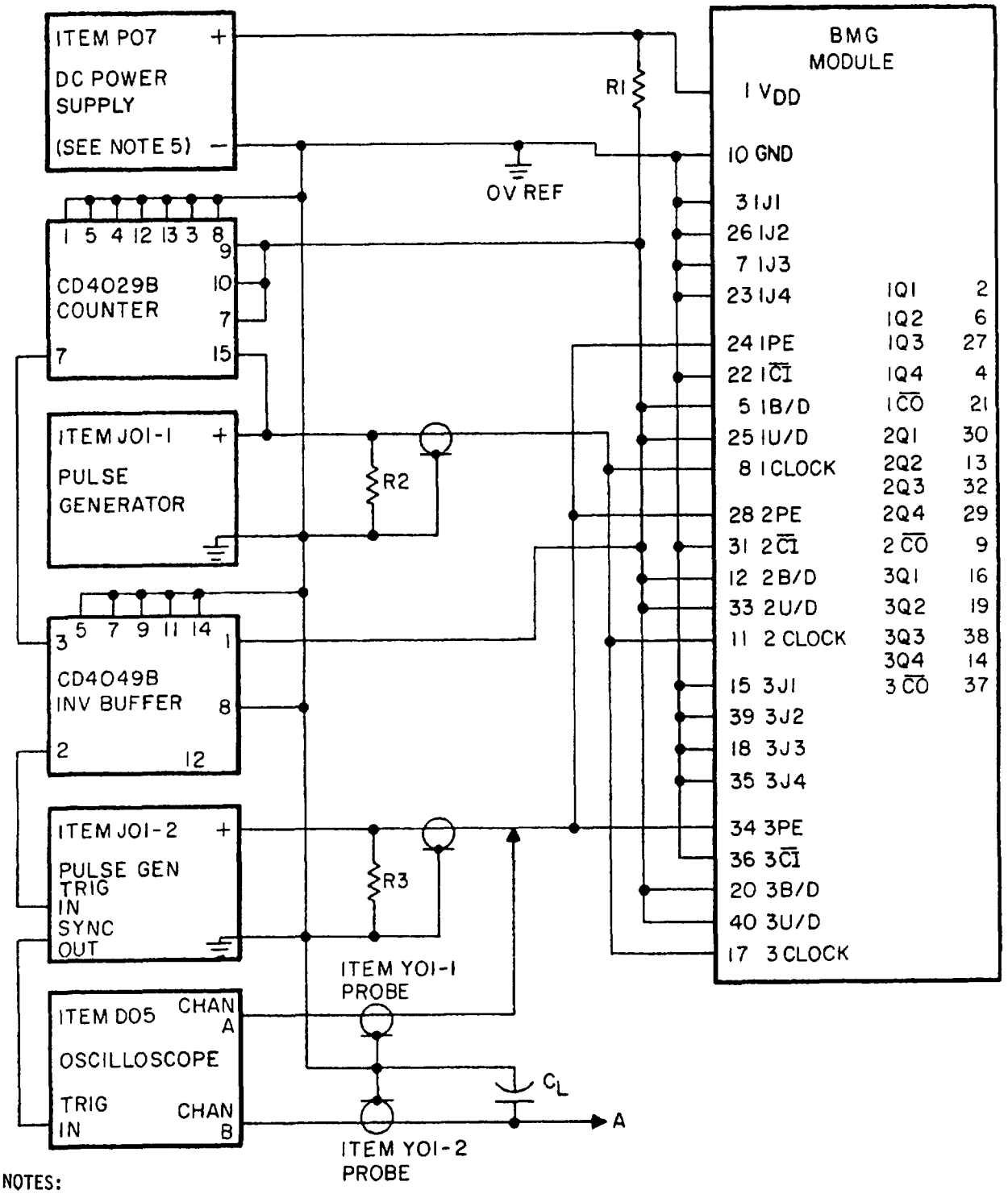
Test condition	Parameter		Connect A to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
DD6	2Q2	Propagation delay time (t_{PHL}), PE to Q, C_L = 150 pF	13	92	920	ns	28,29
DD7	2Q3	"	32	"	"	"	"
DD8	2Q4	"	29	"	"	"	"
DD9	3Q1	"	16	"	"	"	"
DD10	3Q2	"	19	"	"	"	"
DD11	3Q3	"	38	"	"	"	"
DD12	3Q4	"	14	"	"	"	"
DD13	1$\overline{CO}$	Propagation delay time (t_{PLH}), PE to $\overline{CO}$, C_L = 150 pF	21	102	1,020	"	"
DD14	2$\overline{CO}$	"	9	"	"	"	"
DD15	3$\overline{CO}$	"	37	"	"	"	"
EE1	1Q1	Propagation delay time (t_{PHL}), PE to Q, C_L = 50 pF	2	28	280	"	"
EE2	1Q2	"	6	"	"	"	"
EE3	1Q3	"	27	"	"	"	"
EE4	1Q4	"	4	"	"	"	"
EE5	2Q1	"	30	"	"	"	"
EE6	2Q2	"	13	"	"	"	"
EE7	2Q3	"	32	"	"	"	"
EE8	2Q4	"	29	"	"	"	"
EE9	3Q1	"	16	"	"	"	"
EE10	3Q2	"	19	"	"	"	"
EE11	3Q3	"	38	"	"	"	"
EE12	3Q4	"	14	"	"	"	"

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TABLE XXIV. Propagation delay time (t_{PHL} and t_{PLH}) test - Continued.

Test condition	Parameter		Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	Min	Max		
EE13	$1\overline{CO}$	Propagation delay time (t_{PLH}), PE to $\overline{CO}$, C_L = 50 pF	21	33	330	ns	28,29
EE14	$2\overline{CO}$	"	9	"	"	"	"
EE15	$3\overline{CO}$	"	37	"	"	"	"
FF1	1Q1	Propagation delay time (t_{PHL}), PE to Q, C_L = 50 pF	2	22	220	"	"
FF2	1Q2	"	6	"	"	"	"
FF3	1Q3	"	27	"	"	"	"
FF4	1Q4	"	4	"	"	"	"
FF5	2Q1	"	30	"	"	"	"
FF6	2Q2	"	13	"	"	"	"
FF7	2Q3	"	32	"	"	"	"
FF8	2Q4	"	29	"	"	"	"
FF9	3Q1	"	16	"	"	"	"
FF10	3Q2	"	19	"	"	"	"
FF11	3Q3	"	38	"	"	"	"
FF12	3Q4	"	14	"	"	"	"
FF13	$1\overline{CO}$	Propagation delay time (t_{PLH}), PE to $\overline{CO}$, C_L = 50 pF	21	26	260	"	"
FF14	$2\overline{CO}$	"	9	"	"	"	"
FF15	$3\overline{CO}$	"	37	"	"	"	"

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NOTES:
1. R1 = 1.5 kΩ ±1%, 1/8 W.
2. R2 = R3 = 51Ω ±5%, 1/2 W.
3. C_L = 50 ±5 pF (including probe and parasitic capacitance) for test conditions CC1-CC15, EE1-EE15, and FF1-FF15.
4. C_L = 150 ±15 pF (including probe and parasitic capacitance) for test conditions DD1-DD15.
5. Set item P07 voltage as follows:
   Test conditions CC1-CC15, DD1-DD15 = 5 V ±1%.
   Test conditions EE1-EE15 = 10 V ±1%.
   Test conditions FF1-FF15 = 15 V ±1%.

FIGURE 28. Propagation delay time (t_{pHl} and t_{plH}) test circuit for table XXIV.

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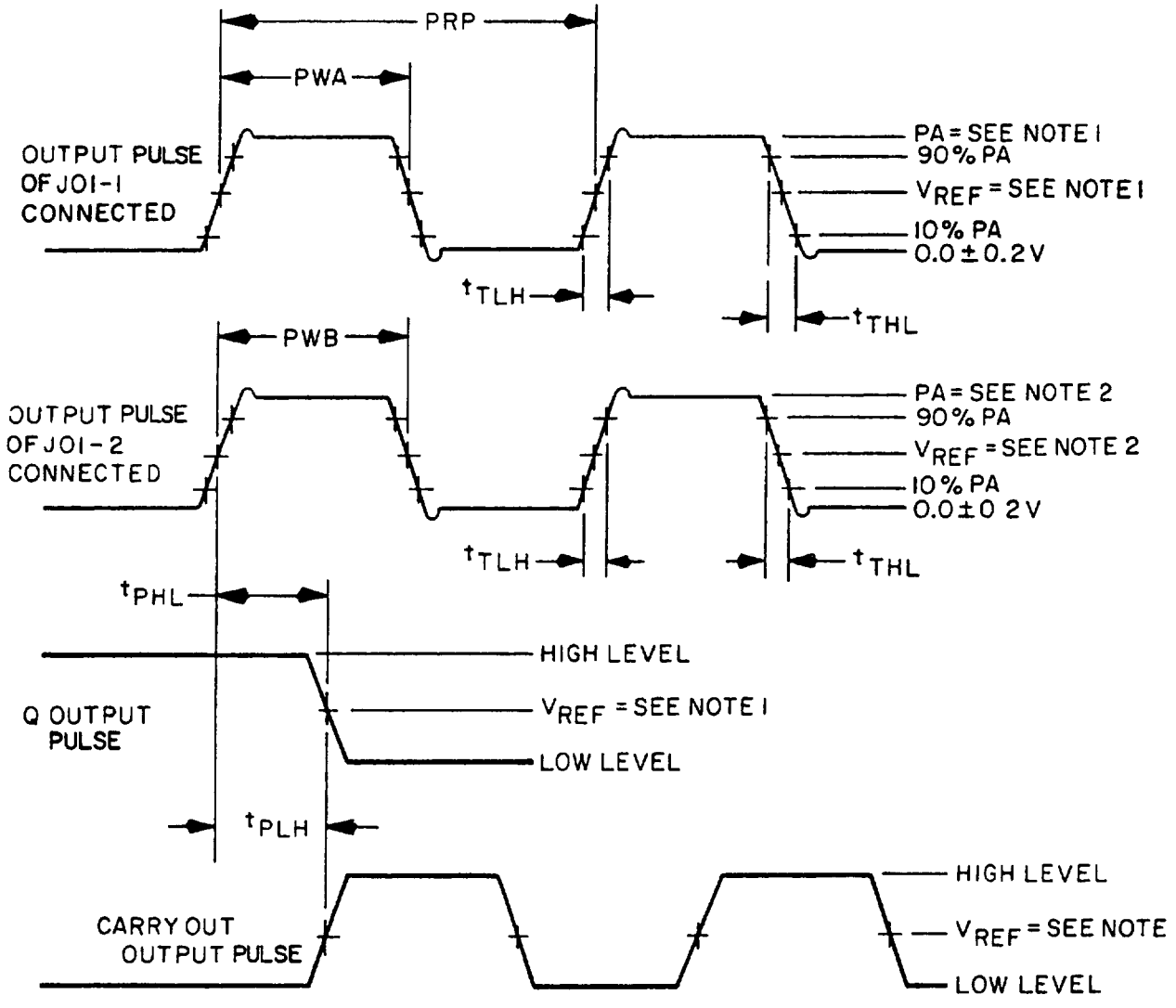


FIGURE 29. Propagation delay time (t_{PHL} and t_{PLH}) test circuit waveforms for table XXIV.

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NOTES:

1. Input pulse parameters for item J01-1:

Test conditions CC1-CC15, DD1-DD15	Test conditions EE1-EE15	Test conditions FF1-FF15
PA = 5.0 ±0.1 V	PA = 10.0 ±0.1 V	PA = 15.0 ±0.1 V
V_{REF} = 2.5 ±0.1 V	V_{REF} = 5.0 ±0.1 V	V_{REF} = 7.5 ±0.1 V
PWA = 600 ±60 ns	PWA = 600 ±60 ns	PWA = 600 ±60 ns
PRP = 1,200 ±120 ns	PRP = 1,200 ±120 ns	PRP = 1,200 ±120 ns
$t_{TLH} = t_{THL}$ = 10 ±2 ns	$t_{TLH} = t_{THL}$ = 10 ±2 ns	$t_{TLH} = t_{THL}$ = 10 ±2 ns

2. Input pulse parameters for item J01-2:

Test conditions CC1-CC15, DD1-DD15	Test conditions EE1-EE15	Test conditions FF1-FF15
PA = 5.0 ±0.1 V	PA = 10.0 ±0.1 V	PA = 15.0 ±0.1 V
V_{REF} = 2.5 ±0.1 V	V_{REF} = 5.0 ±0.1 V	V_{REF} = 7.5 ±0.1 V
PWB = 340 +30, -0 ns	PWB = 170 +15, -0 ns	PWB = 140 +15, -0 ns
$t_{TLH} = t_{THL}$ = 10 ±2 ns	$t_{TLH} = t_{THL}$ = 10 ±2 ns	$t_{TLH} = t_{THL}$ = 10 ±2 ns

3. Before making the measurements specified, verify that the output pulse high-level and the low-level voltage is as follows:

For test conditions	High-level	low-level
CC1-CC15, DD1-DD15	$\geq$ 4.6 V	$\leq$ 0.4 V
EE1-EE15	$\geq$ 9.5 V	$\leq$ 0.5 V
FF1-FF15	$\geq$ 13.5 V	$\leq$ 1.5 V

4. The lead length of electrical conductors between resistors R2 and R3 and input pins and items Y01-1 and Y01-2 to input pin shall be less than 1 inch (25.4 mm).
5. All voltages are referenced to ground.

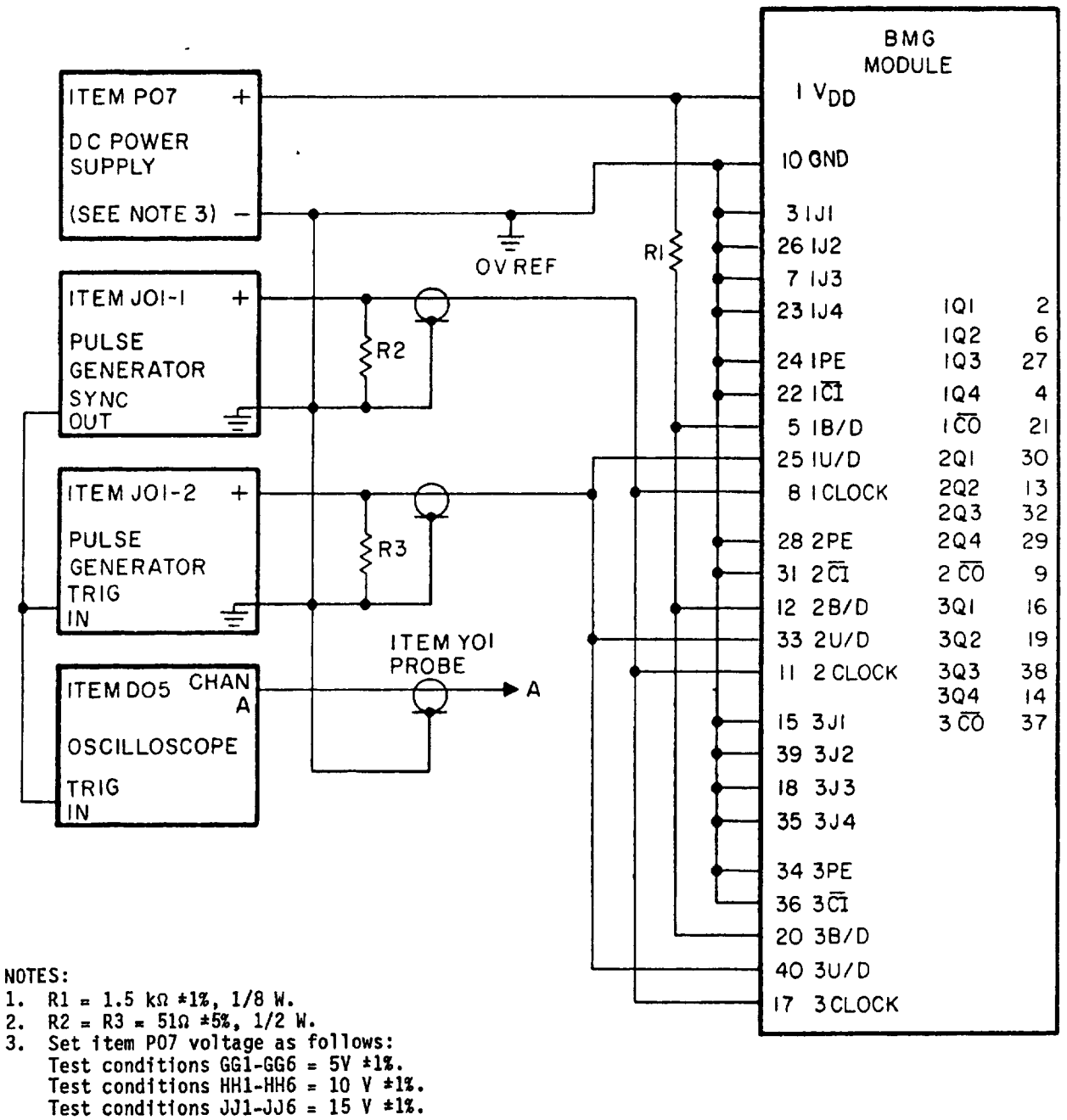
FIGURE 29. Propagation delay time (t_{PHL} and t_{PLH}) test circuit waveforms for table XXIV - Continued.

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TABLE XXV. Pulse width setup time (countup and countdown) test.

Test condition	Parameter		Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	Min	Max		
GG1	1Q2	Pulse width setup time, countup	6	3,600	4,400	ns	30,31
GG2	2Q2	"	13	"	"	"	"
GG3	3Q2	"	19	"	"	"	"
GG4	1Q2	Pulse width setup time, countdown	6	"	"	"	"
GG5	2Q2	"	13	"	"	"	"
GG6	3Q2	"	19	"	"	"	"
HH1	1Q2	Pulse width setup time, countup	6	1,800	2,200	"	"
HH2	2Q2	"	13	"	"	"	"
HH3	3Q2	"	19	"	"	"	"
HH4	1Q2	Pulse width setup time, countdown	6	"	"	"	"
HH5	2Q2	"	13	"	"	"	"
HH6	3Q2	"	19	"	"	"	"
JJ1	1Q2	Pulse width setup time, countup	6	"	"	"	"
JJ2	2Q2	"	13	"	"	"	"
JJ3	3Q2	"	19	"	"	"	"
JJ4	1Q2	Pulse width setup time, countdown	6	"	"	"	"
JJ5	2Q2	"	13	"	"	"	"
JJ6	3Q2	"	19	"	"	"	"

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NOTES:
1. R1 = 1.5 kΩ ±1%, 1/8 W.
2. R2 = R3 = 51Ω ±5%, 1/2 W.
3. Set item P07 voltage as follows:
   Test conditions GG1-GG6 = 5V ±1%.
   Test conditions HH1-HH6 = 10 V ±1%.
   Test conditions JJ1-JJ6 = 15 V ±1%.

FIGURE 30. Pulse width setup time (countup and countdown) test circuit for table XXV.

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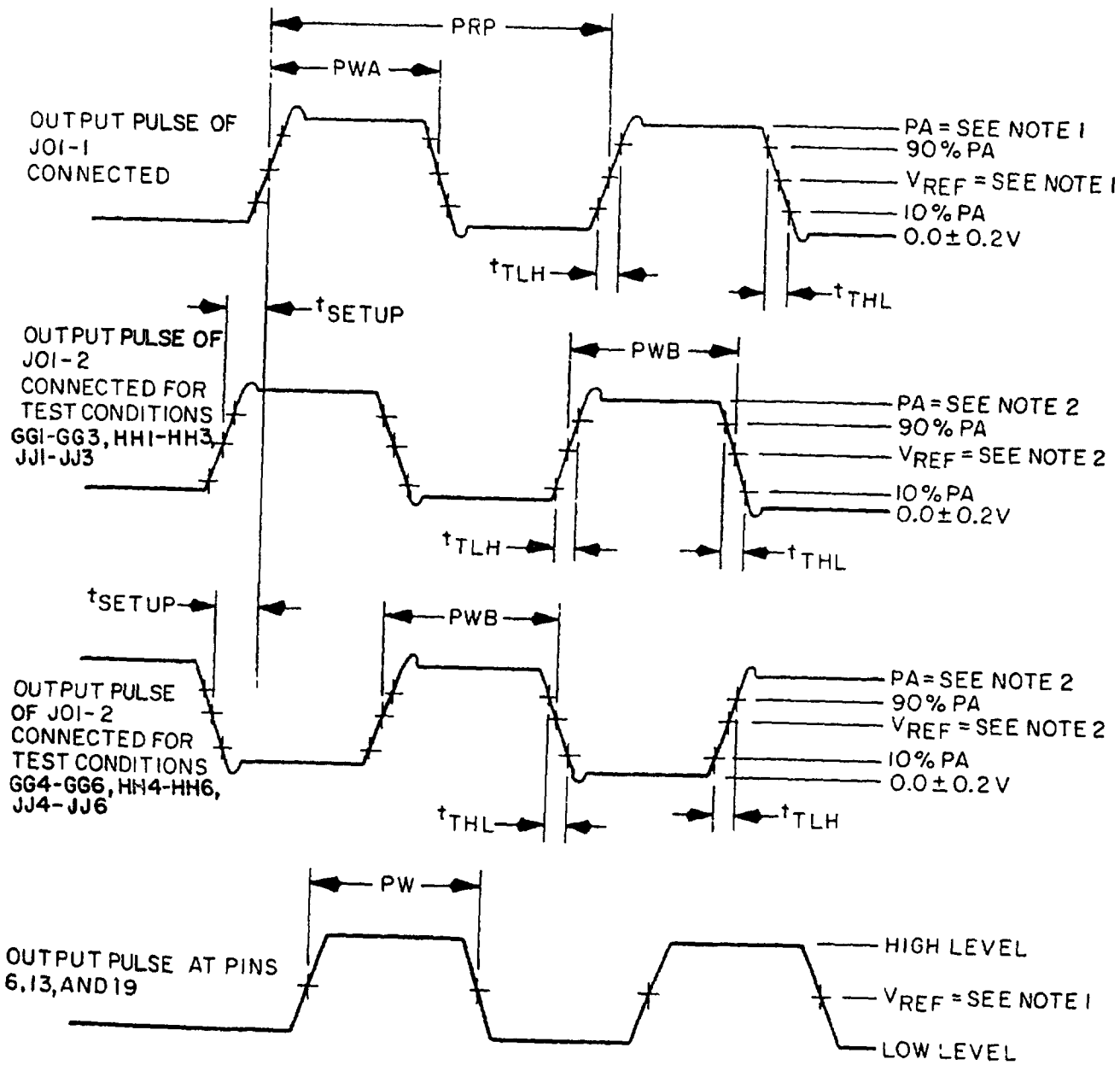


FIGURE 31. Pulse width setup time (countup and countdown) test circuit waveforms for table XXV.

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NOTES:
1. Input pulse parameters for item J01-1:

Test conditions GG1-GG6	Test conditions HH1-HH6	Test conditions JJ1-JJ6
PA = 5.0 ±0.1 V	PA = 10.0 ±0.1 V	PA = 15.0 ±0.1 V
V_{REF} = 2.5 ±0.1 V	V_{REF} = 5.0 ±0.1 V	V_{REF} = 7.5 ±0.1 V
PWA = 1,000 ±100 ns	PWA = 500 ±50 ns	PWA = 500 ±50 ns
PRP = 2,000 ±200 ns	PRP = 1,000 ±100 ns	PRP = 1,000 ±100 ns
$t_{TLH} = t_{THL}$ = 10 ±2 ns	$t_{TLH} = t_{THL}$ = 10 ±2 ns	$t_{TLH} = t_{THL}$ = 10 ±2 ns

2. Input pulse parameters for item J01-2:

Test conditions GG1-GG6	Test conditions HH1-HH6	Test conditions JJ1-JJ6
PA = 5.0 ±0.1 V	PA = 10.0 ±0.1 V	PA = 15.0 ±0.1 V
V_{REF} = 2.5 ±0.1 V	V_{REF} = 5.0 ±0.1 V	V_{REF} = 7.5 ±0.1 V
PWB = 500 ±50 ns	PWB = 500 ±50 ns	PWB = 500 ±50 ns
$t_{TLH} = t_{THL}$ = 10 ±2 ns	$t_{TLH} = t_{THL}$ = 10 ±2 ns	$t_{TLH} = t_{THL}$ = 10 ±2 ns
t_{SETUP} = 650 +65, - 0 ns	t_{SETUP} = 230 +20, - 0 ns	t_{SETUP} = 170 +15, -0 ns

3. Before making the measurements specified, verify that the output pulse high-level and the low-level is as follows:

For test conditions	High-level	Low-level
GG1-GG6	≥ 4.6 V	≤ 0.4 V
HH1-HH6	≥ 9.5 V	≤ 0.5 V
JJ1-JJ6	≥13.5 V	≤ 1.5 V

4. The lead length of electrical conductors between resistors R2 and R3 and input pins and item YO1 and output pin shall be less than 1 inch (25.4 mm).

5. All voltages are referenced to ground.

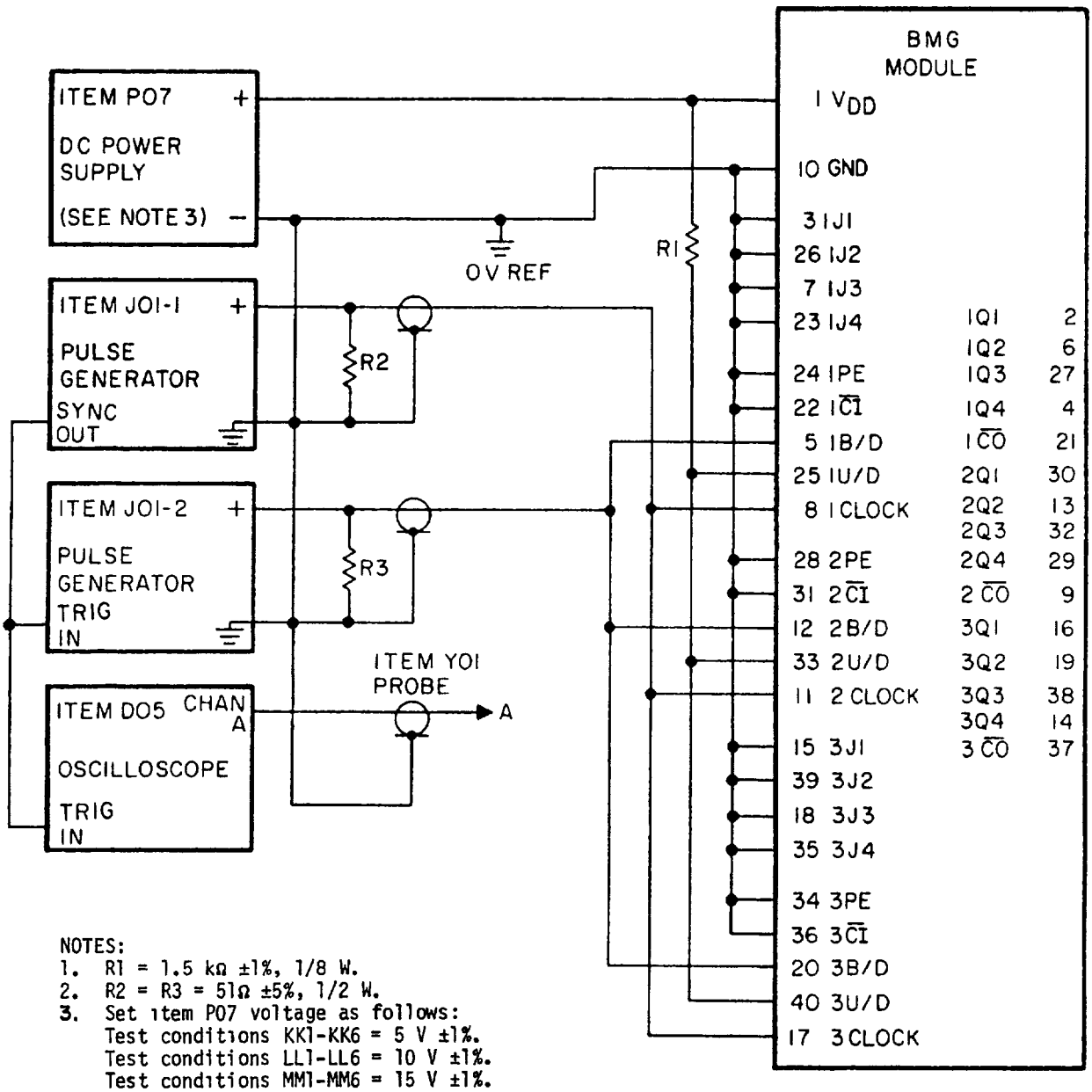
FIGURE 31. Pulse width setup time (countup and countdown) test circuit waveforms for table XXV - Continued.

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TABLE XXVI. Pulse width setup time (binary and decade) test.

Test condition	Parameter		Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	Min	Max		
KK1	1Q2	Pulse width setup time, binary	6	3,600	4,400	ns	32,33
KK2	2Q2	"	13	"	"	"	"
KK3	3Q2	"	19	"	"	"	"
KK4	1Q2	Pulse width setup time, decade	6	"	"	"	"
KK5	2Q2	"	13	"	"	"	"
KK6	3Q2	"	19	"	"	"	"
LL1	1Q2	Pulse width setup time, binary	6	1,800	2,200	"	"
LL2	2Q2	"	13	"	"	"	"
LL3	3Q2	"	19	"	"	"	"
LL4	1Q2	Pulse width setup time, decade	6	"	"	"	"
LL5	2Q2	"	13	"	"	"	"
LL6	3Q2	"	19	"	"	"	"
MM1	1Q2	Pulse width setup time, binary	6	"	"	"	"
MM2	2Q2	"	13	"	"	"	"
MM3	3Q2	"	19	"	"	"	"
MM4	1Q2	Pulse width setup time, decade	6	"	"	"	"
MM5	2Q2	"	13	"	"	"	"
MM6	3Q2	"	19	"	"	"	"

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NOTES:
1. R1 = 1.5 kΩ ±1%, 1/8 W.
2. R2 = R3 = 51Ω ±5%, 1/2 W.
3. Set item P07 voltage as follows:
   Test conditions KK1-KK6 = 5 V ±1%.
   Test conditions LL1-LL6 = 10 V ±1%.
   Test conditions MM1-MM6 = 15 V ±1%.

FIGURE 32. Pulse width setup time (binary and decade) test circuit for table XXVI.

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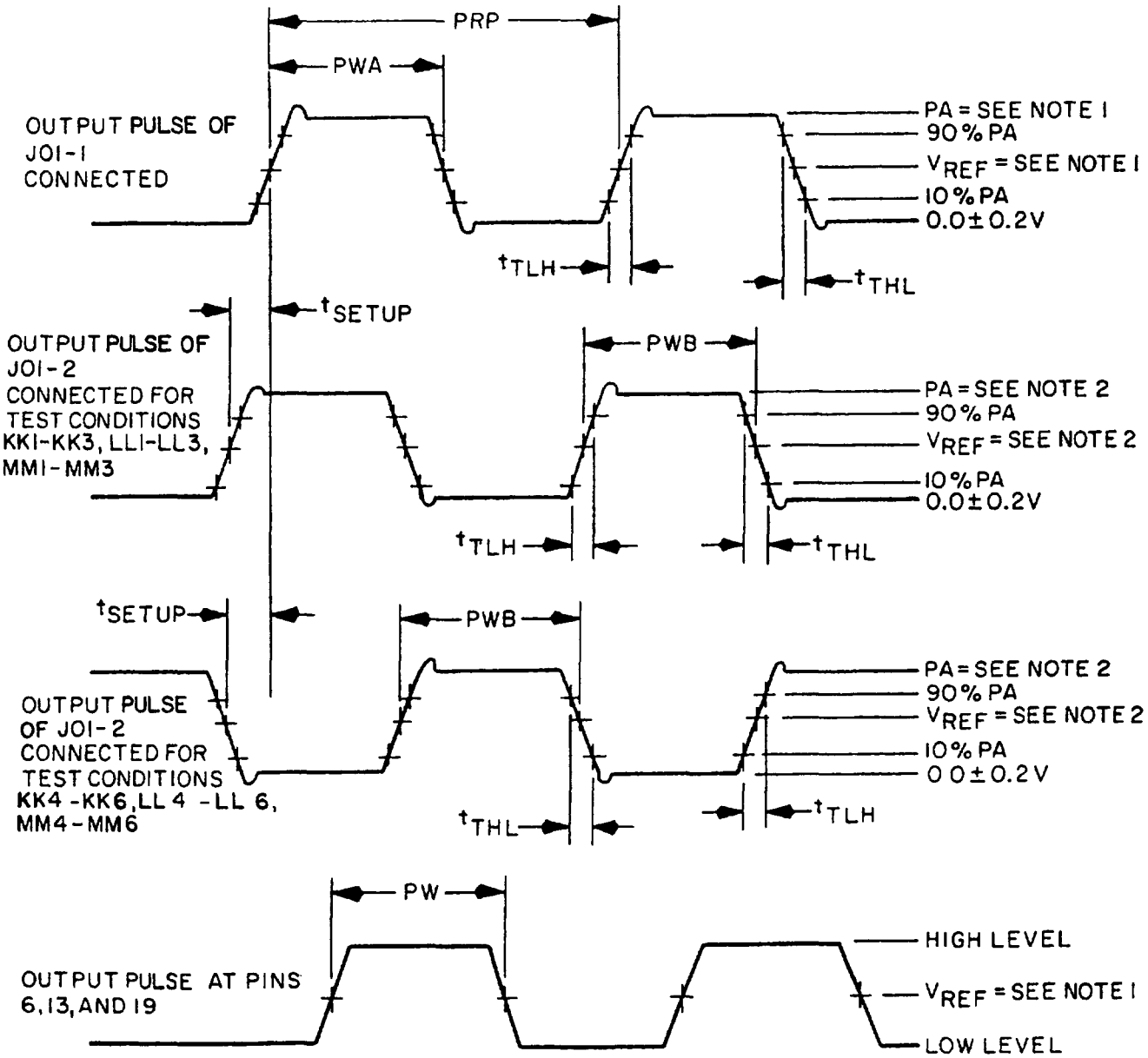


FIGURE 33. Pulse width setup time (binary and decade) test circuit waveforms for table XXVI.

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NOTES:

1. Input pulse parameters for item J01-1:

Test conditions KK1-KK6	Test conditions LL1-LL6	Test conditions MM1-MM6
PA = 5.0 ±0.1 V	PA = 10.0 ±0.1 V	PA = 15.0 ±0.1 V
V_{REF} = 2.5 ±0.1 V	V_{REF} = 5.0 ±0.1 V	V_{REF} = 7.5 ±0.1 V
PWA = 1,000 ±100 ns	PWA = 500 ±50 ns	PWA = 500 ±50 ns
PRP = 2,000 ±200 ns	PRP = 1,000 ±100 ns	PRP = 1,000 ±100 ns
$t_{TLH} = t_{THL}$ = 10 ±2 ns	$t_{TLH} = t_{THL}$ = 10 ±2 ns	$t_{TLH} = t_{THL}$ = 10 ±2 ns

2. Input pulse parameters for item J01-2:

Test conditions KK1-KK6	Test conditions LL1-LL6	Test conditions MM1-MM6
PA = 5.0 ±0.1 V	PA = 10.0 ±0.1 V	PA = 15.0 ±0.1 V
V_{REF} = 2.5 ±0.1 V	V_{REF} = 5.0 ±0.1 V	V_{REF} = 7.5 ±0.1 V
PWB = 500 ±50 ns	PWB = 500 ±50 ns	PWB = 500 ±50 ns
$t_{TLH} = t_{THL}$ = 10 ±2 ns	$t_{TLH} = t_{THL}$ = 10 ±2 sn	$t_{TLH} = t_{THL}$ = 10 ±2 ns
t_{SETUP} = 650 +65 , - 0 ns	t_{SETUP} = 230 + 20 , -0 ns	t_{SETUP} = 170 +15 , -0 ns

3. Before making the measurements specified, verify that the output pulse high-level and the low-level is as follows:

For test conditions	High-level	Low-level
KK1-KK6	≥ 4.6 V	≤ 0.4 V
LL1-LL6	≥ 9.5 V	≤ 0.5 V
MM1-MM6	≥13.5 V	≤ 1.5 V

4. The lead length of electrical conductors between resistors R2 and R3 and input pins and item Y01 and output pin shall be less than 1 inch (25.4 mm).

5. All voltages are referenced to ground.

FIGURE 33. Pulse width setup time (binary and decade) test circuit waveforms for table XXVI - Continued.

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TABLE XXVII. Preset enable setup and removal time test.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
NN1	Preset enable setup and removal time	2	900	1,100	ns	34,35
NN2	"	30	"	"	"	"
NN3	"	16	"	"	"	"
PP1	"	2	"	"	"	"
PP2	"	30	"	"	"	"
PP3	"	16	"	"	"	"
QQ1	"	2	"	"	"	"
QQ2	"	30	"	"	"	"
QQ3	"	16	"	"	"	"

TABLE XXVIII. Power supply filter capacitance test.

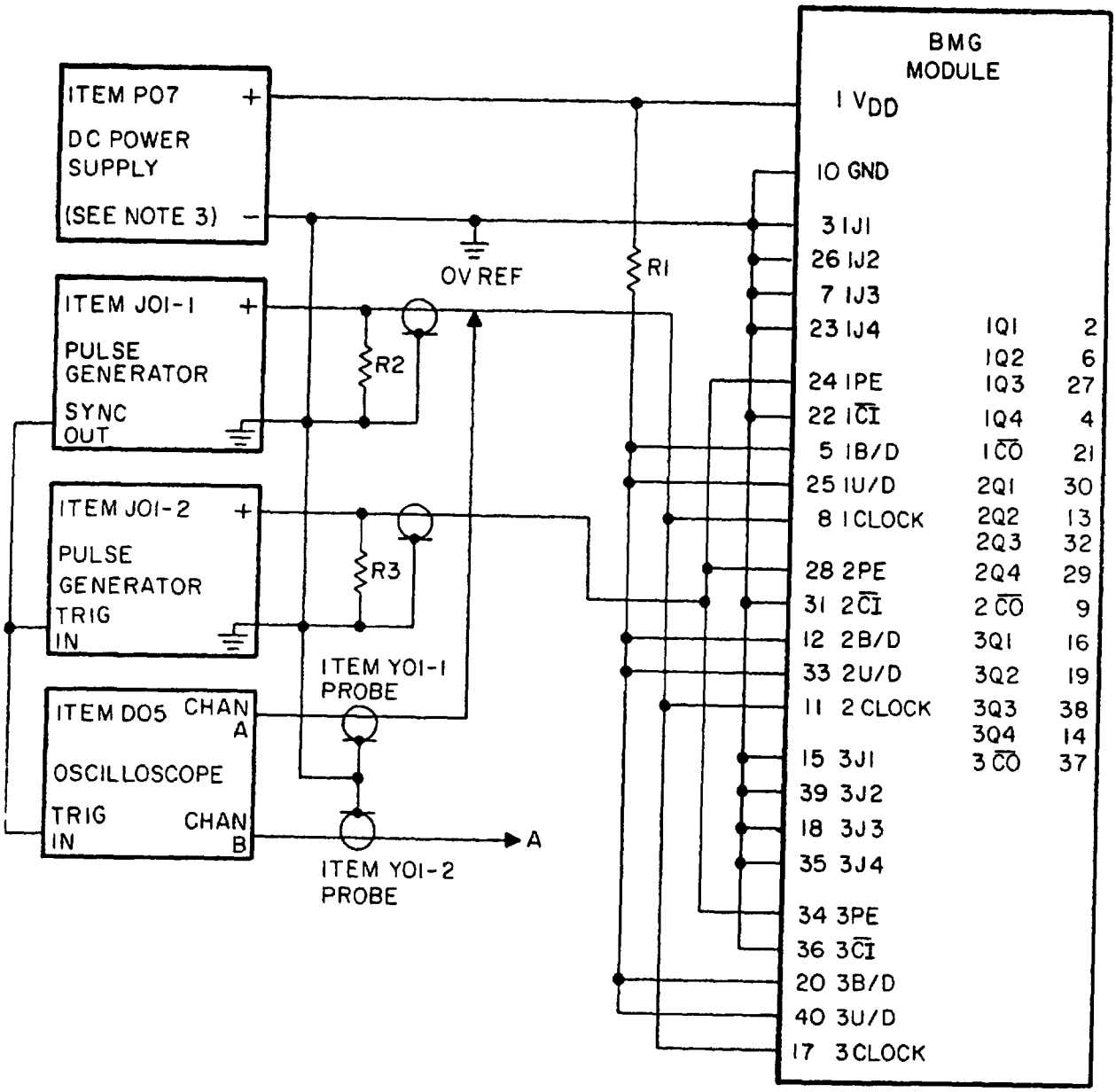
Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
RR	Power supply filter capacitance	1	Open	0.007	0.013	µF	36

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TABLE XXIX. Input capacitance test.

Test condition	Parameter		Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
SS1	1J1	Input capacitance	Open	3	2	15	pF	36
SS2	1J2	"	"	26	"	"	"	"
SS3	1J3	"	"	7	"	"	"	"
SS4	1J4	"	"	23	"	"	"	"
SS5	1PE	"	"	24	"	"	"	"
SS6	$1\overline{CI}$	"	"	22	"	"	"	"
SS7	1B/D	"	"	5	"	"	"	"
SS8	1U/D	"	"	25	"	"	"	"
SS9	1 Clock	"	"	8	"	"	"	"
SS10	2 PE	"	"	28	"	"	"	"
SS11	$2\overline{CI}$	"	"	31	"	"	"	"
SS12	2B/D	"	"	12	"	"	"	"
SS13	2U/D	"	"	33	"	"	"	"
SS14	2 Clock	"	"	11	"	"	"	"
SS15	3J1	"	"	15	"	"	"	"
SS16	3J2	"	"	39	"	"	"	"
SS17	3J3	"	"	18	"	"	"	"
SS18	3J4	"	"	35	"	"	"	"
SS19	PE3	"	"	34	"	"	"	"
SS20	$3\overline{CI}$	"	"	36	"	"	"	"
SS21	3B/D	"	"	20	"	"	"	"
SS22	3U/D	"	"	40	"	"	"	"
SS23	3 Clock	"	"	17	"	"	"	"

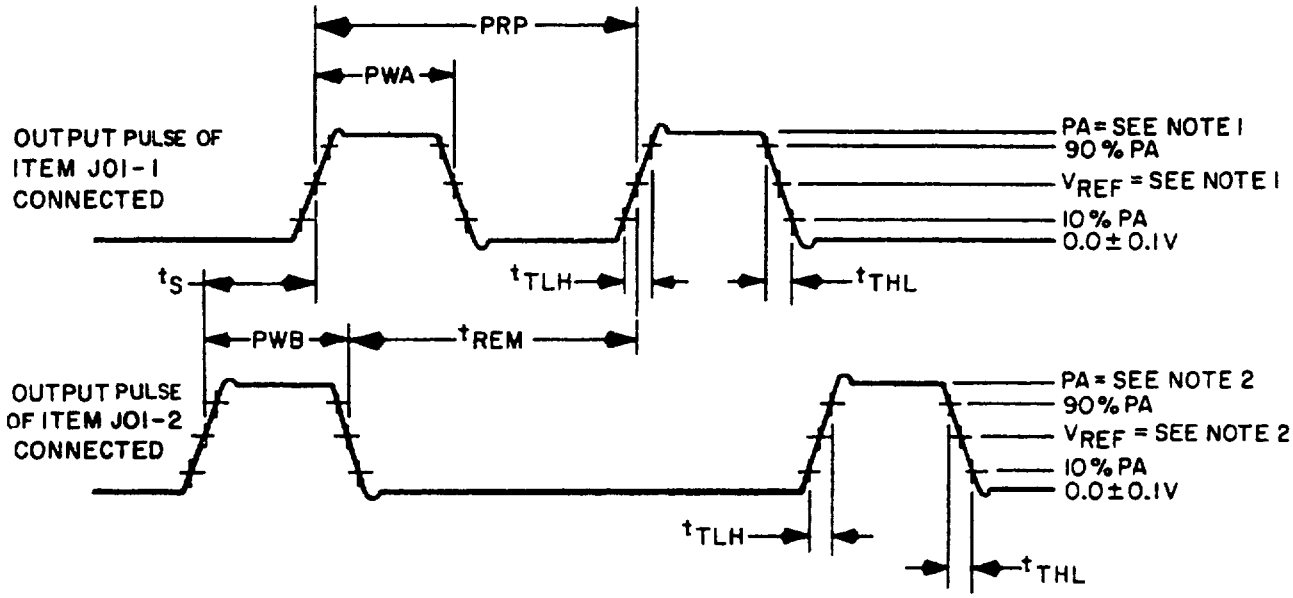
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:19Z
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NOTES:
1. R1 = 1.5 kΩ ±1%, 1/4 W.
2. R2 = R3 = 51Ω ±5%, 1/2 W.
3. Set item P07 voltages as follows:
   Test conditions NN1-NN3 = 5 V ±1%.
   Test conditions PP1-PP3 = 10 V ±1%.
   Test conditions QQ1-QQ3 = 15 V ±1%.

FIGURE 34. Preset enable setup and removal time test circuit for table XXVII.

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NOTES:

1. Input pulse parameters for item J01-1:

Test conditions NN1-NN3	Test conditions PP1-PP3	Test conditions QQ1-QQ3
PA = 5.0 ±0.1 V	PA = 10.0 ±0.1 V	PA = 15.0 ±0.1 V
V_{REF} = 2.5 ±0.1 V	V_{REF} = 5.0 ±0.1 V	V_{REF} = 7.5 ±0.1 V
PWA = 500 ±50 ns	PWA = 500 ±50 ns	PWA = 500 ±50 ns
PRP = 1,000 ±100 ns	PRP = 1,000 ±100 ns	PRP = 1,000 ±100 ns
$t_{TLH} = t_{THL}$ = 10 ±2 ns	$t_{TLH} = t_{THL}$ = 10 ±2 ns	$t_{TLH} = t_{THL}$ = 10 ±2 ns
Double pulse mode	Double pulse mode	Double pulse mode

2. Input pulse parameters for item J01-2:

Test conditions NN1-NN3	Test conditions PP1-PP3	Test conditions QQ1-QQ3
PA = 5.0 ±0.1 V	PA = 10.0 ±0.1 V	PA = 15.0 ±0.1 V
V_{REF} = 2.5 ±0.1 V	V_{REF} = 5.0 ±0.1 V	V_{REF} = 7.5 ±0.1 V
PWB = 500 ±50 ns	PWB = 500 ±50 ns	PWB = 500 ±50 ns
t_S = 325 +30, -0 ns	t_S = 115 +10, -0 ns	t_S = 100 +10, -0 ns
t_{REM} = 650 +65, -0 ns	t_{REM} = 220 +20, -0 ns	t_{REM} = 180 +20, -0 ns
$t_{TLH} = t_{THL}$ = 10 ±2 ns	$t_{TLH} = t_{THL}$ = 10 ±2 ns	$t_{TLH} = t_{THL}$ = 10 ±2 ns

3. Before making the measurements specified, verify that the output pulse high-level and the low-level is as follows:

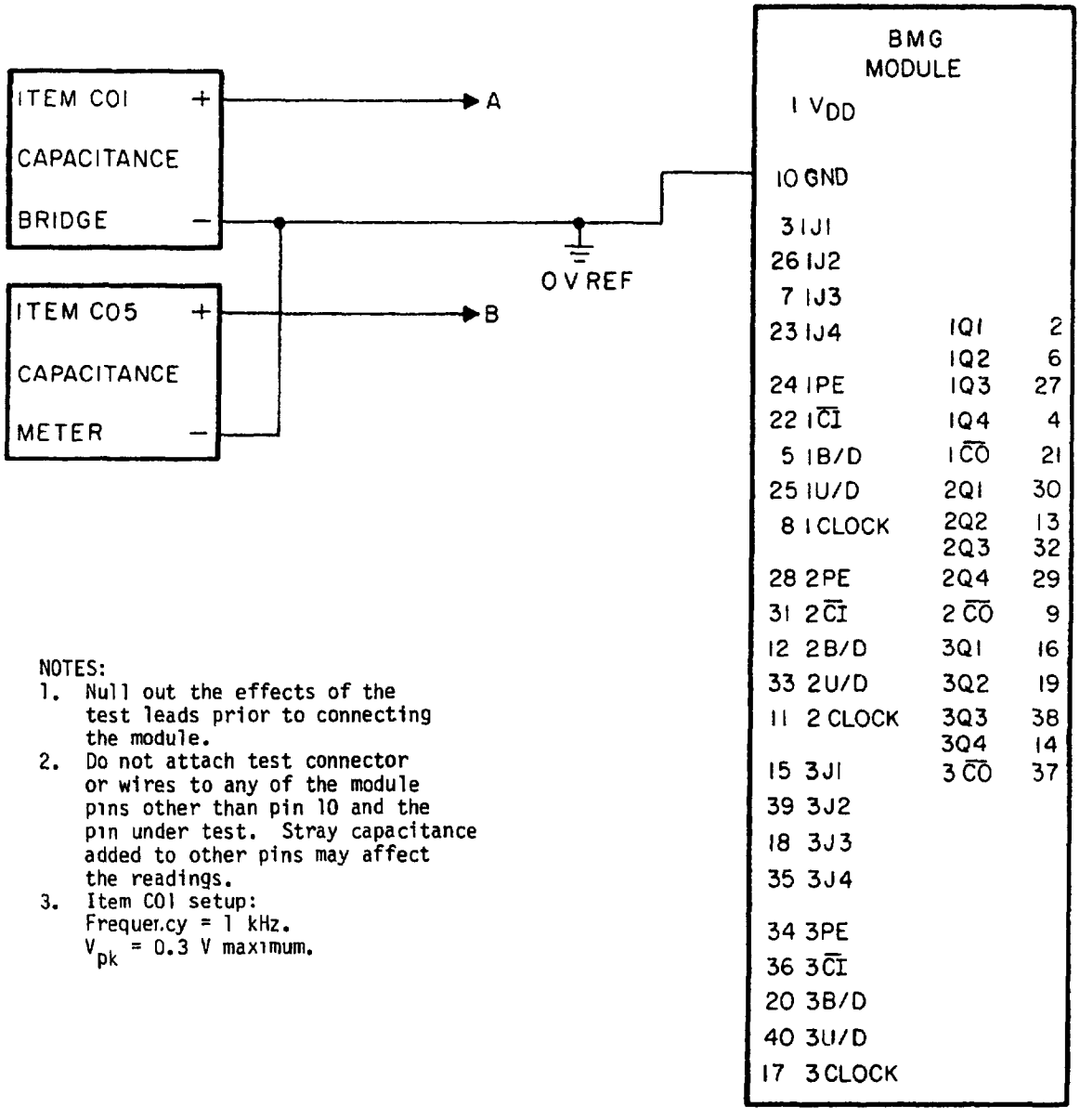
For test conditions:	High-level	Low-level
NN1-NN3	≥ 4.6 V	≤ 0.4 V
PP1-PP3	≥ 9.5 V	≤ 0.5 V
QQ1-QQ3	≥ 13.5 V	≤ 1.5 V

4/ The lead length of electrical conductors between resistors R2 and R3 and input pin and item Y01-1 to input pin shall be less than 1 inch (25.4 mm).

5. All voltages are referenced to ground.

FIGURE 35. Preset enable setup and removal time test circuit waveforms for table XXVII.

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NOTES:
1. Null out the effects of the test leads prior to connecting the module.
2. Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test. Stray capacitance added to other pins may affect the readings.
3. Item C01 setup:
   Frequency = 1 kHz.
   V_{pk} = 0.3 V maximum.

FIGURE 36. Power supply filter and input capacitance test circuit for tables XXVIII and XXIX.

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